

THE ECOLOGY OF LENTOGENIC NEWCASTLE DISEASE VIRUS IN PASSERINES

by

ANDREA JOSEFINA AYALA

(Under the Direction of Sonia M. Hernandez)

ABSTRACT

Newcastle disease virus (NDV), also known as avian orthoavulavirus-1, is among the most significant of poultry pathogens. In the United States, free-roaming backyard chickens are rarely subjected to biosecurity that reduces the transmission of pathogens such as NDV among wild birds. Resources such as chicken feed, water, and coops attract passerines, increasing the likelihood of transmission between them two groups. In the field and under biosafety laboratory - 2 (BSL-2) conditions, we assessed passerine susceptibility, transmission capabilities, and behaviors conducive to infection with lentogenic NDV from backyard chickens. In BSL-2, Northern Cardinals (*Cardinalis cardinalis*), Brown-headed Cowbirds (*Molothrus ater*), American Goldfinches (*Spinus tristis*) and House Finches (*Haemorrhous mexicanus*) shed virus when directly inoculated with the LaSota vaccine virus, although antibody responses and shedding patterns varied. The same species were co-housed for 14 days with specific-pathogen free (SPF) chickens inoculated with LaSota. While Northern Cardinals and Brown-headed Cowbirds remained uninfected, 20% of House Finches (1/4) and (1/5) American Goldfinches tested positive by real-time PCR and virus isolation. In addition, we collected 1,574 observations of 72 species across three sites associated with backyard chickens. One site was managed as a free-ranging flock, one site primarily cooped their chickens, and the third site served as the control, without chickens. Fourteen species were repeatedly observed performing behaviors conducive to NDV transmission, such as chicken feed consumption. Chipping Sparrows

(*Spizella passerina*), Northern Cardinals, and Tufted Titmice (*Baeolophus bicolor*) accounted for the greatest number of visits. To understand the spillover dynamics of lentogenic NDV from backyard chickens to wild birds, we inoculated both backyard chicken flocks with the LaSota vaccine virus and surveyed the third site as the control. At the two inoculated sites, thirteen species of passerines either developed or demonstrated increases in NDV antibody titers, whereas all passerines at the control site remained seronegative. As interactions between wild birds and chickens frequently occur, we must educate backyard chicken owners on how to mitigate disease risks such as NDV.

INDEX WORDS: Newcastle disease virus, LaSota, avian orthoavulavirus-1, avian paramyxovirus-1, wild birds, passerines, songbirds, Northern Cardinals, House Finches, Brown-headed Cowbirds, Tufted Titmice, Carolina Chickadees, Chipping Sparrows, Eastern Phoebe, hemagglutination inhibition assay, virus isolation, reverse-transcriptase real-time PCR, serology, backyard chickens, biosecurity, bioexclusion, husbandry, avian community disease risk, behavioral susceptibility, spillover, transmission, RNA virus, viral shedding, lentogenic, avian disease, poultry disease

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ANDREA JOSEFINA AYALA

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ANDREA JOSEFINA AYALA

Major Professor:	Sonia M. Hernandez
Committee:	Michael J. Yabsley
	Kaori Sakamoto

Electronic Version Approved:

Ron Walcott
Interim Dean of the Graduate School
The University of Georgia
May 2020

DEDICATION

I dedicate this to Poohbear, Belle, Bianca and Patches. You sat mesmerized as I regularly recounted tales of recalcitrant birds, unobliging viruses, and incredulous colleagues. Together we chased North Carolina waterfalls, Tennessee smoky mountains, Kansas sunsets, and California dreams. Poohbear and Belle, I miss you.

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CHAPTER 1

INTRODUCTION

The appearance of pathogens in new hosts is increasing as a result of human-induced habitat perturbations (Parker, Whiteman, and Miller 2006; Lebarbenchon et al. 2008).

Anthropogenic modifications such as habitat encroachment, urbanization, and non-native species introductions significantly influence the natural world, and the organisms in those habitats. In terms of pathogen emergence and re-emergence, these activities also often work in concert to facilitate the establishment, host-shifting, or virulence of avian pathogens (Bradley, Gibbs, and Altizer 2008; Bradley and Altizer 2007; Boal and Mannan 1999; Atkinson et al. 2000; Ricklefs 2010; Friend and Trainer 1970).

Concurrent with habitat encroachment, the ownership of backyard poultry is on the rise across urban, suburban, and rural areas. Backyard flocks have been implicated in major epidemics of both Newcastle disease virus (NDV) and highly pathogenic avian influenza virus (HPAI) within the commercial poultry industry. For the most part, these epidemics have been attributed to a combination of limited biosecurity and vaccination compliance (Tiensin et al. 2007; Crespo et al. 1999; Breitmeyer, Whiteford, and Shere 2004; Capua et al. 2002; Garber et al. 2007; USDA 2010). In addition, approximately thirty percent of backyard poultry owners maintain wild bird feeders on their properties (USDA 2010), increasing contact rates to free-ranging birds in a potentially infectious environment. ‘Poultry’ pathogens isolated from

backyard poultry and domesticated turkeys also identified in wild birds include *Bordetella avium* (Raffel et al. 2002), avian encephalomyelitis virus (Padilla et al. 2003), hemorrhagic enteritis virus (Schrenzel et al. 2005; Sileo et al. 1983), infectious bronchitis virus (Hughes et al. 2009), immunosuppressive infectious bursal disease virus (Wilcox et al. 1983; Ogawa et al. 1998), and Newcastle disease virus (OIE 2012; Diel et al. 2012; Miller and Afonso 2011). In addition, novel pathogen host shifts that originated from poultry to wild birds include the epizootic spread of *Mycoplasma gallisepticum* in House Finches (*Haemorrhous mexicanus*) and related passerines (Delaney et al. 2012). The viral re-assortment (mutation) of low pathogenic avian influenza to highly pathogenic influenza in poultry is a consistent risk, given that the virus may spill back into waterfowl and shorebirds as virulent and lethal highly pathogenic influenza strains (Olsen et al. 2006; USGS 2013; Sturm-Ramirez et al. 2004; Aly, Arafa, and Hassan 2008).

Backyard poultry and live bird markets remain ‘hotspots’ of transmissible pathogens to free-ranging wild birds (Songserm et al. 2006; Aly, Arafa, and Hassan 2008; Marangon and Busani 2007). An extensive outbreak of highly pathogenic avian influenza in Houston, Texas, originated from a live bird market as a result of surveillance of the virus for backyard poultry (Elbakidze 2008). In eastern Texas, a survey of backyard poultry flocks found that 100% of each flock harbored individuals with antibodies to infectious bursal disease virus (Millar and Naqi 1980). Moreover, millions of migratory birds utilize the four major flyways of North America each year. In order to prevent the transmission to pathogens from backyard chickens to highly

mobile wild birds, education policies and extension services are imperative for residents owning backyard poultry proximal to migration routes or stopovers.

Backyard poultry flocks serve as a primary source of protein within some industrializing nations. However, in areas where backyard chickens free-roam with wild birds, they are also considered high risk sources of pathogens to Neotropical or island endemic birds (Gottdenker et al. 2005; Hernandez-Divers et al. 2008; Hernandez-Divers et al. 2006). A cross-sectional surveillance study of backyard poultry in the Galápagos Island of Santa Cruz resulted in a high seroprevalence for six common avian cross-species pathogens in free-roaming chickens. Surprisingly, all wild birds tested from the same sites were seronegative. Thus, the authors concluded that the highly endemic Galápagos avifauna were too moribund for capture, or quickly succumbed upon transmission (Soos et al. 2008). This observation is not only significant for conservation and the repatriation of endangered species into Galápagos National Park, but for all wildlife refuges, where the risk of disease transmission transcends the borders of protected habitats (Cunningham 1996; Snyder et al. 1996).

Disease transmission from backyard poultry and live bird markets to wild birds has generally only been causally linked. However, it has become clear that backyard poultry play a pivotal role in the transmission of potentially virulent pathogens (Causey and Edwards 2008; Fischer et al. 1997). Specifically, the absence of natural cross-species barriers between backyard poultry and native birds may increase pathogen transmission as a result of high-density poultry

and native bird flocking (Lebarbenchon et al. 2006). In addition, backyard poultry and wild birds may come in contact with pathogens through the use of wild bird feeders (Dobson and Foufopoulos 2001), poor hygienic practices, and the lack of biosecurity such as coop enclosures (Arzey and Littleton 2007; Songserm et al. 2006; Davies and Breslin 2003). The foraging behaviors of wild birds attracted to chicken feed and water sources may increase environmental sources of risk both wild birds and backyard chickens (Hasan et al. 2012). My doctoral research examined the role that backyard poultry husbandry may play upon lentogenic Newcastle disease virus transmission into wild birds.

In Chapter 1, we reviewed the literature to identify poultry pathogens that have spilled over into backyard chickens and wild birds. In Chapter 2, we performed an experimental inoculation study under laboratory conditions to determine whether four species of wild birds were susceptible to the Newcastle disease virus LaSota vaccine virus as a result of direct inoculation. In the same study, we then examined whether wild birds were susceptible to infection with the LaSota vaccine virus as a result of exposure to chickens that were shedding the LaSota vaccine virus. In Chapter 3, we examined whether the LaSota vaccine virus could be transmitted from infectious backyard chickens, and the role that husbandry plays in influencing those viral transmission dynamics. Finally, in Chapter 4, we performed an experimental supplemental feeding experiment followed by observations at three sites associated with backyard chickens in Athens, Georgia. One site allowed their chickens to free-roam during the

day, the second site cooped their chickens all day, and the third site served as the control, with the chickens having been recently removed. At each site, we identified the species of wild birds a) that shared habitat with the chickens b) for whom a frequency of detection could be calculated and c) that encountered the chicken coops.

It is also intended that the novel use of the vaccine as the model pathogen will also yield transmission data for similar RNA viruses of respiratory and enteric importance, such as avian influenza (Alexander 2009).

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CHAPTER 2

A REVIEW OF PATHOGEN TRANSMISSION AT THE BACKYARD CHICKEN – WILD
BIRD INTERFACE ¹

¹ Ayala, A.J, M.J. Yabsley, and S.M. Hernandez. 2020. Submitted to *Frontiers in Veterinary Medicine*, 03/03/2020

Abstract

Habitat conversion and the expansion of domesticated, invasive species into native habitats are increasingly recognized as drivers of pathogen emergence at the agricultural-wildlife interface. Poultry agriculture is one of the largest subsets of this interface, and pathogen spillover events between free-ranging poultry and wild birds are becoming more commonly reported. Native wild bird species are under numerous anthropogenic pressures, but the risks of pathogen spillover from domestic chickens have been historically underappreciated as a threat to wild birds. Now that the backyard chicken industry is one of the fastest growing industries in the world, it is imperative that the principles of biosecurity, specifically bioexclusion and biocontainment, are legislated and implemented.

We reviewed the literature on spillover events of pathogens historically associated with poultry into wild birds. We also reviewed the reasons for biosecurity failures in backyard flocks that lead to those spillover events and provide recommendations for current and future backyard flock owners.

Introduction

Transboundary emerging and re-emerging infectious diseases are now increasingly recognized as interconnected threats to public health, livestock, and wildlife communities [1-3]. In North America, almost 80% of World Organization for Animal Health (OIE) reportable pathogens require at least one wildlife species to complete their life cycles, and half of those are

zoonotic [4]. Livestock losses to pathogens such as Newcastle disease virus (NDV) and highly pathogenic avian influenza (HPAI) not only impact international trade, but they also increase food insecurities among impoverished peoples, in regions where protein sources are limited [5-7]. Pathogens that we associate with human populations, such as measles, influenza A, and pertussis, have been known to man since antiquity; yet are now known to have originated within domesticated species [8].

Almost two decades ago, A Dobson and J Foufopoulos [9] defined emergent pathogens as those whose “geographical range, host range, and/or prevalence” are expanding. As transboundary emerging pathogens continuously broaden their ranges, it is no coincidence that the abiotic and biotic ecosystem barriers which have impeded transmission are also dwindling [10, 11]. Pathogens are exploiting anthropogenic encroachment into wildlife dominated terrain, which facilitates the co-mingling of once separate wildlife and domestic species [10, 12-14]. Such activities include the wildlife trade [15], urbanization [16], and agricultural overlap into wildlife landscapes [17, 18].

Pathogen emergence at the global livestock-wildlife interface

The livestock-wildlife interface is a landscape now associated with rising incidences and expanding distributions of OIE-reportable pathogens [17, 19-21]. The convergence of food animal production activities with wildlife habitats form optimal circumstances for multi-host pathogen and zoonosis emergence [22-25]. A classic example of this phenomenon is

Mycobacterium bovis in the UK and the Republic of Ireland, whereby cattle and wild European Badgers (*Meles meles*) remain locked in a spillover and spillback transmission loop that remains unsolved with contemporary disease control measures [26-28].

Pathogen emergence at the livestock wildlife interface is the consequence of simultaneous perturbations such as pathogen pollution, climate change, biodiversity loss, habitat fragmentation, and agricultural sprawl [29-35]. When perturbations become permanent, disruptions to the ecosystem services of disease regulation are the result [32]. In many systems, the origin of the wildlife-livestock interface starts with landscape-level processes, such as globalization, which facilitate the translocations of livestock, wildlife, and their microbes into new areas [16, 36, 37]. Habitat for livestock is then converted to into farmland, undergoing wetland drainages for irrigation and deforestation for crops [21, 35]. Subsequently, this may lead to the disturbance of hibernating reservoirs, i.e. bat and rodent species [38, 39], as well as changes to the distributions and activities of local vectors [33].

At a community level, localized biodiversity loss is a consequence of land-use change, which may result in a disproportionate number of competent or amplifying hosts [29, 40]. Biodiversity loss in conjunction with invasive species that outcompete native taxa may also increase the densities of vector species such as ticks, mosquitoes, and bats [41-43]. Fundamentally, these processes are ultimately shaped through the interactions of the most significant components of the interface, first, the stability of the wildlife community [44],

second, the type of domestic animal production [45, 46], and finally, the level of ecosystem fragmentation [47, 48].

In short, pathogen emergence at the livestock-wildlife interface is the result of any one or many mechanisms. Nonetheless, those factors generally fall into three broad categories: changes among reservoir host and recipient host densities [19], the rate of habitat transformation [49], and the presence of multi-host pathogens [50]. Although one category (host, habitat, or pathogen) may appear to dominate in a given scenario, understanding how infectious diseases emerge from the livestock-wildlife interface requires a holistic approach of the host-habitat-pathogen triad [51-53].

Poultry Agriculture and Wild Birds

The pathogen dynamics of poultry based agriculture are unique among animal-based farming systems [54-57]. Birds are often owned as pets as well as for food [58, 59], they have a unique physiology [60-62], and have an extraordinary taxonomic diversity [63-65]. Many wild bird habitats also now suffer consistent encroachment or habitat conversion from the expansion of the poultry industry, which introduces new microbes into the wild bird community [66-69]. In 2018, Birdlife International listed agricultural expansion and intensification as a major threat to 74% of the world's 1,469 globally threatened bird species [70]. Global threats also include the introduction of invasive species, habitat conversion, pathogens, and logging for small-scale subsistence farming [70]. As early as the mid-20th century, emerging and re-emerging

infectious diseases were increasingly recognized as threats to avian populations around the globe [71-73].

Historically, avian infectious diseases were not appreciated for their ability to influence populations, and relatively neglected for their part in causing species declines [74, 75]. However, within the last twenty years, up to 18 species of primarily endemic island-species were declared extinct, or extinct in the wild due to infectious diseases coupled with invasive species displacement [76, 77]. Currently 98 wild bird species are at considered at risk by the World Conservation Union (IUCN) and Birdlife International due to infectious diseases [70]. It is only recently that the mechanisms contributing to disease emergence in free-ranging birds have been investigated [78]. In particular, endangered species and the role that avian diseases present in regulating their populations have garnered specific attention from conservationists and wildlife epidemiologists [77, 79-87].

Pathogens have been implicated in more than just mortality events. They have also exhibited sublethal, population level consequences to native birds such as fluctuations in breeding success and reductions to fecundity [88, 89]. Pathogens have also been demonstrated to inhibit territorial defense mechanisms, and allow non-native species to outcompete native birds from their habitats [88-91]. Unfortunately, public perception also plays a role on the effects that pathogens have on avian populations. In particular, the attention directed towards RNA viruses of public health concern, primarily the H5 and H7 variants of HPAI that arise from Southeast

Asian agricultural practices have at times led to the culling of migratory waterfowl and shorebirds in those locations [92, 93].

The spillover of poultry pathogens are reported regularly [94-97]; unfortunately, much remains unknown regarding which wild species are consistently affected and the frequency with which it occurs. The intensive practices that allow the poultry industry to produce more chickens, e.g. high-density rearing, also result in the maintenance of pathogens, and increased pathogen transmission [55, 98]. Poultry vaccines have contributed to the efficient, high-density and high-output avian production that comprises the commercial poultry industry [99-101]. However, the same vaccines may also provide optimal conditions for pathogens to spread rapidly due to a lack of sterilizing immunity [100, 102-107]. In commercial systems, producers routinely apply live, killed or in-vivo vector-based vaccines to counter high-profile viruses and reduce economic losses [99]. Economically important viruses in which mass vaccination practices are applied include: infectious bursal disease virus (IBDV), infectious bronchitis virus (IBV), infectious laryngotracheitis, and Newcastle disease virus (NDV) [108-111].

High-density commercialization is not the only form of poultry production, nor is it likely the most dominant worldwide. Village poultry remain the primary source of protein among some impoverished peoples [6, 112-115], while urban chicken-keeping is expanding among industrialized nations [116-118]. In some European nations, backyard chickens are even outcompeting commercial industries [119]. Overall, poultry production has increased over the

last 70 years, and the growth of the industry is unlikely to slow in the foreseeable future [115, 120, 121].

This recent rise in backyard poultry ownership or ‘micro-livestock’ among developed nations such as the U.S. is a unique subsection of the agricultural-wildlife interface, with tremendous implications for multispecies transmission [122-132]. Common avian pathogens or antibodies against avian pathogens isolated from urbanized backyard poultry and domesticated turkeys include infectious bronchitis [123, 133], Marek’s disease virus [134], infectious bursal disease virus [123, 133, 134], *Mycoplasma* spp.[123, 133, 134], NDV [123, 133, 135], *E. coli* [134], and *Salmonella* spp.[122, 123, 134].

A meta-analysis performed by AK Wiethoelter, D Beltrán-Alcrudo, R Kock and SM Mor [17] reported that the avifaunal-poultry interface was either the most highly reviewed or studied interface in relation to the worldwide risks of emergent pathogens. This is partly due to the evidence supporting that hobby and village poultry are now established within the world’s costliest cities [130, 136-138] to the edges of protected areas in developing nations [139-142]—suggesting that few terrestrial habitats have remained untouched and without risk from poultry-origin outbreaks.

Hobby poultry’s explosive growth as an industry results from increasing consumer demands for organic, humane meat and eggs [143-145], and the growing desire for a sustainable, farm to table food source [116]. Many backyard chickens are kept as pets, due to animal welfare

efforts towards humanely rehoming ‘spent’ or non-laying hens [146, 147]. Lastly, in many areas, there remains a desire to continue long-standing cultural practices of maintaining family flocks [124-126]. What remains unresolved are the potential long-term effects of the backyard poultry industry’s unregulated biosecurity and limited veterinary care [148, 149] on our native wild bird species. Poor pathogen management, coupled with the overlap of native wild bird habitats, serves to bring together the microbial communities endemic to domestic chickens, [150, 151] and those endemic to native birds [152, 153].

Pathogen Transmission Between Wild and Domestic Birds

Pathogen transmission dynamics at the poultry-avifaunal interface are not only subject to within-host ecological and evolutionary pressures [154], but most importantly, are frequently bidirectional in nature [46, 155, 156]. The establishment of “open-air” or “free-range” poultry habitats that overlap with wild bird habitats is a key step in the loss of sympatric species contact barriers; which may facilitate the transmission of opportunistic microbes [153, 157-159]. The introduction of high densities of poultry into a diverse, susceptible avian community are likely to accelerate wild and domestic bird encounter rates with generalist pathogens. For example, the most commonly cited group of viruses that follow this model are the highly pathogenic avian influenza (HPAI) subtypes [160, 161]. Repeated transmission events of generalist, multi-host pathogens into a recipient species, in this case, domestic chickens, from wild bird spillover hosts may lead into pathogen establishment and spillback from backyard poultry [162]. This has the

potential to establish a positive feedback cycle among susceptible wild birds and domestic chickens, such has occurred over the last two decades with the H5N1 goose/Guangdong (GsGD) lineage of highly pathogenic avian influenza virus [163-166].

It is well-established that backyard poultry may serve as pathogen reservoirs to the commercial poultry industry [167, 168], and that the most likely mechanism of spillover involves wild birds [155]. For example, Lebarbenchon et al. [169] hypothesized that small passerines served as bridge hosts for the H7N2 virus low-pathogenic avian influenza virus (LPAI) from infectious waterfowl to U.S. commercial turkey houses. However, the wild bird species that may serve as spillover hosts must also be considered from a conservation standpoint; in particular, species of special concern as defined the World Conservation Union (IUCN).

Newcastle disease (ND), caused by virulent Newcastle disease virus (NDV) strains, is a highly contagious, acute and systemic illness primarily known to afflict poultry, however, clinical symptoms have also been documented among bird species outside of the Galliformes family [170, 171]. NDV has been isolated from a broad range of avian hosts, and it is now generally presumed that all bird species are susceptible to the replication, shedding, and transmission of the virus [172-174]. In a review of Newcastle disease virus vaccine spillover events, Ayala et al. [175] found that 9.3% percent of the vaccines were isolated from an IUCN listed species. In Latin America, free-ranging backyard flocks have been investigated as potential sources of pathogen spillover into Neotropical or island endemic birds, including NDV [150,

151, 176]. Across four African countries, Cappelle et al. [177] found that 8.9% of the species that tested positive for NDV by real-time PCR were listed by the IUCN as either vulnerable or near threatened.

On the Galápagos Island of Santa Cruz, Soo et al. [142] found that backyard poultry on the island yielded a high prevalence of seropositive chickens for six common ‘poultry’ pathogens, whereas all nearby wild birds that were tested were seronegative. As it was unlikely that all wild birds that were truly negative to exposure from poultry pathogens, it was concluded that the highly endemic Galápagos avifauna potentially succumbed too quickly after infection to seroconvert [142].

More recently, a similar study was conducted on the Galapagos island of Floreana, which has a much longer history of anthropogenic and agricultural modification than Santa Cruz. In this case however, endemic birds yielded positive serological tests against NDV, avian poxvirus and avian adenovirus-2 [140]. The contrasting results between the two islands may be explained by the evolution of pathogen resistance in Floreana birds, which have been exposed to these viruses over a longer period [140]. Pathogen resistance has been proposed as a mechanism for the survival of the Hawaiian Amakihi (*Hemignathus virens*), which was nearly extirpated by avian malaria, specifically *Plasmodium* spp. [178]. However, Deem et al. [140] concluded that any reintroduction efforts of endangered wild bird species into Galápagos National Park would have a high likelihood of failure due to the risk of pathogen transmission. These observations are

significant for the endangered species repatriation across all wildlife refuges, as pathogen transmission from backyard chickens has the potential to transcend the borders of protected habitats [179, 180].

‘Poultry’ pathogens detected from wild birds have been reviewed previously by Gottdenker et al. [141]. However, some notable examples from the literature include the gram-negative bacterium *Bordetella avium* in waterfowl, psittacines, and passerines [181], a pathogen most commonly associated with bordetellosis in commercial turkeys [182, 183]. Avian encephalomyelitis virus is a picornavirus with a worldwide distribution that infects juvenile chickens, pheasants, quail and turkeys [184]. It has also been reported in songbirds of the Paridae family [184], wild turkeys (*Meleagris gallopavo*) in the Southeastern United States [185] and Rock Pigeons (*Columba livia*) in Turkey [186]. Corvids appear to be particularly susceptible to infection with avian reovirus [187], while two die-offs of American Woodcocks (*Scolopax minor*) were attributed to the virus [188].

Avian lymphoid leukosis virus has been consistently isolated from both captive and wild bird species, including passerines, Columbids, waterfowl and psittacines [189-193]. Fowl adenovirus-4 (FAdV-4), an emerging pathogen of poultry, was isolated from Rock Pigeons of Hong Kong and Black Kites (*Milvus migrans*) of Kashipur, India [194-196]. Infectious bronchitis virus, a gamma-coronavirus of poultry, has been isolated from waterfowl and wading birds of Beringia, Alaska, and the nation of Poland [197]. Infectious bursal disease virus (IBDV)

is an immunosuppressive virus that targets B-lymphocytes and associated tissues of the immune system [108]. It has been isolated from a wide variety of bird species, including waterfowl, Columbids, passerines, Galliformes, and members of the Charadriiformes [198-201].

The bacterium *Mycoplasma gallisepticum* spilled over from poultry in 1994 into House Finches (*Haemorrhous mexicanus*) and continues to establish itself into North American passerine species [202, 203]. A related species in the same genera known as *Mycoplasma synoviae* has also occasionally been isolated from passerines and near passerine species [204, 205]. In one instance, an isolate with genetic similarities to an *M. synoviae* vaccine strain was isolated from a captive Lesser Flamingo (*Phoeniconaias minor*) in Northern Italy [206]. Marek's disease virus is a neoplastic virus in the Herpesviridae family, which is ubiquitous among chicken flocks worldwide [207]. The virus or antibodies against the virus have been detected in captive members of the Galliformes and wild waterfowl, including endangered Lesser White-fronted Geese (*Anser erythropus*) [208-211]. Reticuloendotheliosis viruses (REV) are a group of retroviruses and the causative agent of reticuloendotheliosis, an immunosuppressive and neoplastic disorder of chickens [212]. The virus has been isolated from Galliformes, waterfowl, Columbids and endangered Attwater's Prairie Chickens (*Tympanuchus cupido attwateri*) [213-215].

MacQueen's Bustards (*Chlamydotis macqueenii*) are categorized as vulnerable by the IUCN [216]. Illegally trapped and transported individuals were found to have been exposed to

NDV, avian poxvirus, and *Chlamydia* spp. [217]. In Burkina Faso, three Hooded Vultures (*Necrosyrtes monachus*), a Western Plaintain-eater (*Crinifer piscator*) and an Ovambo Sparrowhawk (*Accipiter ovampensis*) were infected with various combinations of HPAI, NDV, and infectious bronchitis virus [218]. Given that Hooded Vultures are categorized as critically endangered by the IUCN, this finding is extremely significant [219]. Similarly, Eurasian Oystercatchers (*Haematopus ostralegus*) are listed by the IUCN as vulnerable in Europe [220]. In the UK, 12% of sampled Eurasian Oystercatchers, along with various waterfowl and shorebirds were found to be infected with infectious bronchitis virus [221]. Several species of Antarctic penguins, including the near threatened Emperor Penguin (*Aptenodytes forsteri*), have been found to be infected with *Chlamydia* spp., LPAI, NDV, and IBDV [222]. A die-off of reintroduced, endangered Whooping Cranes (*Grus americana*) in Florida was attributed to IBDV, serotype 2 [223]. Lastly, a low-virulent NDV strain was isolated from two Bald Eagles (*Haliaeetus leucocephalus*) and one Great Horned Owl (*Bubo virginianus*) in Minnesota during the winter of 2009 [224].

Even given all these reports, pathogen transmission between backyard poultry and native birds remain only causally linked, and likely underreported [175]. Moreover, the mechanisms of transmission differ across pathogens and avian host species [225, 226]. Consequentially, conclusive evidence for pathogen host-shifts have generally only been reconstructed with molecular techniques after enzootics have resulted in high mortality rates, i.e. *Mycoplasma*

gallisepticum in Fringillids [203]. The same molecular techniques have also clarified the origins of pathogens once attributed to chickens. For example, Canarypox has been identified as the causative agent of disease in Hawaiian and Galápagos avifauna as opposed to chicken-origin Fowlpox viruses [227, 228].

It has become clear that backyard poultry do play a role in the transmission of potentially virulent pathogens [75, 229, 230]. Specifically, is it the absence of previously described crossspecies barriers between backyard poultry and native birds that may exacerbate pathogen transmission. When gregarious and social native bird species overlap in habitat with high-density poultry, open-air operations, the ecological barriers to pathogen transmission are lost [231]. This scenario pertains to peridomestic birds that consume free-ranging backyard chicken feed and water sources, which may then interact with other wild bird species that would have otherwise remained unexposed [232-234].

Risk assessment of backyard chickens

Backyard flocks are implicated in maintaining enzootics of two critically important RNA viruses around the globe, NDV and highly pathogenic avian influenza (HPAI) [235-243]. This is largely due to essential differences in how backyard flocks and commercial flocks are managed, primarily concerning biosecurity [225, 244]. For instance, while the commercial industry practices high containment and mass immunization against NDV, backyard flocks are subject to

little to no biosecurity compliance, and only three to ten percent of flocks are immunized for common poultry pathogens [124, 245, 246].

In areas where vaccination, hygienic measures, and biosecurity compliance remain logistically prohibitive, backyard poultry and live bird markets are sources of transmissible pathogens [247, 248]. Moreover, limitations in husbandry practices are sometimes the result of finite resources. In these cases, clinical illness may be overlooked due to limited experience with disease, or otherwise unreported, for fear of mandatory flock culling [99, 125]. Even more worrisome is the practice of covertly transporting and selling sick fowl or discarding infected eggs and carcasses into the environment. Such practices have been attributed to maintaining cyclic NDV among unvaccinated flocks [249].

Surveys of backyard chicken owners show that people do not understand disease transmission associated with various practices, such as carcass disposal, coop cleaning regimens, and the proximity of wild birds [118, 137, 250]. Approximately 30% of backyard poultry owners maintain wild bird feeders, increasing contact rates to free-ranging birds within infectious environments [246]. Backyard flock owners span socioeconomic, geographical, and community types, i.e. urban, rural, suburban [250], and the average flock size per household varies from 25 to 49 birds [117, 251]. With the exception of local zoning regulations, backyard chicken laws remain primarily unregulated and/or unenforced. In addition, poultry specialists, extension

veterinarians, and community educators increasingly lack the resources needed to monitor the continuously expanding and vulnerable backyard poultry sector [251].

Doubtlessly, hobby chickens are increasing in popularity, and between 2% and 7.4% of U.S. households without chickens plan to own them within five years [117]. As backyard chickens continue to expand, the issues are pressing further to the surface. Practices such as inconsistent husbandry management [124], poor vaccination compliance [246], and a reduction in national poultry ‘herd immunity [252]’ perpetuate disease vulnerability to the commercial food supply and to native bird communities. Wet markets, the illicit pet bird trade, and poultry trafficking have instigated low-pathogenic avian influenza outbreaks in the Northeastern U.S. [253, 254], and were responsible for the 1972 [255], 2002 [256] and current virulent Newcastle disease virus outbreaks in California [257]. In 2013, the Food and Agriculture Organization of the United Nations identified “smallholder livestock systems” as disproportionately large facilitators of infectious disease spread [258].

Critical events, such as the 2014-2015 avian influenza H5Nx (HPAI) crisis in the United States not only leads to the loss of public confidence in food security, but also further decimates the agricultural economy through mass depopulation, quarantines, and trade embargos [259, 260]. In some cases, wild avian species may even undergo mass mortality events [261-264]. For example, in the 2014-2015 H5Nx outbreak, some North American waterfowl, shorebirds, and

even raptors did not survive infection with avian influenza due to circulating HPAI strains that were poorly adapted to these species [244].

In the United States, outbreaks of low-pathogenic avian influenza (LPAI) H7 variants in live bird (wet) markets were sporadically detected in the mid-Atlantic and Northeastern areas between 1986 and 2004. These occurred primarily in New York, New Jersey, Massachusetts, Rhode Island, Connecticut and Pennsylvania [253, 254, 265]. An extensive outbreak of HPAI in Houston, Texas in 2004 also originated from a live bird market [266]. In eastern Texas, an earlier survey of backyard poultry flocks found that 100% of each flock harbored individuals with antibodies to infectious bursal disease virus [267]. Such events demonstrate that backyard chickens and the movement of live poultry not only pose a risk to native birds, but to the commercial poultry industry as well [137].

Conclusions and Recommendations

Although spillover and spillback at the poultry-avifaunal interface has been documented, for many pathogens, the underlying reasons that facilitate spillover and spillback remain confounded. Recommendations are provided here, but they are not exhaustive, and should be tailored to the local circumstances, flock types, laws and regulations [268]. Arzey et al. [268] provides a thorough review of backyard poultry recommendations for small-scale producers which are easily adapted to the backyard flock owner.

Pathogen spillover may be a result of poor bioexclusion, such as the failure to quarantine new flock individuals or inadequate coop enclosures that do not separate wild birds from backyard chickens [256, 269]. For instance, biosecurity and quarantine protocols are two of the most common measures undertaken by the poultry industry to reduce pathogen transmission from backyard chickens and wildlife [270]. Restricting the movements of infectious birds or equipment, including transport, is imperative to reduce transmission to susceptible individuals – both into chickens and wild bird species [268, 271]. Poor vaccine compliance is also a failure of disease management, specifically because vaccination can reduce clinical signs, and the amount of virus shed from an infected chicken that was previously vaccinated [111, 272, 273].

Backyard flock owners should not maintain wild bird feeders on their properties, thereby inhibiting visits from large flocks of multiple wild avian species which may include susceptible or even infectious wild birds [124]. Reducing the number of domestic flocks that contain both poultry and waterfowl in backyards and wet markets would reduce cross-species ‘pathogen-swapping’; which is critical to the control of LPAI and HPAI in domestic birds [247, 274]. Removing contaminated water sources, insects, and rodents reduces point sources of pathogen contamination, not just to other chicken flocks but also to wild birds [268]. Increased attention to owner and visitor hygiene, i.e. the changing of footwear when visiting different flocks, and limiting the number of visitors to backyard coops, are important principles of bioexclusion and biocontainment [124, 274]. Lastly, dead birds, or eggs that are suspected of contamination

should be disposed of in a manner that complies with local guidelines, for example, incineration or if applicable, burial, according to environmental guidelines [268]. If disease is suspected in a mortality event, veterinary colleges in the United States will often accept well-packaged and safely shipped carcasses for necropsy [275].

Backyard flock owners increasingly report emotional ties to their chickens [246, 276]. Thus, the inclusion of standard avian veterinary care, and the management of common chicken pathogens into flock routines are realistic objectives in nations like the United States [129, 132]. However, in areas where backyard chickens are kept strictly as livestock, these recommendations may become impractical to the small-scale flock owner [225]. Effectively implementing biosecurity beyond the community level likely includes a regional assessment of critical backyard chicken pathogens, their mechanisms of spread, and a methodology to interrupt those pathways [57]. Such threat assessments and the resources to implement them may be beyond the means of single flock pastoralist [6]. Thus, regional biosecurity planning would likely involve educational, financial, and regulatory interventions at a governmental level.

As previously discussed, pathogen spillover is ultimately driven through the ecological loss of species barriers. When multiple species of the same phylogenetic order are carrying individual host pathogen biomes, and are then thrust into sharing the same habitat, spillover between those species is likely to occur. Moreover, certain behaviors in domestic and wild birds lend themselves to increasing that probability, such as ground foraging, or the propensity to

congregate in flocks. However, in the literature on the poultry-avifaunal interface, there remains a paucity of experimentally and empirically derived field data regarding the bidirectional potential of poultry-avifaunal pathogens outside of HPAI.

Bidirectional potential includes the ability of the collective wild and domestic avian host community to maintain a pathogen traditionally associated with poultry above the threshold levels required to continue an outbreak. Some pathogens have been extensively studied in the field, laboratory, and through mathematical models, i.e., avian influenza. This is likely due to the zoonotic potential and public health risk of the virus, in addition to the economic burden following a commercial or village poultry outbreak [277-280]. However, even current surveillance methods for HPAI among reservoir avian species have been reported as ‘unsatisfactory’ when contrasted against the efforts applied towards understanding human to human transmission [281]. Further research is needed into the poultry-avifaunal interface, especially near sites such as Important Bird Areas (IBPs), where the conservation of declining species is a priority.

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CHAPTER 3

EXPERIMENTAL INFECTION AND TRANSMISSION OF A NEWCASTLE DISEASE
VACCINE VIRUS IN FOUR WILD PASSERINES ¹

¹ Ayala, A.J., S.M Hernandez, T.L. Olivier, C.N. Welch, KM.Dimitrov, I.V. Goraichuk, C.L. Afonso, P.J. Miller. *Avian Diseases* 63: 389-399. Reprinted here with permission of the publisher.

SUMMARY

Our prior work has shown that live poultry vaccines have been intermittently isolated from wild birds sampled during field surveillance studies for Newcastle disease virus (NDV). Thus, we experimentally investigated the susceptibility of four native agriculturally-associated wild bird species to the NDV LaSota vaccine and evaluated the shedding dynamics, potential transmission from chickens, and humoral antibody responses. To test susceptibility, we inoculated wild caught, immunologically NDV naïve House Finches (n = 16), Brown-headed Cowbirds (n = 9), Northern Cardinals (n = 6), and American Goldfinches (n = 12) with 0.1 ml (106.7EID₅₀/ml) of NDV LaSota vaccine via the oculo-nasal route. To test transmission between chickens and wild birds, adult SPF White Leghorn chickens were inoculated similarly and co-housed in separate isolators with two to five wild birds of the species listed above. This design resulted in three treatments: wild bird direct inoculation (5 groups), and wild bird exposure to one (2 groups) or two inoculated chickens (6 groups), respectively. Blood and oropharyngeal and cloacal swabs were collected before and after infection with the live vaccine. All wild birds that were directly inoculated with the LaSota vaccine shed virus as demonstrated by virus isolation (VI). Cardinals were the most susceptible species based on shedding viruses from 1 to 11 days post-inoculation (dpi) with titers up to 104.9EID₅₀/ml. Although LaSota viruses were shed by all inoculated chickens and were present in the drinking water, most non-inoculated wild birds co-housed with these chickens remained uninfected for 14 days as evident by VI. However, one American

Goldfinch tested positive for vaccine transmission by VI at 7 dpi and one House Finch tested positive for vaccine transmission by rRT-PCR at 13 dpi. Only one directly inoculated cowbird (out of three) and two cardinals (out of two) developed NDV specific hemagglutination inhibition (HI) antibody titers of 16, 16, and 128, respectively. No clinical signs were detected in the chickens or the wild birds post inoculation.

Key words: Newcastle disease virus, NDV, LaSota, passerines, experimental transmission, chickens

Abbreviations. $\bar{x}$ = mean; AHY = after hatch- year; Anatidae = waterfowl including ducks; geese; and swans; ASY = after second-year; BHI = brain-heart infusion; BSL-2 = biosafety level-2; cm = centimeters; Columbids = species of the pigeon and dove family; CT = cycle threshold of fluorescent signal; df = degrees of freedom; dpi = days post-inoculation; ECE = embryonating chicken eggs; EID₅₀ = mean embryo infectious dose; g = grams, HA = hemagglutination assay; HI = hemagglutination inhibition; ICPI = intracerebral pathogenicity index; LaSota = LaSota/1946; ml = milliliters; mm = millimeters; ND = Newcastle disease; NDV = Newcastle disease virus; vNDV = virulent Newcastle disease virus; OIE = World Organization for Animal Health; PROC GLM = statistical test for the means of multiple data sets; PROC MIXED = statistical test for non-linear data; Psittacines = species of parrot family; RNA = Ribonucleic acid; rRT-PCR = real-time reverse transcription polymerase chain reaction; SAS = Statistical Analysis System/SAS Institute; sd= standard deviation; SPF = specific

pathogen free; SY = second-year; U.S. = United States; UK = United Kingdom; USFWS = United States Fish and Wildlife Service; USGS = United States Geological Survey; VI = virus isolation

Introduction

Newcastle disease (ND), caused by infections with virulent strains of avian avulavirus 1 (also known as virulent Newcastle disease virus or vNDV), is a highly contagious and systemic infection of chickens (*Gallus gallus*) that remains a worldwide economic threat to poultry producers (2, 5, 6, 40). Avian avulavirus 1 is a non-segmented, single-stranded, enveloped RNA virus within the Avulavirus genus of the Paramyxoviridae family (1, 48). Genotypically distinct vNDV strains are consistently isolated from large-scale epizootics in South and Central America, the Middle East, Southeast Asia, and Western and Central Africa (25, 26, 55, 58, 78). Previous ND outbreaks in the U.S. resulted in the depopulation of millions of birds and significant containment costs for the industry (65, 90). The ongoing 2018-2019 ND outbreak in U.S. domestic chickens has been attributed to the exhibition of backyard chickens in California, and has also been recently confirmed in commercial flocks (16, 85).

NDV has been isolated from a broad range of avian hosts, and it is generally presumed that all bird species are susceptible to the replication, shedding, and transmission of the virus (4, 39, 76). Virulent NDV is sporadically isolated from waterbirds such as Double-crested Cormorants (*Phalacrocorax auratus*), while NDV of low virulence (lentogenic strains) are

commonly found in migratory waterfowl such as Mallards (*Anas platyrhynchos*) and Green-winged Teals (*Anas carolinensis*) (69, 80, 92). Columbids, such as Rock Pigeons (*Columba livia*), are considered NDV carriers due to their intermittent and extended shedding patterns of virulent strains that vary in their pathogenicity (23, 41). NDV has also been isolated from species far from anthropogenically modified habitats, such as Adelié Penguins (*Pygoscelis adeliae*) in the Antarctic (82) and Bar-tailed Godwits (*Limosa lapponica*) in Mauritania (13).

Although the susceptibility of wild birds to NDV has been largely accepted, their role in the short-term spread of the virus, as maintenance hosts, or in fulfilling both roles does remain unclear (13, 27, 76). It is, however, known that some species of wild birds, such as Double-crested Cormorants, Psittacines, Rock Pigeons, and members of the Anatidae may transmit NDV to other wild bird species or to poultry while presenting with variable clinical signs (12, 42, 44, 73). Conversely, wild birds are also susceptible to NDV infection from poultry (9, 14, 42, 77). Thus, biosecurity is stringently practiced within the commercial poultry industry through barriers to limit the contact between domestic and wild birds (49). Well-executed vaccination programs also play an important role in reducing clinical disease and the quantity of virulent virus shed from infected, but vaccinated poultry (26).

On the other hand, biosecurity in the backyard poultry sector is limited (84), and the attraction of ‘feeder birds,’ i.e. passerines and near-passerines, to chicken feed is a potential link for NDV transmission. For example, backyard poultry flocks serving as a primary source of

protein in Latin American countries are considered high risk sources of disease to Neotropical songbirds (34, 36-38). In urban and suburban areas of the U.S., the continued rise in the ownership of backyard flocks and lack of oversight concerning vaccination and biosecurity protocols have the potential to leave agriculturally associated passerines and the commercial poultry industry vulnerable to NDV outbreaks (32, 43, 83).

Prior work has already demonstrated that wild-type NDV and vaccine strains of poultry-origin can spill back into wild birds; however, these studies have been largely cross-sectional and unable to assess the susceptibility of passerines to lentogenic NDV strains over an infectious period (9, 15, 77). Unlike virulent NDV, which is likely to cause rapid mortality in susceptible species, lentogenic NDV may stimulate self-limiting infections as seen in chickens (53), suggesting infected passerines may shed viruses unnoticed into the environment.

As backyard chicken-passerine interactions become more common, it has become increasingly important to evaluate the susceptibility and the transmission potential of frequently encountered agriculturally associated passerines to poultry-adapted NDV strains. In the southeastern U.S., House Finches (*Haemorrhous mexicanus*), American Goldfinches (*Spinus tristis*), Northern Cardinals (*Cardinalis cardinalis*) and Brown-headed Cowbirds (*Molothrus ater*) are native songbirds that often share habitat with backyard chickens (31, 45). These species regularly exploit bird feeders or backyard chicken feed, making them excellent epidemiological models for testing passerine susceptibility to exposure to NDV vaccines.

In this study, we conducted three experimental inoculation treatments using groups of House Finches, American Goldfinches, Northern Cardinals, and Brown-headed Cowbirds either directly exposed to the lentogenic live NDV LaSota vaccine or otherwise exposed to vaccinated Leghorns under Biosafety-level 2 (BSL-2) conditions. We measured each species' shedding titer, duration, and antibody response over a two-week period. We discuss the patterns these species exhibit, and the implications this may have for the epidemiology of Newcastle disease vaccine viruses in wild birds.

MATERIALS AND METHODS

Wild bird capture and handling. Wild birds were captured in Georgia, U.S. following standard methodology with mist nets from one site in Bishop, Oconee county, and three sites in Athens, Clarke county in May 2017 (24). Capture sites were chosen based on habitat suitability and eBird accounts of species distributions that reported where each target species was found in high abundance (81). To attract sufficient numbers of birds, feeders filled with wild bird blends were maintained over a two-month period prior to capture. Used feeders were rotated weekly with clean feeders and sanitized with 10% bleach to reduce incidental pathogen transmission (59). Mist nets were erected near feeders and all target and non-target wild birds, i.e. species other than House Finches, American Goldfinches, Brown-headed Cowbirds, and Northern Cardinals, were immediately extracted and samples, demographic information, and standard morphological measurements were collected (24). After weighing, up to 1% of the bird's body mass of blood

was collected from the brachial vein (62) into Sarstedt Microvette® Z-Gel 200ul serum tubes (Sarstedt, Nümbrecht, Germany). Cloacal and oropharyngeal swabs using Purflock Ultra® polyester swabs (Puritan Medical Products Company LLC, Guilford, ME, USA) were also collected from all wild birds.

Age and sex were assessed by plumage and/or molt limit patterns (67). Morphological measurements collected from each bird included fat scores (scale of 0 to 7) and degree of wear on the primaries (scale of 0 to 5), where a score of 0 indicated new, crisp feathers and a score of 5 indicated primaries that were extremely abraded, torn, and/or with exposed rachis (68). We also recorded the length of the wing chord (mm), which measured as the length of the longest un-flattened primary, and lastly, mass in grams (data not shown) (68). The biometric data ensured that target birds fell within the statistical mean ($\bar{x}$) for the morphological parameters of their respective species, reducing the potential for bias in susceptibility to the vaccine virus. This data can be found in table 3.5 (Table 3.5) (29, 63, 67). All birds deemed reproductively active by the presence of a brood patch or cloacal protuberance were processed and released. After processing, non-target birds were banded with a federal USGS band and released. Target birds were banded with unique color-band combinations, and were placed in small covered cages with species-specific food and water until mid-morning transport to the U.S. National Poultry Research Center's BSL-2 Laboratory. Wild bird capture, banding, and captivity were performed under USGS Permit Number 23450, USFWS Permit Number MB78901A-1, and Georgia Department

of Natural Resources Permit Number 29-WJH-14-241. All wild bird procedures performed in the field were approved by Institutional Animal Care and Use Committee at the University of Georgia (Animal Use Protocol number A2016 12-006-Y2-A6).

Swab samples were placed in 2.0 mL cryovials (Corning Inc., Corning, NY, USA) containing 1.5 mL of Brain-Heart Infusion (BHI) (Becton Dickinson, San Jose, CA, USA) with Gentamicin (200 µg/mL), Penicillin (2000 units/mL), and Amphotericin-B (4 µg/mL). Serum tubes and swabs were maintained in a cooler with ice packs in the field, transported to the lab, and stored at -20°C and -80°C, respectively. Blood was centrifuged at 13,000 rpm for 5 minutes (g-force = 11,300), which allowed for the separation of the blood components and their storage in the original collection tubes.

Eggs and Chickens

The specific pathogen free (SPF) White Leghorn (hereafter, Leghorn) flock of the U.S. National Poultry Research Center was the source of embryonating chicken eggs (ECE) for VI and the 17-week-old chickens used in the BSL-2 experiments.

Experimental design

In BSL-2, wild birds (n = 43) were assigned to groups that were sorted by species into wire-mesh cages inside negative pressure isolators and were acclimated with conspecific cage-mates for one week prior to the start of the experiment. Group sizes of wild bird ‘flocks’ were calculated to maximize space and accommodate species-specific behaviors such as sociality (30). American

Goldfinches (n = 7) were placed into one flock of five birds and the remaining two birds were paired. House Finches (n = 16) were divided into four flocks of four birds, Brown-headed Cowbirds (n = 9) were divided into three groups of three birds, and Northern Cardinals (n = 6) were paired off into three groups of two birds. Groups were then allocated into one of three treatments. Five wild bird groups were assigned to direct inoculation with the LaSota vaccine virus (Table 3.1). Two groups were assigned to interspecific exposure with a single LaSota-inoculated Leghorn, and six groups were assigned to interspecific exposure to two LaSota-inoculated Leghorns, for a total of 13 experimental Groups. All birds were provided species-specific food and water on an ad-libitum basis during captivity (30).

After acclimation, 14, 17-week-old SPF Leghorns were co-housed with captive wild bird groups. Leghorns of laying age were used to mimic transmission conditions between mature backyard layers and the target wild bird species. Leghorns were placed only in the isolators containing wild bird groups intended for testing cross-species transmission from inoculated chickens, as described above. Wild bird groups were housed inside a wire-mesh cage that was placed within each isolator which protected the wild birds assigned to inter-specific transmission groups from injury from Leghorns. Wire-mesh cage size and shape were designed to provide sufficient space for wild bird natural behaviors and accommodate exposure between chickens and wild birds. Refuges, variable size perches and nest boxes were provided in each mesh-cage as escapes from conspecifics. Cages were further modified so that chickens and wild birds were

able to share the same water source and food bowl (Fig. 1). Isolators each measured 88.9 cm wide x 59.7 cm deep x 53.3 cm height. Mesh-cages consisted of 1.3 cm galvanized wire spacing and dimensions measuring 45.7 cm wide x 45.7 deep cm x 45.7 cm height. All BSL-2 experimental procedures were approved by Institutional Animal Care and Use Committee at the U.S. National Poultry Research Center (Animal Use Protocol number USNPRC-2018-013).

Virus

The NDV vaccine LaSota strain commonly used in poultry (9, 20, 52, 54, 75, 94), and commercially available (Zoetis, Parsippany, NJ, USA), was used (103). The vaccine was reconstituted according to manufacturer instructions and titrated prior to its use to ensure a proper and consistent titer throughout each experiment.

Experiment I

Direct inoculation study. Northern Cardinals (n = 2), Brown-headed Cowbirds (n = 3), and House Finches (n = 4) and two groups of American Goldfinches (n = 2 and n = 5 birds) were directly inoculated by the oculo-nasal route with 0.1 ml of 106.7 EID₅₀/ml of LaSota vaccine virus.

Experiment II. Transmission Study from One Inoculated Chicken to Wild Birds

House Finches (1 group of n = 4) and Brown-headed Cowbirds (1 group of n = 3) were each co-housed for fourteen days with a single SPF chicken inoculated as described in

Experiment I

Although the wire-mesh wild bird cage within each isolator prevented direct contact, each chicken and wild bird group shared a single water source and a shared food bowl. In addition, each isolator was bottom lined with non-absorbent Benchkote™ paper (GE Healthcare Bio-Sciences, Pittsburgh, PA, USA) upon which chicken droppings accumulated. After one day of exposure to chicken droppings, this paper was rotated 180 degrees beneath the wire-mesh cage, exposing wild birds to the chicken feces as an added medium of exposure to NDV excreted from the cloaca. The Benchkote™ paper was replaced every three days and rotated on the fourth day.

Experiment III. Transmission Study from Two Inoculated Chicken to Wild Birds

To determine whether doubling the number of infected Leghorns would increase the likelihood of transmission, we housed Northern Cardinals (2 groups of $n = 2$ birds), House Finches (2 groups of $n = 4$ birds), American Goldfinches (1 group of $n = 5$), and Brown-headed Cowbirds (1 group of $n = 3$) with two inoculated Leghorns. Replicating the same design as Experiment II, each Leghorn was inoculated with the LaSota vaccine. This experiment also tested the susceptibility of American Goldfinches and Northern Cardinals to the LaSota vaccine virus when exposed to an increased density of infectious hosts, as these wild bird species are more likely to exhibit refuge behaviors in response to large birds by hiding inside substrates such as nest boxes.

All Bird Experiments

Wild birds and Leghorns were monitored twice daily for post-vaccinal reaction signs, such as generalized respiratory signs, ocular discharge and conjunctivitis, greenish, watery droppings, and lethargy (35, 86). Two swabs, prewet with BHI media, were collected, one from the oral cavity and one the cloaca from each bird at days 1, 2, 3, 4, 7, 9, 11 and 13 or 14 post-inoculation (dpi). In addition, 0.5 ml of each bird group's drinking water was collected on the same days that birds were swabbed. At the termination of the experiment, wild birds underwent the same morphological measurements (fat, mass, wing chord and feather wear) to compare with the time zero measurements collected in the field (File S1). Blood collection for the detection of anti-NDV antibodies by hemagglutination inhibition (HI titer) was performed on both wild birds and Leghorns prior to euthanasia at 13 or 14 dpi and was centrifuged and stored as described in the field methods.

Virus Isolation and Titration

Virus isolation and titration followed OIE standards (60). Oral and cloacal swabs collected from target and non-target wild birds in the field were assayed first to ensure wild birds were not actively shedding NDV. Post-inoculation oral and cloacal swabs and water samples from experimental wild birds and chickens followed. In brief, all field-collected and post-inoculation oral and cloacal swab media (200 uL) and water samples (200 uL) were inoculated into 9–11-day-old SPF ECE in BSL-2. After one week of incubation, harvested allantoic fluids were tested using the hemagglutination assay (HA). Negative allantoic fluid samples underwent

a second passage into ECE and only those demonstrating HA activity were considered positive for VI. Viral titers for all positive oral swabs were assessed via 1:10 serial dilutions (10⁻¹ to 10⁻⁵), as these dilutions are more sensitive in detecting the smaller quantities of virus likely excreted from songbirds weighing <40 g. For each serial dilution, five SPF ECE were inoculated with 100 uL of swab media, incubated for a week, then evaluated with the HA test. Viral mean embryo infectious doses (EID₅₀) for each oral swab were calculated using the Spearman-Kärber method (8, 88).

Serological assays

HI assays (3) were used to test serum samples to determine the prevalence of NDV exposure in the target and non-target wild birds captured in Athens, GA. Prior to assaying, wild bird serum was pre-treated to remove natural serum agglutinins through heat-inactivation in a 56°C water bath for thirty minutes (10) and a 10% packed SPF rooster chicken erythrocyte wash (64). Standard HIs were used when 25 ul of serum was available after centrifuging. As some species weighed <20 g, 1:10 dilutions incorporating 5 ul of serum were used when serum did not meet the 25 ul threshold. Chicken and wild bird sera collected at 13 or 14 dpi were also tested by HI. All serum underwent HI testing using 4 HA units of homologous antigen, the LaSota vaccine virus. Samples that demonstrated complete inhibition of red blood cell agglutination at the serum dilution of >16 were considered positive for anti-NDV antibodies.

Real-time reverse transcription PCR

The interspecific transmission of the LaSota vaccine virus to birds from inoculated chickens was further evaluated by real-time reverse transcription PCR (rRT-PCR) performed on oropharyngeal wild bird swab samples collected at days 7, 9, 11, and 13 or 14 dpi. Cloacal swabs were not evaluated due to the VI results suggesting infection by LaSota was limited to the respiratory tissues. Viral RNA was extracted from swab media using the Magmax-96 viral AI/ND Viral RNA isolation kit (Ambion, Austin, TX, USA) and the KingFisher Flex Purification System (ThermoFisher Scientific, Waltham, MA, USA). One-tube reactions were used for reverse transcription and amplification using the AgPath-ID One Step RT-PCR kit (ThermoFisher Scientific, USA) in the AB7500 Fast real-time PCR system (Applied Biosystems, USA) using M gene primers and probe and a modified protocol by Wise, Suarez, Seal, Pedersen, Senne, King, Kapczynski and Spackman (91). The cycling profile for reverse transcription took place 45°C for 10 minutes, followed by a second cycle of 95°C for 10 minutes, and denaturation, annealing, and elongation were 40 cycles of 94°C for 10 seconds, 56°C for 30 seconds, and 72°C for 10 seconds. Cycle threshold (CT) values < 37 were considered positive for NDV.

Statistical Analyses

All statistical analyses were performed using SAS 9.4. To determine which species was the most susceptible to the LaSota vaccine virus as defined by viral titers and number of shedding days, we used PROC MIXED, a generalized mixed model approach to analyzing repeated measures which uses the maximum likelihood to explore the best fit of the model

parameters (19). This allowed us to compare multiple growth curves between species, specifically the increases over time of the viral titers and the number of shedding days between species while also incorporating correlated data points. Our second analysis involved PROC GLM, a generalized linear model for unbalanced data and unequal variances (93) to examine whether a relationship existed between percent mass lost by each species during captivity and its treatment type, direct inoculation with the vaccine or indirect exposure.

RESULTS

Virus Isolation and Serology from Non-Target Wild Birds

An additional 110 non-target wild birds, representing 22 passerine and near-passerine species, were captured in mist nets, banded, swabbed and released during the capture period. From these, a total of 57 blood samples were also collected; 22 samples had sufficient serum for pre-experimental HI testing. All swabs and sera from the non-target birds were VI, and HI negative, respectively.

Virus Isolation and Serology from Target Wild Birds at Initial Capture

A total of 43 target wild birds representing four agriculturally-associated passerine species were captured in mist nets, blood-sampled, color-banded and swabbed. Oropharyngeal and cloacal swabs collected from target birds in the field were all negative for NDV by VI. Twenty-three of the 43 target birds provided sufficient sera for pre-experimental HI testing, and these were negative.

Experiment I Results. Direct inoculation study

A total of sixteen wild birds representing six groups were directly inoculated with $10^{6.7}$ EID₅₀/ml of the NDV LaSota vaccine virus. Mean oral viral shedding titers of the wild birds in this experiment are presented in one table for the comparison of the mean oral viral titers of directly inoculated Leghorns from experiments II and III (Table 3.2). Of the two (n = 2) Northern Cardinals in this experiment, both had negative HI titers at 0 dpi HI. After direct inoculation with LaSota, one had a post-inoculation HI titer of 128 at day 14 dpi, and the second developed an HI titer of 16 at day 14 dpi. Virus isolation revealed both birds were shedding from the oropharyngeal cavity for six out of the eight post-inoculation days that birds were sampled ($\bar{x}$ = 6 days, sd=0). Cloacal shedding from either bird was not detected. Virus was detected in one water sample at 4 dpi with a titer of $10^{1.5}$ EID₅₀/ml. The Cardinal that developed the HI titer of 128 shed a maximum titer of $10^{3.9}$ EID₅₀/ml at 7 dpi, while the Northern Cardinal with a titer 16 shed a maximum titer of $10^{2.9}$ EID₅₀/ml at 11 dpi.

Two Cowbirds had negative HI titers at 0 dpi, however the third Cowbird was not tested prior to inoculation due to insufficient sera. Within the Brown-headed Cowbird group (n = 3), only one bird developed a positive HI titer of 16 by day 14 dpi. Similar to the pattern demonstrated by the Cardinals, the Cowbird with the highest HI titer also had the highest shedding titer of $10^{3.7}$ EID₅₀/ml at 1 dpi in contrast to its cage-mates, and also shed for the longest duration for up to 7 dpi. The remaining two birds shed for four days and one day each, for an overall group mean

shedding duration of 3.3 days ($\bar{x}=3.3, sd=1.8$), and a maximum oral viral shedding titer of $10^{1.7}$ EID₅₀/ml. Cloacal shedding was not detected from the Cowbirds by VI, nor was virus recovered from the drinking water for the duration of the experiment.

Directly inoculated American Goldfinches were divided into two experimental units, one paired ($n = 2$) and one group consisting of the maximum number of Goldfinches that could be accommodated by the available space ($n = 5$). The HI titers for all Goldfinches were negative at 0 dpi and at 13 dpi. The maximum viral oral titer from a Goldfinch was $10^{3.1}$ EID₅₀/ml at 2 dpi from a bird housed with the larger group of five. This bird and only one other within the same experimental unit had detectable viral shedding from the cloaca for a single day each, at four and seven dpi. The experimental group of consisting of two Goldfinches had the longest overall duration of shedding from the oropharyngeal route for a total of four and 11 days, while the group of five Goldfinches shed for zero, one, four ($n = 2$) and five days of the eight total sampling dates. The mean duration of shedding for both groups combined was 3.7 days ($\bar{x}=3.7, sd=2.4$). No virus was recovered from the drinking water shared by the group of two birds, although virus was recovered from the water shared by the group of five birds at 9 dpi with a viral titer of $10^{1.5}$ EID₅₀/ml.

Three of the four of directly inoculated House Finches had negative HI titers at 0 dpi and 14 dpi. The maximum viral oral titer from a House Finch was $10^{1.9}$ EID₅₀/ml at day 4 dpi. Overall, the number of days Finches were positive for shedding from the oropharyngeal route

ranged from one, two, four to and five days of the eight total sampling dates ($\bar{x}$ =3.0,sd=1.8). The Finch that shed for two days from the oral route was positive for cloacal shedding at day 7 dpi with a titer of $10^{1.5}$ EID₅₀/ml. No other birds were positive for cloacal shedding. Virus was recovered from the drinking water of the House Finch group for a single day at 3 dpi, with a viral titer of $10^{1.5}$ EID₅₀/ml.

No clinical signs were detected in any of directly inoculated wild birds throughout the experiment. Nonetheless, changes in body condition parameters within the experimental endpoint measurements were detected in comparison to the parameters collected in the field at time zero. These parameters include fat and flight feather wear scores, length of the wing chord, mass in grams, and percent changes in wing chord length and mass (Table 3.5).

Experiment II Results. Transmission Study from One Inoculated Chicken to Wild Birds

A total of seven wild birds and two Leghorns representing two experimental units comprised the single inoculated chicken interspecific transmission experiment. This experiment evaluated the susceptibility of House Finches (n = 4) and Brown-headed Cowbirds (n = 3) to lentogenic NDV from a single Leghorn directly inoculated with the LaSota vaccine virus. The Leghorn co-housed with the House Finches developed an HI titer of 256 by day 14 dpi. Three of the four of House Finches had negative HI titers at 0 dpi and 14 dpi. The third Finch was not tested at 0 dpi due to insufficient sera. The Leghorn housed with the Finches began shedding at 1 dpi and at 4 dpi resulted in a titer of $10^{5.7}$ EID₅₀/ml. No shedding was detected on

the next sampling date of 7 dpi from the chicken, nor was cloacal shedding detected by VI. All swab samples collected from House Finches during the experimental period remained negative for NDV transmission by VI. Shared drinking water samples were also negative for the presence of virus throughout the experiment.

The Leghorn co-housed with the Cowbirds developed a HI titer of 128 by day 13 dpi. Only one of the three Cowbirds had sufficient sera for analysis at 0 dpi, and resulted in a negative HI titer. Post-inoculation HI titers were negative for all three Brown-headed Cowbirds by day 13 dpi. All Cowbirds remained negative for viral transmission as measured by VI for the entirety of the experiment, yet virus was isolated from the oropharyngeal samples of the chicken at 1dpi – 4dpi, excreting oral viral titers ranging from $10^{5.1}$ to $10^{5.7}$ EID₅₀/ml. All chicken cloacal swabs were negative by VI. Virus was isolated from the shared drinking water three out of the eight sampling days, with a maximum virus titer of $10^{1.9}$ EID₅₀/ml at 3 dpi.

No clinical signs were detected in either the wild birds or the Leghorns throughout the experiment. Body condition parameter changes were measured at the end of the experiment and calculated to permit for cross-species comparisons (File S1).

Experiment III Results

Transmission Study from Two Inoculated Chickens to Wild Birds. A total of 20 wild birds and 12 Leghorns representing six experimental units comprised the interspecific transmission experiment with two inoculated chickens per group. This experiment evaluated the

susceptibility of Northern Cardinals (2 groups of $n = 2$), House Finches (2 groups of $n = 4$), Brown-headed Cowbirds (1 group of $n = 3$), and American Goldfinches (1 group of $n = 5$) to vaccines from Leghorns directly inoculated with the LaSota vaccine virus. Based on VI and rRT-PCR, transmission did not occur from inoculated chickens to any wild birds except for one American Goldfinch and one House Finch.

In Group 1 (Table 3.1), the two Leghorns directly inoculated with the LaSota vaccine that were co-housed with a pair Northern Cardinals each developed antibody titers of 128. One Leghorn shed for four of the eight sampling dates from 1-4 dpi, while the second Leghorn shed for five sampling days, from 1-4 and 7 dpi. Both Leghorns demonstrated a maximum oral viral shedding titer of $10^{6.3}$ EID₅₀/ml at days 3 and 4 dpi. Both Leghorns were negative for cloacal shedding according to VI. One Cardinal died of unknown causes overnight between day 2 and 3 dpi, although oral and cloacal swabs for this bird were VI negative. Blood for HI testing was not collected due to post-mortem blood coagulation; however, a 0 dpi serum sample had a negative HI titer. The second Cardinal had negative HI titers at 0 dpi and 14 dpi, and was also negative for VI for all swab sampling dates. Virus was detected in the shared drinking water samples on days 3 and 4 dpi with a viral titer of $10^{1.5}$ EID₅₀/ml.

Group 8 also consisted of two inoculated Leghorns that were co-housed with two Northern Cardinals. One Leghorn developed an HI titer of 256, and the second developed an HI titer of 128 by day 13 dpi. Group 8 Leghorns shed for 1-4 dpi, and both birds excreted a

maximum oral viral titer of $10^{5.7}$ EID₅₀/ml. On day 4 dpi, the Leghorn with the HI titer of 128 shed from the cloaca, and the amount of virus was titrated to $10^{1.5}$ EID₅₀/ml. Although the HI titer status for these birds were unknown for 0 dpi, similar to the Cardinals in Group 1, both Cardinals in Group 8 had negative HI titers at the end of the experiment. All swabs taken from the Cardinals were VI negative. Shared drinking water samples were positive for two out of the eight sampling days, with a maximum virus titer of $10^{1.5}$ EID₅₀/ml at 1 and 4 dpi.

Group 2 was comprised of two inoculated Leghorns that were co-housed with four House Finches. By day 14 dpi, both Leghorns developed humoral antibody responses with HI titers of 256. Each Leghorn shed via from the oral route for days 1-4 dpi, and both shed the greatest amount of virus at 4 dpi with a titer of $10^{5.9}$ EID₅₀/ml. Cloacal shedding from both Leghorns was negative by VI. The three of the four House Finches with sufficient sera for analysis in this group had negative HI titers at 0 dpi and all four were HI negative at the end of the experiment. All contact House Finches remained negative as all oral and cloacal swab samples were VI negative. However, one House Finch did test positive for NDV by rRT-PCR via the oropharyngeal route at 13 dpi with a CT value of 29.9. Shared drinking water samples also remained negative for virus throughout the experimental period.

The second replicate of interspecific transmission from two inoculated Leghorns to four House Finches (Group 9) exhibited a similar pattern to Group 2. One Leghorn developed an HI titer of 128 and the second an HI titer of 256 by day 13 dpi. This pair of Leghorns shed for three

out of the eight sampling days, and the Leghorn who developed an HI titer of 256 shed on days 1, 3, and 4 dpi with a maximum oral viral titer of $10^{5.7}$ EID₅₀/ml. The second Leghorn shed consistently from days 1-3 dpi with peak oral viral titer of $10^{5.3}$ EID₅₀/ml and was also positive for shedding from the cloaca on day 4 dpi with a viral titer of $10^{3.3}$ EID₅₀/ml. Only one House Finch had sufficient sera for testing, and had negative HI titers at 0 dpi. All four House Finches remained negative with negative HI titers, and all also remained negative for shedding of the virus throughout the experimental period. Shared drinking water samples were positive on days 2 and 3 dpi, with viral titers of $10^{1.5}$ EID₅₀/ml on both days as measured by VI and titration.

Group 11 consisted of two inoculated Leghorns and five American Goldfinches. One inoculated Leghorn developed an HI titer of 128, whereas the second developed an HI titer of 64. Both Leghorns shed on days 1-4 dpi, excreting orally with a maximum viral titer of $10^{5.7}$ EID₅₀/ml. Both Leghorns were negative for cloacal shedding by VI. The one American Goldfinch with enough sera to test had a negative HI titer at 0 dpi. Three Goldfinches in this group died or were euthanized as a result of injury prior to the experimental endpoint. One Goldfinch was found dead due to injury overnight between days 2-3 dpi. All swabs from this bird were negative for virus, however, a blood sample could not be collected. A Goldfinch died on day 7 dpi for unknown reasons, and had negative HI titers at 0 dpi and 7 dpi. All post-exposure swabs from this bird were negative for NDV. One Goldfinch was euthanized for severe monocular conjunctivitis on day 9 dpi. All swabs from this individual were negative for NDV, and the

serum resulted in a negative HI titer at 9 dpi. Of the remaining two Goldfinches, both had negative HI titers at the experimental endpoint. One of the two Goldfinches developed a positive oropharyngeal titer on day 7 dpi of $10^{1.5}$ EID₅₀/ml as measured by virus titration, while the second bird remained negative for interspecific transmission throughout the experiment.

The Group 12 interspecific transmission experiment between two inoculated Leghorns and three Cowbirds resulted in positive humoral antibody responses for both Leghorns, one Leghorn with an HI titer of 128, and the other Leghorn with an HI titer of 256 by day 13 dpi. The Leghorn with the HI titer of 256 shed from days 1-4 dpi, peaking at day 2 dpi with an oral viral titer of $10^{6.3}$ EID₅₀/ml. The second Leghorn was positive for virus on the sampling dates 1-4 dpi and 7 dpi. Both Leghorns were negative for cloacal shedding according to VI. All exposed Cowbirds remained negative for humoral antibody responses with negative HI titers at 0 dpi and 13 dpi. All 24 oral or cloacal samples collected from these three wild birds throughout the experiment were negative by VI.

No clinical signs related to the LaSota infection were exhibited by any of the wild birds or Leghorns in this experiment. The mean oral viral titers for all combined Leghorns from 1-4 dpi and 7, 9, 11, and 13 or 14 dpi from experiment two and three are presented in Table 3.2. Morphometric data results are presented in File S1.

Real-time RT-PCR Results

Viral RNA was detected from the oropharyngeal swab of one House Finch from experiment three by rRT-PCR at 13 dpi. This bird originated from Group 9, an experimental group of four House Finches exposed to two inoculated Leghorns. The positive CT value for this sample was 29.9.

Statistical Test Results

We examined whether directly inoculated wild bird species demonstrated significantly different shedding patterns. Using PROC MIXED, we constructed a repeated measures model in which the dependent variable was the clustered data consisting of oropharyngeal viral titers measured for each day post-inoculation, 1-13 or 14 dpi, in comparison to our independent variable, species. To account for individual, within-species differences, we added a random effect for each individual bird. The model compared growth curves consisting of oral viral titer increases throughout the experimental period between each directly inoculated wild bird species using the maximum likelihood approach.

The overall Type III model for the fixed effect of species was significant ($F = 3.35$, density $df = 112$, $df = 3$, $p = 0.0216$), suggesting that the differences for each species shedding patterns were significant at the 0.05 threshold. Table 3.3 provides the breakdown for the fixed effect of species, which incorporates the random effects of each individual bird. The low sample size of Northern Cardinals excluded them from analysis, however the p-values for the remaining species demonstrate significant differences in oral shedding titer patterns, specifically the height

of each species' shedding growth curve and when peak oral shedding occurred. Brown-headed Cowbirds ($p = 0.0139$, Table 3.3) demonstrated a significantly different pattern of shedding in starting earlier and peaking sooner than House Finches ($p = 0.0026$, Table 3.3) and American Goldfinches ($p = 0.0111$, Table 3.3). Statistical analysis results of morphometric data are presented in supplementary table 3.5.

DISCUSSION

To our knowledge, this is the first study to recover NDV, specifically LaSota vaccine virus, from inoculated House Finches, American Goldfinches, Northern Cardinals and Brown-headed Cowbirds, and it is also the first study to test the transmission of NDV from experimentally infected chickens to wild birds. Historically, passerines, apart from House Sparrows (*Passer domesticus*) (Miller, unpublished data), have been largely neglected among NDV experimental infection studies. This is likely due to the relatively fewer passerine-origin NDV isolates recovered in comparison to waterfowl, shorebirds, and gulls (27).

While NDV strains of varying virulence are occasionally isolated from wild passerines (15, 17, 28, 51), whether those infections were the result of spillover from infectious chickens or these shedding individuals represented a carrier state remains unresolved. Most of the information regarding prior NDV exposure in passerines comes from seroprevalence surveys (7, 17, 61, 66). Humoral antibody responses vary across species, and antibodies mounted against

NDV cannot distinguish between the virulence of the source virus or the length of time since exposure (40, 74).

We speculated that all four species would be susceptible to direct inoculation with lentogenic NDV due to the wide host range of the virus (39), where we defined susceptibility as viral shedding at or above a titration of $10^{1.5}$ EID₅₀/ml and an immunogenic response above an HI titer of 16. While all four species were susceptible, the data suggest that Northern Cardinals were the most susceptible to direct inoculation with NDV LaSota. This species had the longest shedding duration, the highest viral titers, and the highest humoral antibody response. Even within this species, a degree of individual heterogeneity between humoral antibody responses and viral shedding was detected. The individual with an HI titer of 128 had an eight-fold increase in humoral antibody response in contrast to its cage-mate with a titer of 16, with a ten-fold difference in shedding, i.e., $10^{3.9}$ EID₅₀/ml at 7 dpi versus $10^{2.9}$ EID₅₀/ml at 11 dpi. Both individuals were after-hatch year (AHY) male birds captured from the same site, and therefore the underlying cause driving these differences to infection remains unclear.

Similar to the other directly inoculated species, no cloacal shedding in Northern Cardinals was detected, suggesting a restriction to respiratory tissues. Outside of this experiment, we found only one other instance of members of the Cardinalidae family, three naturally infected Ultramarine Grosbeaks (*Cyanocompsa brissonii*), that were confirmed as positive for NDV of an

unspecified virulence (23). Immunohistochemistry revealed from five birds that 2 birds positive for NDV in the trachea, and one bird positive in both trachea and spleen (22).

Brown-headed Cowbirds demonstrated moderate susceptibility to direct inoculation, with oral viral shedding lasting up to four days, and peaking to a moderate of titer of $10^{3.7}$ EID₅₀/ml. Contrary to expectations, only one bird out of the three directly infected birds developed a positive humoral antibody response of 16. As with Cardinals, no obvious differences between individuals were detected. The Cowbirds in this group were all aged and sexed as after-hatch year/second-year (AHY/SY) female birds. They were also captured from two sites less than one kilometer apart. Members of the blackbird family, however, may be more resistant to lentogenic NDV in contrast to Cardinals. Vickers and Hanson (87) used a nebulizer to expose Red-winged Blackbirds (*Agelaius phoeniceus*), another member of the Icteridae family, to an aerosol of live LaSota NDV with a high mean embryo infectious dose (10^9 EID₅₀/ml). LaSota was re-isolated from tracheal, but not cloacal swabs, from days 2 and 5 dpi from only half of the birds. Swabs from 9 and 11 dpi were VI negative. The geometric mean HI titers for the 11 Blackbirds were 6.3 and 0.6 at 3- and 6-weeks post infection, with the highest HI titer recorded titers of 16 and 4, respectively. Spalatin, Turner and Hanson (79) also concluded that non-neutralizing antibodies were detected and that the HI response was of a short duration (87).

House Finches and American Goldfinches, both members of the Fringillidae, or New World finch family, were the least susceptible to direct inoculation with LaSota. House Finches

began shedding the latest of all species, between days three to seven dpi. Nonetheless, they excreted so little virus that seven of the eleven total positive isolates, out of 32 total swabs titrated to $<10^{1.5}$ EID₅₀/ml. Although Finches are often associated with NDV or other avian avulaviruses (33, 46, 50, 70, 72), those finch species generally correspond to the Old World Estrildidae family (61, 73, 87). A few exceptions include velogenic NDV isolated from quarantined, pet-trade Fringillids originating from Western Europe and held in close quarters during transport to North America (18, 70). During a seroprevalence survey of periurban birds in Ohio in 1998, Morishita, Aye and Harr (57) found Purple Finches (*Haemorhous purpureus*), House Finches and Goldfinches, negative for anti-NDV antibodies.

Overall, directly inoculated Leghorns demonstrated the highest mean oral titers, but exhibited a truncated shedding duration in comparison to Northern Cardinals and American Goldfinches (Tabl3.2). The most interesting result of this study was the minimal transmission between the inter-specific transmission groups as determined by VI and rRT-PCR, excepting one Goldfinch and one House Finch, respectively. The low transmission rate demonstrated by the data may be related to the fact that the LaSota vaccine is a lentogenic vaccine, even though the strain it is derived from has an ICPI value (i.e. 0.4) higher than the other live lentogenic ND vaccines (56, 75). Our reasoning for using the LaSota vaccine virus was the assumption that wild birds would be likely to encounter it under present-day field conditions (15, 47), as it is the most commonly isolated NDV vaccine strain reported in wild birds (9, 71, 77). For instance, 51% of

the 54 NDV vaccine virus spillover events in wild birds documented by our prior work (9) were associated with the LaSota vaccine virus. This is not surprising, considering how frequently it is applied in the commercial poultry industry and its accessibility to smallholder flock owners for purchase (52, 75).

We also speculated that some of the wild birds that were co-housed with two inoculated Leghorns would demonstrate more evidence of transmission by the end of the experimental period. Wild birds may have been exposed via several infectious routes, such as shared circulating air, water, Leghorn feces, and a shared feed tray. Moreover, two 17-week old Leghorns in a single isolator with a transmission group of 2 to 5 naïve birds is a high infectious host-to-area ratio. Lack of transmission may be explained by the lack of virus excreted via the cloacal routes from the Leghorns, with only two birds excreting 101.5 to 103.3 EID₅₀/ml. In addition, negative pressure isolators undergo multiple changes in the volume of circulating air hourly, likely reducing the efficiency of aerosol transmission.

Other laboratories have documented the LaSota strain as having a high tropism for avian respiratory epithelium (89). This reported tropism is also consistent with our data demonstrating 56/112 total days of oral shedding for all Leghorns, in comparison to 2/112 of cloacal shedding. These observations may also explain why the movement of the Benchkote™ paper with the Leghorn droppings did not enhance transmission of NDV within the interspecific transmission groups. In a prior study, Spalatin, Turner and Hanson (79) reported a similar pattern within their

experimental inoculations of commercial chickens with a series of vaccine strains, including LaSota. While they did not utilize SPF birds, the chickens tested negative for *Salmonella* spp., NDV, and *Mycoplasma* spp. Of the 50 chickens in the study that were inoculated with 0.2 ml of $10^{7.3}$ EID₅₀/ml of the LaSota vaccine virus, cloacal shedding was detected in 10% of the birds by day 5 dpi. In the Spalatin, Turner and Hanson (79) study, the Hitchner B1 NDV vaccine was recovered from 7% of the 44 inoculated chickens. Our data, too, suggest very limited shedding from the cloaca in two out of five directly inoculated American Goldfinches, and one out of four directly inoculated House Finches as determined by VI. Interestingly, the transmission American Goldfinch, an after-second year (ASY) female, and a SY male House Finch that tested positive for NDV did so at 7 dpi and 13 dpi, which correlate to the later shedding patterns demonstrated in their directly inoculated counterparts.

House Finches and Brown-headed Cowbirds are particularly aggressive and will seek shelter within human-made structures such as commercial or backyard chicken coops. We expected that their aggressive behavior would make them more susceptible to infection than Northern Cardinals and American Goldfinches due to reduced time spent in refuges, and more time feeding and drinking near the single inoculated Leghorn in experiment two. We hypothesized that because they would be likely to increase their proximity to the source of exposure, transmission to these wild bird species would generate a frequency-dependent pattern of transmission (11). However, prior mathematical modelling work has shown that transmission

of NDV is largely density-dependent (21). Given that experiment III yielded only two transmission positive wild birds out of twenty passerines exposed to two Leghorns, and zero transmission positive birds in Experiment II, these data are more supportive of a density-dependent basis of transmission under experimental conditions, although more study is needed under experimental and field conditions.

It is unknown whether stress may have played a role among the interspecific transmission birds that died prior to the experimental endpoint. A male AHY Cardinal from Group 1 died overnight at 2 dpi from an unknown etiology, as the necropsy was inconclusive. In Group 11, one Goldfinch possibly died due to positional asphyxiation from an immobilized tarsometarsus at 3 dpi, while its cage mate was euthanized at 9 dpi due to monocular conjunctivitis. A third Goldfinch in the same group died possibly due to handling stress on day 7 dpi. In addition, it is also important to note that while physiological measurements were not collected for comparison from the Leghorns, they also demonstrated moderate stress behaviors consistent with neophobia upon co-housing with the wild birds, until the Leghorns acclimated 24-48 hours after placement into the isolator.

Given our results, the implications of our study indicate that the four species in this study would likely demonstrate heterogeneous susceptibility and transmission patterns upon exposure to lentogenic NDV in the field. Although laboratory conditions are difficult to extrapolate into a broader ecological context, our data also suggest that direct exposure to an infectious medium,

such as water laden with vaccine or a wild-type lentogenic virus that is directly consumed by passerines, may provide a more efficient mode of transmission as opposed to direct contact with chickens alone. Small birds such as American Goldfinches and House Finches can readily access chicken coops, thus wild bird proofing should consider small passerines for biosecurity purposes. Although our sample size was small, and our species breadth limited, our results are highly relevant to the current dilemmas faced by the wild bird-backyard chicken-commercial poultry triad. As free-roaming chickens continue to increase in popularity, passerines sharing their habitats will likely gain more significance in epidemiological studies of NDV. The four wild bird species were chosen for this study due to their high abundance in the poultry-dominated southeast and co-habitation of the same habitats where backyard flocks are often established. Further experimental studies in conjunction with field studies centered on backyard chicken farms should examine the susceptibilities of more peridomestic passerines, and their transmission potential to the avian community and nearby poultry operations.

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Table 3.1. Experimental design in tabular form. Each group number corresponds to a separate isolator in which birds were held, for a total of 13 wild bird groups divided into three different experiment types. The total birds per experiment number equals the number of wild birds per group plus the number of chickens assigned to each treatment. Three groups of Northern Cardinals were represented by one group in Experiment I, and two groups in Experiment III. Three Brown-headed Cowbird groups were each placed into Experiments I, II, and III. Three groups of American Goldfinches were represented by two groups in Experiment I and two groups in Experiment III. Four House Finch groups were allocated into a single group in Experiment I, one group in Experiment II, and two groups into Experiment III.

Species	Group Number	Number of Wild Birds	Experiment Number	Treatment	Total Birds Per Experimental Unit
Northern Cardinal	4	n = 2	1	Direct Inoculation	n = 2
Brown-headed Cowbird	5	n = 3	1	Direct Inoculation	n = 3
American Goldfinch	6	n = 2	1	Direct Inoculation	n = 2
House Finch	7	n = 4	1	Direct	n = 4

				Inoculation	
American Goldfinch	13	n = 5	1	Direct Inoculation	n = 5
House Finch	3	n = 4	2	Transmission – One Inoculated Chicken	n = 5
Brown-headed Cowbird	10	n = 3	2	Transmission – One Inoculated Chicken	n = 4
Northern Cardinal	1	n = 2	3	Transmission – Two Inoculated Chickens	n = 4
House Finch	2	n = 4	3	Transmission – Two Inoculated Chickens	n = 6
Northern Cardinal	8	n = 2	3	Transmission – Two Inoculated Chickens	n = 4
House Finch	9	n = 4	3	Transmission – Two Inoculated Chickens	n = 6
American Goldfinch	11	n = 5	3	Transmission – Two Inoculated Chickens	n = 7
Brown-headed Cowbird	12	n = 3	3	Transmission – Two Inoculated Chickens	n = 5

Table 3.2. Mean oral viral shedding titers and standard deviations. Tabular representation of the mean oral titers for each date of sampling post-inoculation for all directly inoculated species. Values were pooled across experiments to represent the shedding patterns of directly inoculated birds. Leghorns from Experiments II and III are represented, and Northern Cardinals, Brown-headed Cowbirds, American Goldfinches, and House Finches from Experiment I are represented.

Day Post- Inoculation	Leghorn Mean Oral Viral Titer $10^{\times}\text{EID}_{50}/\text{ml}$ and Standard Deviation ($n = 14$)	Northern Cardinal Mean Oral Viral Titer $10^{\times}\text{EID}_{50}/\text{ml}$ and Standard Deviation ($n = 2$)	Brown- headed Cowbird Mean Oral Viral Titer $10^{\times}\text{EID}_{50}/\text{ml}$ and Standard Deviation ($n = 3$)	American Goldfinch Mean Oral Viral Titer $10^{\times}\text{EID}_{50}/\text{ml}$ and Standard Deviation ($n = 7$)	House Finch Mean Oral Viral Titer $10^{\times}\text{EID}_{50}/\text{ml}$ and Standard Deviation ($n = 4$)
1	5.0 ± 0.4	1.5 ± 0.0	1.8 ± 1.5	1.2 ± 0.8	0.0 ± 0.0
2	5.0 ± 1.4	1.3 ± 1.3	1.2 ± 0.9	1.3 ± 1.2	0.0 ± 0.0
3	5.3 ± 0.4	2.2 ± 0.7	1.1 ± 0.8	1.2 ± 1.1	0.4 ± 0.6
4	4.8 ± 1.5	2.6 ± 0.7	1.0 ± 0.7	1.3 ± 0.9	0.9 ± 0.9
7	0.2 ± 0.6	2.8 ± 1.1	0.6 ± 0.8	0.7 ± 0.8	1.3 ± 0.7

9	0.0 ± 0.0	2.2 ± 0.5	0.0 ± 0.0	0.5 ± 0.7	0.4 ± 0.6
11	0.0 ± 0.0	1.5 ± 1.5	0.0 ± 0.0	0.7 ± 0.8	1.1 ± 0.6
End	0.0 ± 0.0	0.0 ± 0.0	0.5 ± 0.7	0.0 ± 0.0	$0.8 \pm 0.8)$

Table 3.3. Solution for the mixed model's fixed effects of the wild bird species viral titers over time in Experiment I. The model demonstrates that at least one species exhibited statistically different oral viral titer curves throughout the sampling period. Similar to a repeated measures analysis, independent variables in this mixed model were represented by the categorical variable “species,” and each dependent variable equaled the mean oral viral titer for each wild bird per date of sampling, where y1 = 1 dpi, y2 = 2 dpi, y3 = 3 dpi up to the last date of sampling where yEND equaled the final date of viral sampling at day 13 dpi or day 14 dpi.

Effect	Species	Estimate	Standard Error	DF	Pr > t
Intercept		1.7500	0.3049	12	<.0001
Species	AMGO	-0.8929	0.3458	112	0.0111
Species	BHCO	-0.9833	0.3937	112	0.0139
Species	HOFI	-1.1500	0.3735	112	0.0026
Species	NOCA	0	.	.	.

Fig. 3.4. Experimental setup prior to inoculation. Chickens and songbirds subjected to experimental infection trials were housed in an isolator such that the songbirds were enclosed in a wire mesh cage and there was sufficient space outside of the wire-mesh cage for one to two chickens. Wild birds and chickens shared food and water sources for the duration of the experimental period. Note that this image was captured from the vantage point facing the interior of the isolator through an open door, and has been created to approximate the housing conditions.



Table 3.5. Body condition changes. The changes in body condition parameters for captive birds from measurements recorded in the field at time zero to the final date of their respective experiments. Negative decreases in fat scores indicate that on average, the birds gained in visible fat stores when compared to scores recorded during field capture, while positive decreases mean the birds decreased their stores of visible fat. Score increases in flight feather wear indicate that the flight feathers- consisting of the primaries, secondaries and tertials exhibited visible decreases in feather quality in comparison to the flight feather score given in the field. Loss of wing chord length is the recorded change in millimeters from the longest un-flattened primary, and percent loss in wing chord length converts the change in millimeters to the number of millimeters lost as a percentage of the length of the wing chord upon initial capture. Loss of mass in grams is the difference between the weight in grams collected in the field at initial capture and the weight recorded at the end of the experiment, while percent loss of mass is the weight change calculated to represent the percent loss of weight in grams from the original weight recorded in the field.

Species	Score	Score	Loss in	Percent		
	Decrease in	Increase	Wing	Loss in	Loss in	Percent
	Fat	in Flight	Chord	Wing	Mass (g)	Loss in
		Feather	Length	Chord		Mass

Wear (mm)						
House						
Finches	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 4$)	$\bar{x} = 2.3, sd$	$= 1.5, sd$	$= 5.8, sd$	$= 7.3, sd$	$= 2.0, sd$	$\bar{x} = 9.0, sd$
Direct	$= 2.2$	$= 1.3$	$= 4.5$	$= 5.8$	$= 1.1$	$= 4.6$
Brown-headed						
Cowbirds	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 3$)	$= -2.0, sd$	$= 1.0, sd$	$= 2.3, sd$	$= 2.3, sd$	$= 4.4, sd$	$= 11.8, sd$
Direct	$= 2.0$	$= 1.0$	$= 1.5$	$= 1.5$	$= 0.2$	$= 0.2$
Northern						
Cardinals	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 2$)	$\bar{x} = 0.5, sd$	$= 1.0, sd$	$= 7.0, sd$	$= 7.5, sd$	$= 0.9, sd$	$\bar{x} = 2.1, sd$
Direct	$= 0.7$	$= 0$	$= 1.4$	$= 1.2$	$= 0.4$	$= 0.8$

American

Goldfinches	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	
($n = 7$)	$= -0.1, sd$	$= 1.0, sd$	$= 1.9, sd$	$= 2.6, sd$	$= 0.2, sd$	$\bar{x} = 1.8, sd$
Direct	$= 1.1$	$= 0.8$	$= 1.6$	$= 2.2$	$= 0.4$	$= 3.6$

House

Finches	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	
($n = 4$)	$= -0.5, sd$	$= 0.8, sd$	$= 2.3, sd$	$= 2.8, sd$	$= 1.4, sd$	$\bar{x} = 6.9, sd$
One	$= 1.0$	$= 0.5$	$= 1.7$	$= 2.1$	$= 0.8$	$= 4.0$
Leghorn						

Brown-
headed

Cowbirds	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 3$)	$= -3.3, sd$	$= 1.3, sd$	$= 0.7, sd$	$= 0.7, sd$	$= 3.9, sd$	$= 10.6, sd$
One	$= 0.6$	$= 1.2$	$= 1.2$	$= 1.2$	$= 1.6$	$= 3.9$
Leghorn						

Northern						
Cardinal	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 3$)	$= -0.3, sd$	$= 1.3, sd$	$= 3.0, sd$	$= 3.3, sd$	$= 4.1, sd$	$= 10.0, sd$
Two	$= 0.6$	$= 0.6$	$= 1.7$	$= 1.9$	$= 1.8$	$= 4.5$
Leghorns						
House						
Finches	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x} = 6.0, sd$
($n = 8$)	$= -0.9, sd$	$= 1.0, sd$	$= 2.0, sd$	$= 2.6, sd$	$= 1.2, sd$	$= 2.8$
Two	$= 2.1$	$= 1.3$	$= 1.9$	$= 2.4$	$= 0.6$	
Leghorns						
American						
Goldfinch	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
($n = 4$)	$\bar{x} = 0.5, sd$	$= 0.8, sd$	$\bar{x} = 0, sd$	$\bar{x} = 0, sd$	$= 1.7, sd$	$= 14.4, sd$
Two	$= 0.6$	$= 1.0$	$= 0$	$= 0$	$= 2.0$	$= 16.2$
Leghorns						
Brown-	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$	$\bar{x}$
headed	$= -0.7, sd$	$= 0.3, sd$	$= 1.0, sd$	$= 1.0, sd$	$= 4.8, sd$	$= 13.0, sd$
Cowbirds	$= 0.6$	$= 0.6$	$= 1.7$	$= 1.8$	$= 1.0$	$= 3.0$

($n = 3$)

Two

Leghorns

CHAPTER 4

SPILOVER OF THE LASOTA VACCINE VIRUS FROM BACKYARD CHICKENS TO
WILD BIRDS ¹

¹ Ayala, A.J., M.J. Yabsley, B.M. Williams², J.I. Dornbush, S.S. Fink, L.K. Haas, J.L. Crane, C.M. Crow, C.A. Camp, S.M. Schappaugh, A.J. Goodman, K.M. Thompson, L.J. Head, J.A. Williams, S.M. Cook, and S.M. Hernandez. To be submitted to the *Proceedings of the Royal Society B: Biological Sciences*.

Abstract:

Newcastle disease virus (NDV), a significant pathogen of poultry, is increasingly being isolated from wild birds such as passerines. Spillover of the virus between chickens and wild bird species is increasingly documented, yet the ecological reasons behind spillover events remains unclear. We examined how lentogenic NDV transmits from infectious chickens to passerine and near passerines using an experimental administration experiment performed under natural field conditions. Using a live NDV vaccine, the LaSota vaccine virus, we inoculated chickens living under two conditions at different sites in Athens, Georgia. The flock at Site S was maintained in a coop both day and night, whereas the flock at Site L was maintained as a flock of free-ranging chickens during the day. As a control, we also surveyed the wild birds at an additional site without backyard chickens. At the two sites in which the chickens were inoculated, we identified 13 wild bird species (of the 46 species captured) that had seroconverted and were identified as ‘high’ risk for NDV transmission from backyard chickens. Across all sites combined, these 13 high-risk species had 197 out of 351 captures (56%) near the backyard chicken coops. In addition, Chipping Sparrows (*Spizella passerina*), Northern Cardinals (*Cardinalis cardinalis*), and Tufted Titmice (*Baeolophus bicolor*) had the highest antibody responses across all three sites. Our results provide evidence that NDV spillover between passerines and backyard chickens occurs, and that inhibiting spillover events may also be crucial to the management of NDV outbreaks in backyard chickens, specifically those that share habitat with wild birds.

Introduction

Avian orthoavulavirus-1 (AOAV-1), or Newcastle disease virus (hereafter, NDV), is increasingly recognized as a pathogen of wild birds as well as poultry [1-9]. One of eight serotypes in the newly named genus *Orthoavulavirus*, NDV is one of 20 species of the *Avulavirinae* subfamily nested within the *Paramyxoviridae* family [10-12]. Highly transmissible and variable in its pathogenicity, NDV is a negative-sense, non-segmented, single-stranded and enveloped RNA virus [13-16]. Strains that are virulent to poultry are the causative agents of Newcastle disease (ND), which induces high morbidity, mortality, and disseminates swiftly among chicken flocks [7, 17-21]. NDV is currently circulating at near panzootic proportions, with 73 countries reporting poultry outbreaks to the World Organization for Animal Health (OIE) in 2018 [17, 20, 22-27].

Currently, NDV is divided into two classes based on phylogenetic characteristics [11, 28, 29]. Class I isolates are generally avirulent, and are most commonly isolated from waterfowl and shorebirds [30-32]. Class II isolates are isolated from a broad host range of wild birds as well as poultry, and are divided into three categories based upon increasing pathogenicity: lentogenic, mesogenic, and velogenic [2, 33, 34]. Strains that are lentogenic or low pathogenic strains cause mild respiratory disease in chickens, whereas mesogenic and virulent isolates present with high morbidity and high mortality, respectively [34-36]. Some lentogenic strains are used today as live vaccines in chickens, whereas mesogenic and virulent strains are the causative agents of

Newcastle disease and are thus reportable to the USDA and OIE [7, 37-39]. Wild birds are also susceptible to Newcastle disease; however, clinical signs are dependent upon the host and properties of the virus. Demographics that may influence the course of the disease include affected host species, host age, host immune status and exposure history, the viral dose, as well as the strain of NDV involved [2, 40-44].

In the United States, stringent biosecurity, biosafety, and vaccination protocols are utilized by the commercial poultry industry and the USDA to mitigate the transmission of pathogens such as NDV [34]. These protocols include sanitation, contained access to poultry houses, and most importantly, enclosures that prevent contact with wild birds or backyard chickens [27, 45, 46]. However, commercial poultry biosecurity has not always been sufficient to prevent outbreaks of NDV in the United States. In the last fifty years, NDV has periodically erupted into costly and devastating outbreaks in the Southwestern United States [47-49]. Curtailing each epidemic required the culling of millions of birds and hundreds of millions of dollars [50-52]. Moreover, each epidemic in the United States was perpetuated from backyard chickens infected with vNDV [47, 49, 53]. Thus, the importance of biosecurity in smallholder backyard flocks, as well as in commercial poultry, cannot be understated.

Backyard chickens are currently one of the fastest-growing sectors amidst the world's livestock agricultural industry [54-56]. Among industrialized countries where commercial chicken facilities are common, backyard chickens are increasingly popular as hobby pets and

environmentally friendly food sources [57-60]. Backyard chicken flocks are now almost equally represented across rural, suburban, and urban housing gradients in the United States. With the exception of local zoning ordinances, backyard chickens also remain unregulated [61, 62]. Thus, a wide diversity of husbandry protocols have been adopted by hobbyists, including whether wild birds are excluded from chicken coops and pens [63, 64]. For backyard chickens without coop exclusions, frequent interactions may occur with wild birds that may influence pathogen transmission [64-67].

Newcastle disease virus has a broad avian host range, and it has been demonstrated that spillover and spillback between chickens and wild bird species does occur [3, 14, 38, 40, 41, 68-74]. Species typically found in open rural landscapes such as Rock Pigeons (*Columba livia*), European Starlings (*Sturnus vulgaris*), and House Sparrows (*Passer domesticus*) are commonly associated with NDV in terrestrial habitats [42, 75-81]. However, outside of the agricultural sector, little is known regarding the peridomestic wild bird species that may become infected when they share habitat with infected backyard chickens (Ayala et al., unpublished data). Such peridomestic wild bird species could serve as bridge or maintenance hosts of NDV, if they visit multiple backyard chicken flocks in an area [82, 83]. Thus, identifying the species that may serve as bridge hosts are potentially key to managing vNDV outbreaks.

For example, during the California vNDV outbreak of 2002-2003, Kinde and Hullinger [65] surveyed wild birds, including American Goldfinches (*Spinus tristis*, Fringillidae), Blue Jays

(*Cyanocitta cristata*), Myrtle Warblers (*Dendroica coronata*), and unspecified members of the Passerellidae and Parulidae. None of these species were positive for vNDV by virus isolation, yet were frequently observed interacting with backyard chickens. In a separate surveillance study conducted by Toro and Hoerr [84], 180 wild bird serum samples were collected in Alabama to document serological exposure to NDV during an outbreak of Pigeon Paramyxovirus-1. Eleven of 14 high-risk peridomestic species, as defined by Ayala *et al.* (unpublished data), were represented in this study. Using competitive ELISAs, antibodies against NDV were detected, but all fell below the positive threshold. Interestingly, Tufted Titmice and Carolina Chickadees had antibody levels up to 30 (units not reported), which in chickens, antibody levels >40 are considered below the positive threshold.

Also unknown is the epidemiology of these spillover events, specifically the point in the backyard chicken NDV shedding cycle that wild birds become infected and seroconvert. Although surveillance studies for NDV in wild birds are commonly performed, surveying wild birds too early or late in the backyard chicken spillover cycle may underestimate both the species and number of wild birds afflicted [85].

During a prior study, we demonstrated the susceptibility of four peridomestic passerines found in backyard chicken habitats to NDV and transmission from chickens [43]. In a separate experiment, we also assessed the role of freely available chicken feed in attracting peridomestic wild birds to backyard chicken coops, where pathogens such as NDV may be concentrated

(Ayala *et al.* unpublished data). To address the knowledge gap regarding field transmission of viruses between peridomestic species and backyard chickens, we performed an experimental inoculation experiment using the LaSota vaccine virus at two sites associated with backyard chickens, and surveyed the wild birds at an additional site without backyard chickens. Our prior work has demonstrated that chickens inoculated with the LaSota vaccine virus shed virus [42, 43].

At each site, we identified the species of wild birds that: a) were captured in the same habitat as the chickens and/or their coops, b) their frequency of capture, c) and the wild birds that seroconverted in response to the LaSota vaccine virus. For the wild bird species that seroconverted, we also measured the antibody titer with which they did so. In chickens, we measured the humoral antibody response to inoculation with the LaSota vaccine virus over five sampling points.

Methods

Ethics Statement

All research was reviewed and approved by the University of Georgia's Institutional Animal Care and Use Committee (A2017 05-023-Y3-A20). Additionally, work was conducted under Georgia Department of Natural Resources Permit Number 1000524490, U.S. Fish and Wildlife Service Permit MB78901A-1, and U.S. Geological Service Master Bander Permit 23450.

Sites

We conducted our experimental vaccine virus inoculation experiment between 2016-2017 in three counties of Northern Georgia, United States. We selected two actively managed backyard flock sites, designated as Site S and Site L, based on whether the chickens were always cooped (Site S) or were able to free-range during the day (Site L). We surveyed an additional site that lacked chickens (Site C); however, an empty coop served as the center of the site. At all sites, wild birds smaller than Mourning Doves (*Zenaida macroura*) were able to enter and exit the chicken pens freely. The habitat of each site was mixed hardwood forest, interspersed by open grassland for an area of approximately 0.004 km².

Chickens

The backyard flock at Site S consisted of nine chickens (n=9) that were six months of age at the start of the experiment, otherwise designated as hatch-year (HY) birds. The free-range flock at Site L consisted of a larger group of chickens (n=16) at the beginning of the field season. The Site L flock included both after-hatch year (AHY, 68.75%) and HY (31.25%) birds. Site C contained no chickens (n=0).

All sites were in three adjoining counties in North Georgia; however, each site was managed using different husbandry techniques. Site L site allowed their chickens to free-roam during the day, Site S cooped their chickens during the day and at night, and Site C served as the control, where no chickens resided, but a chicken coop was present.

Experimental Design

Sites were sampled in order of decreasing chicken numbers to decrease risk of contamination. We sampled wild birds first at Site C (n=0 chickens), then at Site S (n=9 chickens), followed by Site L (n=16 chickens). Equipment was sanitized between farms using a 2% chlorhexidine gluconate solution to reduce microbial contamination. Per manufacturer instructions, chlorhexidine gluconate is effective against NDV. Hand sanitizer and handwashing with chlorhexidine gluconate were used between wild bird captures to reduce cross-contamination, and thus the potential for false positives from wild birds. At Sites S and L, the experiment was divided into three periods based upon the timeline of vaccine administration (Figure 2). As there were no chickens at Site C, the sampling of wild birds at that site consisted only of a single 30-day surveillance period, using the methods described in the pre-inoculation period. During the pre-inoculation period, we began with a chicken flock health assessment at Sites S and L at least one month before administering the LaSota vaccine virus [86-88]. This was followed by a pre-inoculation wild bird capture period, in which we captured wild birds throughout Site S and Site L for 30 days. Next, we performed another chicken health assessment on the date of inoculation (0 DPI). Following inoculation, we performed post-inoculation wild bird captures for 21 days. Throughout this wild bird capture period, our last period included chicken health assessments performed at four days post inoculation (4 DPI), 11 DPI, and 18 DPI, after which the experiment was terminated at 21 DPI.

Pre-inoculation exams - Chickens

We performed initial baseline health assessments, including physical examinations, on each chicken prior to the field inoculation experiment between 40 to 70 days before inoculation. [86-88]. We began each physical exam by observing each chicken for approximately ten minutes to observe their general appearance and natural behaviors, such as movement, foraging, preening, and whether the bird appeared bright, alert, and responsive to other flock members [87]. At this time, we also recorded whether each bird exhibited signs of dyspnea or panting [89], dysstasia from physical trauma or neurological deficits, or other obvious signalments of trauma, stress, or illness such as lethargy, recumbency, cachexia, concealment from flock mates, or fluffed, unpreened feathers [90, 91].

Upon capture and manual restraint, we performed a complete physical examination [87]. We scored the body condition by scoring the pronouncement of the keel on a 6 point, zero to five scale, with a zero indicating no detectable musculature overlaying the sternum, while a five indicated highly robust flight muscles with no discernable keel bone protrusion. Finally, each chicken was banded with an individual color combination on the tarsi for future identification.

We determined the age and sex of each chicken using standard methods [88]. We collected body weight and standard avian morphometric measurements [88, 92] and scored furculum fat on an eight-point scale from zero to seven, where zero indicated no stored fat in the furculum and seven indicated an excessive fat pocket. For all chickens, we utilized mass, fat, and keel scores as determinants of body condition instead of ordinary least squares regression or

weight-to-wing chord, due to the wide variety of breeds, and high variability in size that comprised each flock [42, 93].

Up to three milliliters of blood, based on weight, were collected from the brachial or saphenous veins of each chicken for pathogen antibody testing. Blood was placed in serum separator tubes with Hemogard™ closure (Becton Dickinson, San Jose, CA, USA). Sera were tested for antibodies to NDV and for *Mycoplasma* spp., because co-infection with lentogenic NDV and *Mycoplasma* spp. can increase NDV vaccine reactions, as well as inhibit the NDV humoral response [94, 95].

We also collected oral swabs for real-time reverse-transcription PCR (rt-RT PCR) to determine if chickens were shedding wild-type or vaccine origin NDV [42, 69, 71, 96]. Choanal swabs were placed in 2.0 mL cryovials (Corning Inc., Corning, NY, USA) containing 1.5 mL of Brain-Heart Infusion (BHI) (Becton Dickinson, San Jose, CA, USA) with gentamicin (200 µg/mL), penicillin (2000 units/mL), and amphotericin-B (4 µg/mL), and stored at -80°C [42]. Blood serum tubes and swabs were maintained in a cooler with ice packs in the field. Upon arrival to the lab, all blood tubes were centrifuged at 6,450 x g for five minutes, and sera stored at -20°C.

Pre-inoculation – Wild Birds

During the month prior to deploying the LaSota vaccine virus in chickens, wild birds at Sites S and L were captured to test for the presence of existing NDV antibodies or the shedding

of wild-type or vaccine-origin NDV by rt-RT-PCR [3, 6, 42, 69-71]. Wild birds at Site C were also captured and tested. At all sites, passerines and near passerines were captured using standard mist-netting techniques, with mist-nets concentrated near the chicken coops and pens [97]. Wild birds were processed, banded, and released according to methods established by the Institute for Bird Populations [98, 99]. We recorded age, sex of sexually dimorphic individuals, and body condition indices including wing chord length in mm, the degree of wear on the primaries on a six point scale from zero to five, body mass to the nearest 0.1 g, and the amount of visible furculum fat on an eight-point scale from zero to seven. All birds were banded with a unique USGS band number [100, 101].

Blood samples (< 1% of body weight) were collected from the brachial vein of birds weighing >8 grams [102]. Blood was placed into Sarstedt Microvette® Z-Gel 200 µl serum tubes (Sarstedt, Nümbrecht, Germany) [43, 102]. Choanal and cloacal swabs were collected from each bird [43]. After sampling, wild birds were released, usually within five minutes. Blood and swabs were processed and stored as described above.

Inoculation

Each chicken at Site S (n=9) and Site L (n=15) was inoculated through the oculo-nasal route with 0.1 ml of $10^{6.7}$ EID₅₀/ml of a commercially available B1 Type LaSota vaccine virus (Zoetis, Parsippany, NJ, USA) at 0 DPI, or zero days post-inoculation. The vaccine was reconstituted according to manufacturer instructions and titrated to ensure a homogeneous

challenge titer [43]. Site L was inoculated after the experiment at Site S had been terminated.

All chickens were re-examined and re-sampled as previously described, except for collection for *Mycoplasma* spp. testing.

Post-inoculation - Chickens

The same samples collected on 0 DPI were collected again from all chickens at 4 DPI, 11 DPI, and 18 DPI. Oral and cloacal swabs, and assessments of vaccine reactions were collected on days 1-7 DPI, and 9, 13, 15 DPI.

Post-inoculation – Wild Birds

Beginning on 1 DPI, we continued capturing wild birds each day at Site S and Site L for up to 21 DPI and collected demographic information, and biological samples as previously described. Recaptured wild birds were not re-bled until at least 14 days after their previous draw.

Serological assays

Standard hemagglutination inhibition (HI) tests were used to measure baseline NDV antibody titers using 25 µl of chicken serum [103]. A commercially-available indirect enzyme-linked immunosorbent assay (ELISA) (Biochem Scarborough, ME, USA) was used to test for antibodies against *Mycoplasma gallisepticum* (MG) or *Mycoplasma synoviae* (MS). Results for the MG/MS tests were reported as reactive when the sample/positive ratio (S/P) was ≥ 0.5 , and the optical density (O.D.) was greater than 667. *Mycoplasma* ELISA-positive samples were then tested using a HI test using 10 µl of serum using 4 HA units of MS or MG antigen. Samples that

demonstrated complete inhibition of red blood cell agglutination at a serum dilution of ≥ 80 were considered positive for *Mycoplasma* antibodies.

In chickens, antibody responses to inoculation from the LaSota vaccine virus over the course of the field season were further evaluated by HI from serum samples collected on 0, 4, 11, and 18 DPI. In wild birds, HI tests were also used to determine wild-type NDV exposure rates at Site C, and pre- and post-inoculation exposure rates at Sites S and L. Prior to assaying, all wild bird serum was first pre-treated to remove natural serum agglutinins as described (12, 109). Standard volume HIs were used when 25 μ l of wild bird serum remained after washing and centrifuging. However, given that most of our species weighed less than 20 g, 1:10 dilutions incorporating 5 μ l of serum were used in the HI when insufficient serum volumes were available.

All HI testing for NDV incorporated 4 HA units of homologous antigen, the LaSota vaccine virus. Samples that demonstrated complete inhibition of red blood cell agglutination at a serum dilution of ≥ 16 were considered positive for NDV antibodies.

Real-time reverse-transcription PCR

Chickens inoculated with the LaSota vaccine virus and wild birds from Sites S, L, and C were evaluated by rt-RT PCR performed on selected oropharyngeal chicken and wild bird swab samples collected during the pre-inoculation, inoculation, and post-inoculation periods. Cloacal swabs were not evaluated due to prior virus isolation and rt RT-PCR results that suggested infection by LaSota was limited to the respiratory tissues [43]. Viral RNA was extracted from

swab media using the TRizol LS (Invitrogen, Carlsbad, CA) phenol based RNA extraction method, as previously described [28, 106].

One-tube reactions were used for reverse transcription and amplification using the AgPath-ID One Step RT-PCR kit (ThermoFisher Scientific, USA) in the Bio-Rad CFX96 Touch real-time PCR system (Bio-Rad Laboratories, Inc., Hercules, CA) using M gene primers, probe, and a modified protocol [107]. The cycling profile for reverse transcription took place at 45°C for 10 minutes, followed by a second cycle of 95°C for 10 minutes, and denaturation, annealing, and elongation were 40 cycles of 94°C for 10 seconds, 56°C for 30 seconds, and 72°C for 10 seconds. Cycle threshold (CT) values ≤ 37 were considered positive for NDV.

Statistical analyses

All statistical analyses were performed using R version 3.6.1. First, we examined whether sampling period (time) had a significant effect upon the mean change from pre-inoculation chicken antibody HI titers against NDV. In the same model, we also examined whether the measured HI titer changes in sampling period covaried by site (L or S) and included the individual chickens as a random effect. As our data were non-normally distributed, we utilized the lme4 and lmerTest packages to run our linear mixed effects model [108].

We also examined whether a correlation existed between the pre-inoculation chicken NDV HI titers with the *Mycoplasma* ELISA titers, as serum for both assays were collected on the

same date. We used the non-parametric Spearman's Rank Correlation test to examine the monotonic relationship between the two variables [109].

For all wild birds, we examined whether there were site differences in NDV HI titers using the non-parametric Kruskal Wallis H test [110]. We further tested the differences between individual sites by using post-hoc pair-wise Wilcoxon Rank Sum tests, that were adjusted using Bonferroni corrections. We also examined the relationship between wild bird age and NDV HI titers using the Kruskal Wallis H test to determine whether older birds were more likely to have higher HI titers. Lastly, for wild bird species with a sufficient sample size, we examined the relationship between weight to wing chord (body condition) and NDV HI titers to determine whether a lower body condition was associated with NDV HI titer [42]. We used the Spearman's Rank Correlation test to examine the monotonic relationship between the weight to wing chord and NDV HI titer, testing each species individually.

Results

Pre-inoculation and 0 DPI at Sites S and L

All physical exams performed on the chickens from Site S during the pre-inoculation and inoculation phases were normal. However, at Site L, some chickens had desquamation of the scales covering the tarsi consistent with *Knemidocoptes* spp. infestations, and one hen had a pendulous crop. Mean fat and keel scores for the chickens at Site S were generally higher, but not examined statistically, in comparison to Site L (Table 4.4, Figure 4.3).

4,11,18 days post-inoculation exams on Site S and Site L chickens

Mild vaccine reactions that are commonly reported were first detected in the chickens at Site S at 2 DPI to 3 DPI [39]. The most common clinical signs were discharge from the nares and runny stools. The clinical signs detected among the Site L chickens were similar at 2-3 DPI, although some chickens at this site also had vents stained with urates, suggesting polyuria. Between 4 and 9 DPI, most of the chickens at Site S also had similar clinical signs. At Site L, chickens had the same clinical signs, and 20% of chickens had pyrexia. Clinical signs resolved at Site S for most birds by 9 DPI, whereas the birds at Site L had clinical signs up to 15 DPI, with one bird demonstrating clinical signs up to 18 DPI.

Wild bird captures

At all three sites, 46 different wild bird species were captured (Table 4.5), representing 351 captures. Recapture rates ranged from 9.4% to 23.1% (Table 4.6). Tufted Titmice, Northern Cardinals, and Carolina Wrens were the most frequently captured species overall.

Serology in Chickens at Sites S and L

Mycoplasma serology

All chickens at Site S were below the O.D. of 667 for *Mycoplasma* spp. and were thus negative. At Site L, five individuals had O.D. values below 667, and eleven were above the positive threshold. The follow-up hemagglutination inhibition (HI) assays that were performed on serum samples reactive to the ELISA at Site L were all below the positive threshold of 80.

Statistical analysis – Mycoplasma and NDV serology

A Spearman's Rank Correlation test examined whether a statistically significant correlation existed between the pre-inoculation chicken NDV HI titers and the *Mycoplasma* ELISA titers. We found a weakly negative, but statistically significant relationship between the two variables ($r_s(3363.2) = -0.46, p = 0.023$) (Table 4.2).

NDV serology

Serology for NDV at Site S demonstrated exposure to a strain of NDV prior to vaccine administration. During the pre-inoculation phase, 100% of the chickens at Site S had titers ≥ 16 and a geometric mean titer of 43.54 (Figure 4.4). After administering the LaSota vaccine virus, titers increased until 18 DPI to a geometric mean of 406.37.

During the pre-inoculation phase, seven (47%) chickens at Site L had NDV titers ≥ 16 (geometric mean titer 3.55), and they remained negative until 0 DPI (Figure 4.4). After administering the LaSota vaccine virus, titers increased until 18 DPI to a geometric mean of 168.9.

Statistical analysis-NDV serology

In our linear mixed effects model, we found that there was no significant difference between Site S and Site L concerning pre-inoculation NDV HI titers. However, we did find a significant difference between baseline NDV HI titers and the repeated measures of subsequent NDV HI titers collected at 0, 4, and 18 DPI. Due to missing values, we were unable to include 11 DPI.

When covarying the repeated measures of NDV HI titers by Site, we found no significant difference between Site S and Site L (Table 4.2).

Serology in Wild Birds

During the surveillance period at Site C, we tested 26 of 104 total captured birds (excluding recaptures) by HI, in which a positive HI titer is ≥ 16 . All tested individuals were negative (titers < 10) (Table 4.1, Table 4.7). During the pre-inoculation period at Site S, we tested 19 of 45 captures by HI; seven individuals were positive. Neotropical migrants such as Wood Thrushes (n=2) and an Eastern Wood Pewee (n=1) had the highest HI titers (Table 4.1, Table 4.7). During the post-inoculation period at Site S, we tested 40 birds out 87 captures by HI; 28 birds were positive. (Table 4.6). Resident species such as Tufted Titmice (n=9), Northern Cardinals (n=7), and Chipping Sparrows (n=7) had the highest HI titers (Table 4.1, Table 4.7).

At Site L, 12 of 32 captures were tested during the pre-inoculation period and eight birds were positive. Migratory species, such as a Swainson's Thrush, and residents such as Downy Woodpeckers and Tufted Titmice demonstrated the highest HI titers during this period (Table 4.1, Table 4.7). During post-inoculation period at Site L, we tested 18 of 56 birds and 12 were positive (Table 4.6). Resident species such as Tufted Titmice, Downy Woodpeckers, and Eastern Phoebe had the highest HI titers (Table 4.1, Table 4.7).

Statistical analyses – Wild Bird Serology

Using the non-parametric Kruskal Wallis H test, we examined whether there were site differences in NDV HI titers for all wild birds. We found a significant difference across all sites, including the control Site C, ($\chi^2(2) = 37.129, p = 0.014$). We further tested the differences between individual sites by using post-hoc pair-wise Wilcoxon Rank Sum tests using Bonferroni corrections for p -value adjustments. Site C, the control, demonstrated a significant difference in HI titers when contrasted against Site L (p -value = < 0.0001) and Site S (p -value = < 0.0001), but Site L and Site S did not demonstrate a significant difference between one another (p -value = 1).

To determine whether higher NDV HI titers were stratified by the age of wild birds tested, we used the Kruskal-Wallis H test to determine whether older birds were more likely to have higher HI titers. Although we captured more hatch-year (HY) birds overall, we did find that after-hatch year (AHY) birds had significantly higher NDV HI titers ($\chi^2(1) = 6.5381, p = 0.011$) (Figure 4.6).

We used the Spearman's Rank Correlation test to examine the monotonic relationship between the weight to wing chord ratio (body condition) and NDV HI titer. We tested each species using separate Spearman's Rank Correlation tests, since mean body condition scores varied widely across species. For most species tested, there was no significant relationship between body condition and NDV HI titer. Eastern Phoebe were the only species that demonstrated a strongly negative, significant relationship between the two variables ($r_s(40) = -1, p = 0.01667$) (Table 4.3).

Chicken rt-RT-PCR

At Site S, no viral RNA was detected from oropharyngeal swabs from chickens during the pre-inoculation and 0 DPI periods. Viral RNA was detected at 1 DPI in six of the seven chickens tested (CT values between 27 and 37). At 2 DPI, viral RNA was detected in all five chickens tested (CT values between 25 and 31). At 4 DPI, eight of the nine chickens tested positive for viral RNA (CT values between 24 and 28).

Oropharyngeal swabs were not tested from the chickens at Site L (n=15) during the pre-inoculation phase or upon the date of inoculation. No oropharyngeal swabs at this site were tested during the post-inoculation phase.

Wild Bird rt-RT PCR

At Site C, 101 wild birds were negative for NDV viral RNA by rt-RT PCR. At Site S, 10 wild birds tested were rt-RT PCR negative for NDV viral RNA during the pre-inoculation period. However, during the post-inoculation period, NDV viral RNA was detected in four of 28 (14.3%) wild birds. At 1 DPI and 5 DPI, two Carolina Wrens tested positive. At 6 DPI, one Northern Cardinal was positive, and at 9 DPI, one Chipping Sparrow was positive. At Site L, no wild birds were tested.

Discussion

Although numerous studies have examined the prevalence of NDV in free-roaming wild birds, to our knowledge, this is the first study to test the spillover of lentogenic NDV under field

conditions using a live vaccine [3, 5, 7, 16, 42, 69, 79, 80, 109-115]. Additionally, we performed this transmission study under the natural ecological circumstances that result in backyard chicken-wild bird interactions, which we had previously established (Ayala *et al.* unpublished data). These data suggest that we were able to replicate the circumstances favoring transmission of the LaSota vaccine virus as reported in prior surveillance studies [42, 69, 96].

This finding is especially important given the increasing number of households in the United States that have backyard chickens [116-118]. Oftentimes, owners practice minimal biosecurity or bioexclusion, in that their backyard chickens are able to roam freely or lack enclosures that keep wild birds out of coops and pens [57, 119]. These circumstances diminish natural cross-species barriers between backyard chickens and wild birds, which are crucial to inhibiting the transmission of pathogens, such as NDV [120, 121]. Thus, as backyard chicken ownership continues to rise in popularity, we expect more opportunities for transmission of pathogens and parasites [68, 122-130]. Shared habitat between wild birds and backyard chickens may generate ideal circumstances for the emergence and amplification of multi-host pathogens [131], a phenomenon also referred to as pathogen spillover and spillback dynamics [132, 133]. While examples of spillover effects between backyard poultry and wild birds in the literature are predominantly focused on highly pathogenic avian influenza (HPAI) [134-136], pathogen spillover has been detected for other pathogens [137].

In this study, we aimed to demonstrate that viral pathogens can move from backyard chickens to songbirds and examined whether backyard chicken husbandry played a role in NDV LaSota vaccine virus transmission. During the 4, 11, 18 DPI exams, the Site L chickens demonstrated clinical signs and changes in body condition consistent with NDV vaccine reactions for a longer duration than the Site S chickens [138, 139]. The observed difference in post-vaccine reactions may have been due to the decreased baseline NDV immunity demonstrated by the Site L flock (Figure 4.4).

We explored the possible complicating factor of *Mycoplasma* spp. infections on the NDV humoral response in the chickens. *Mycoplasma* spp. may have played a role in the comparatively higher vaccine reactions at Site L [94]; however, we remain uncertain as to the *Mycoplasma* spp. status of the Site L chickens. All chickens at Site S tested below the ELISA O.D. of 667. In contrast, 11 of the 16 Site L chickens tested positive for *Mycoplasma* spp. antibodies by ELISA but were subsequently all negative by HI. As the ELISA is a highly sensitive screening test, it may be possible that these results were false positives. On the other hand, the ELISA measures multiple antigens expressed by *Mycoplasma* spp., whereas the HI test is specific to the hemagglutinin antigen produced by *Mycoplasma gallisepticum* or *Mycoplasma synoviae*. One potential explanation for the ELISA positive results and the negative HIs may have been due to recent *Mycoplasma* spp. exposure. If the exposure occurred prior to the development of MG or MS hemagglutinin antibodies, a screening test such as the ELISA would be positive; however,

the HIs would fall below the positive threshold [140]. Confirmation of this hypothesis is only possible through PCR. On the other hand, our statistical analysis found evidence that lower NDV HI titers were associated with the higher *Mycoplasma* spp. ELISA titers during the pre-inoculation phase.

Concerning the wild bird captures, we noted interesting patterns. Tufted Titmice, Carolina Wrens, and Northern Cardinals, which are highly abundant resident species in the Athens, Georgia area [141], were the most commonly captured species across all sites. These results complement our prior study, where we quantified the visit rates of wild birds to chicken coops in response to freely available chicken feed. Tufted Titmice, Carolina Wrens, and Northern Cardinals were not only among our most abundant community species but were also categorized as high risk for pathogen spillover from backyard chickens, due to their proclivity to enter chicken coops (Ayala *et al.* unpublished data).

During the surveillance period, all birds at Site C were negative by HI. At Site S and Site L, however, several individuals tested positive for antibodies against NDV during different periods of the experiment. Species that tested positive by HI during the pre-inoculation period included residents such as Carolina Wrens, Tufted Titmice, Eastern Phoebe, and Downy Woodpeckers. Neotropical migrants, such as an Eastern Wood Pewee, a Swainson's Thrush, and two Wood Thrushes, also tested positive by HI. Interestingly, we also captured a suspected pair of overwintering Yellow-bellied Sapsuckers at Site S with a geometric mean HI titer of 80.

According to wild bird serology, we identified 13 species of wild birds in Athens, Georgia that are susceptible to seroconversion in response to exposure to backyard chickens that were shedding the LaSota vaccine virus. Four of those species demonstrated increases in HI titers during the post-inoculation phase in comparison to the pre-inoculation phase, e.g. Carolina Wrens (Sites S and L), Eastern Phoebe (Site S increase), Northern Cardinals (Sites S and L), and Tufted Titmice (Sites S and L). Species such as American Robins (Site S), Chipping Sparrows (Site S), Eastern Bluebirds (Site L), Hermit Thrushes (Site L), Myrtle Warblers (Site L), Northern Mockingbirds (Site S), Slate-colored Juncos (Site S), and White-throated Sparrows (Site S) had positive HI titers only during the post-inoculation phase.

Interestingly, our statistical analyses provided evidence that AHY birds had higher mean NDV HI titers than HY birds. This may be explained by either of two phenomena. The first is that AHY birds were able to mount a significantly stronger immune response to NDV, in comparison to HY birds, potentially as a result of a higher viral dose exposure. The more dominant AHY birds had increased access to the chicken coops for the purposes of consuming chicken feed in comparison to the less dominant HY birds, which may have increased their exposure to the LaSota vaccine virus [142]. The second is that AHY birds were re-exposed to NDV at Sites S and L, and that for HY birds, inoculation with the LaSota vaccine virus during this experiment was their first exposure. Given that only cross-sectional data is available in the

literature regarding NDV exposure and wild bird antibody responses, this suggests an avenue for further research.

Interestingly, Downy Woodpeckers demonstrated a slightly lower mean HI titer during the post-inoculation period than the pre-inoculation period at Site L, with the mean HI titer of 57.98 falling to 51.72. This may be due to sample size, as five individuals were tested by HI during the pre-inoculation phase, and two were tested during the post-inoculation phase. Another interesting result were the two Mourning Doves captured during the post-inoculation period at Site S that were negative by HI. Although the Columbidae are considered competent reservoirs of NDV [30], they may not have been exposed to sufficient virus due to their inability to enter the coops and pens.

In this experiment, we demonstrated that spillover of the NDV LaSota vaccine virus to passerines and near passerines, e.g. backyard ‘birds’ is possible from infectious chickens. Considering what is known of the literature, this result is not surprising. Wild birds in Georgia have consistently tested positive for exposure to ‘poultry’ respiratory pathogens, and this knowledge is significant for the poultry industry, backyard chicken owners, and avian conservationists. For example, the bacterium *M. gallisepticum* has been consistently detected in Northeast Georgia birds since 1996 [143, 144]. Pigeon-paramyxovirus-1, an antigenic variant of NDV adapted to Columbids was detected in 68% of 100 Rock Pigeon serological samples collected in Atlanta, Georgia in 2003 [145]. Avian metapneumovirus subtype C was isolated

from a small sample of American Coots and Canada Geese in Northeast Georgia in 2000 [146]. Other poultry-associated pathogens detected in wild birds from Georgia include *Aspergillus fumigatus* [147], *Mycobacterium avium* subsp. *paratuberculosis* [148], and *Salmonella enterica* serovar Typhimurium [149].

As backyard chickens in the United States are managed under variable husbandry protocols, it is not surprising that ambiguity remains concerning how ‘poultry’ viruses such as NDV are transmitted to wild birds. More specifically, questions remain as to which failure of biosecurity is responsible for a spillover event and/or backyard chicken outbreak . We found no clear answer, as wild bird HI titers were higher at Site L, yet more species were seropositive by HI at Site S. Moreover, our linear mixed model demonstrated no difference between wild bird HI titers at Site S and Site L. Unsurprisingly, this suggests that the mechanisms influencing pathogen spillover and spillback are more complex and multifactorial than can be explained by husbandry alone.

For instance, van Dijk and Verhagen [150] listed nine key host and viral traits governing successful cross-species transmission of low pathogenic avian influenza viruses. These traits can be divided into two basic categories: traits mediated by host-pathogen exposure and traits mediated by host-pathogen susceptibility [150]. Thus, the poultry diseases most likely to spillover from backyard chickens and successfully circulate into wild birds would be multi-host pathogens (host-pathogen susceptibility) [151], whose broad avian host ranges would allow for

increased effective contacts between susceptible community species (host-pathogen exposure) [152].

Given the number of wild birds observed consuming chicken feed, this may have been a significant mode of transmission. Wild birds may undergo temporary, yet artificial, increases in wild bird species richness and abundance resulting from freely available feed, thus increasing pathogen risks to a greater number of individuals [153]. For example, this phenomenon has been detected in local wild bird feeder enzootics of *Salmonella* spp., *Mycoplasma* spp., and Avipoxviruses [153-155]. It is also likely that the same risk factors that broker pathogen transmission at bird feeders also hold true for wild birds attracted to chickens and their feed, near backyard coops. As NDV transmission does not always require bird to bird contact, infectious chickens or wild birds that aggregate at high traffic chicken feeders and waterers may infect other birds through direct contact or by shedding high concentrations of NDV upon shared surfaces [156-158]. This exposure route may come with additional risks to wild birds, such as the consumption of grain toxins, low-quality feed, or insufficient key micronutrients. Poor nutrition may alter immune dynamics in some wild species and, thus, contribute to the development of clinical disease in otherwise healthy birds [159-161].

Survey data from backyard chicken owners have documented that the use of biosecurity practices and flock health protocols, such as microbial bioexclusion, flock containment, vaccination, coop disinfection, disposition of carcasses or waste, and the minimization of traffic

between flocks [55, 57, 62, 64, 119] fall well below commercial poultry industry standards [162]. To prevent spillover events, the management of backyard chickens should be aimed at deterring wild birds from: a) chicken coops, b) chicken feed, and c) waterers [163]. Ultimately, however, whether wild birds and backyard chickens co-mingle and share pathogens relies upon the husbandry and sanitation practices of flock owners [56]. Even more importantly, it relies on the ability of poultry specialists, backyard chicken extension veterinarians, and the avian conservation community to collectively share and disseminate this knowledge to backyard poultry keepers.

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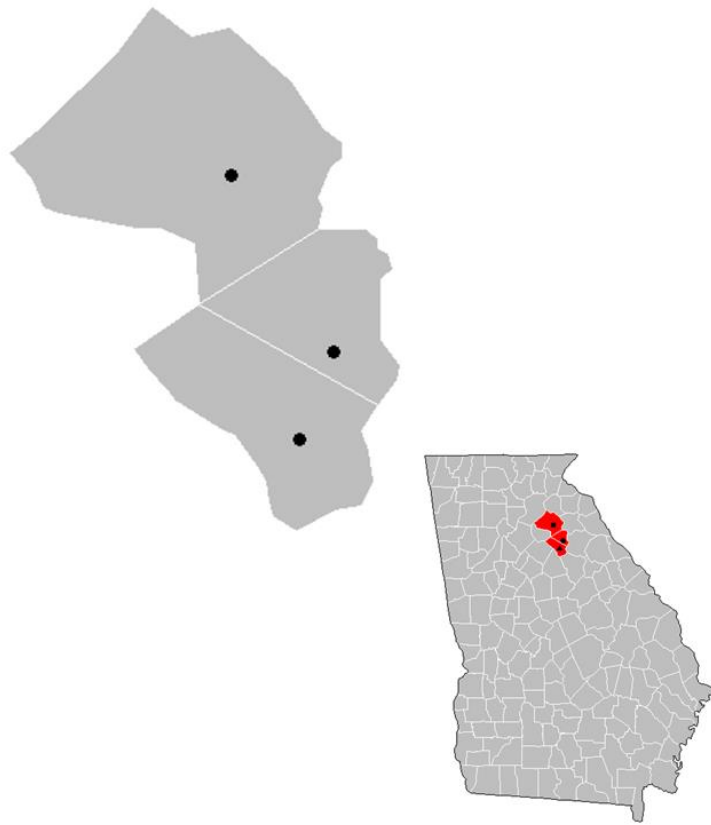


Figure 4.1. Map of the three counties in Georgia with the field sites highlighted in points. The inset map represents the state of Georgia, with Jackson, Oconee, and Athens-Clarke counties highlighted in red. In the county map, Site S, the cooped flock, is represented by the top point, while Site C is represented by the middle point. Site L, the free-range flock, is represented by the bottom point.

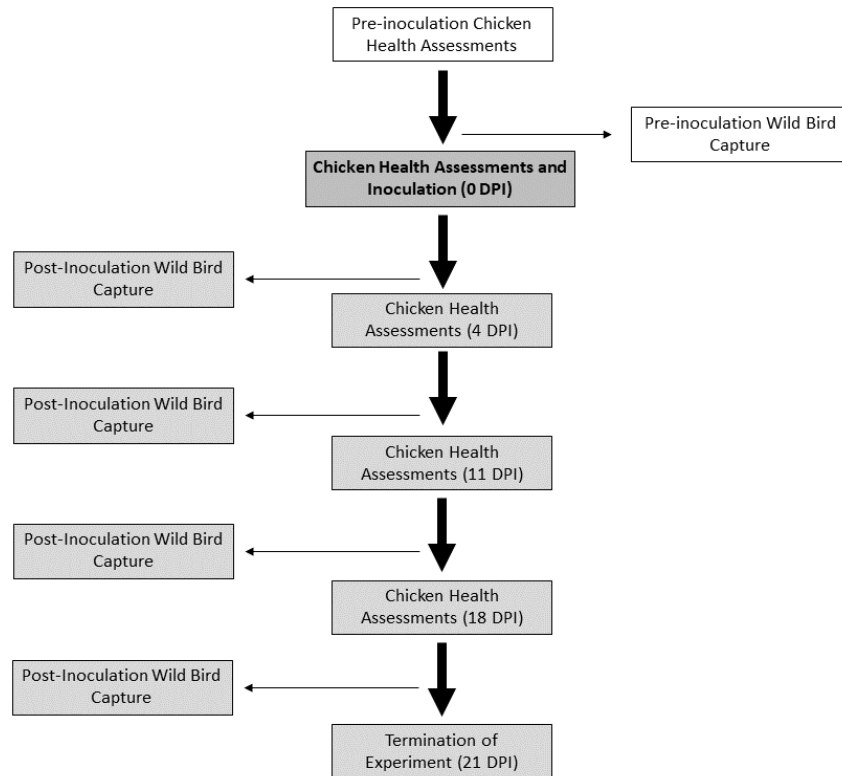


Figure 4.2. Representation of the timeline of the vaccine spillover experiment at Site S between September and November 2017. The pre-inoculation phase includes initial chicken health assessment at the top of the figure, and the pre-inoculation wild bird capture period. The date of inoculation was represented by the 0 DPI chicken health assessment. The post-inoculation phase comprised the post-inoculation wild bird capture period and all of the post-inoculation chicken health assessments at 4, 11, and 18 DPI.

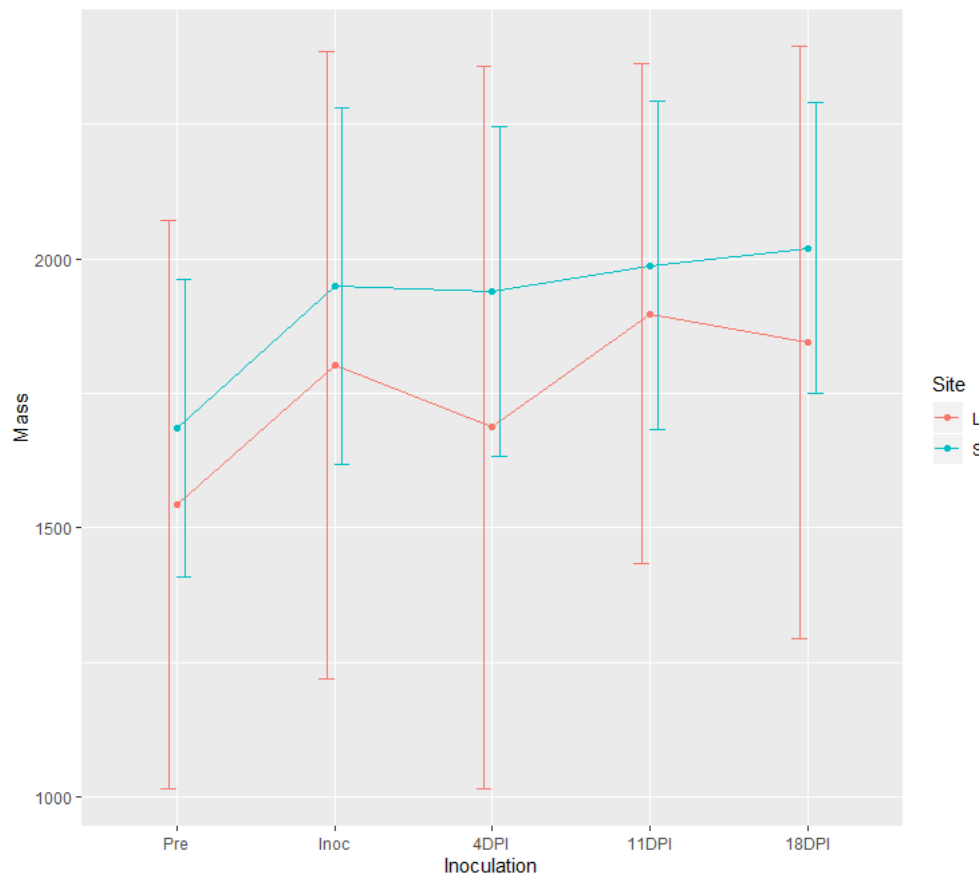


Figure 4.3. Average mass in grams for the chickens at Site S (cooped) and Site L (free-roaming).

Masses were collected over each of the five health evaluation dates, specifically the pre-inoculation period, the date of inoculation (0 DPI), and 4, 11, and 18 days post-inoculation.

Error bars represent the standard deviation.

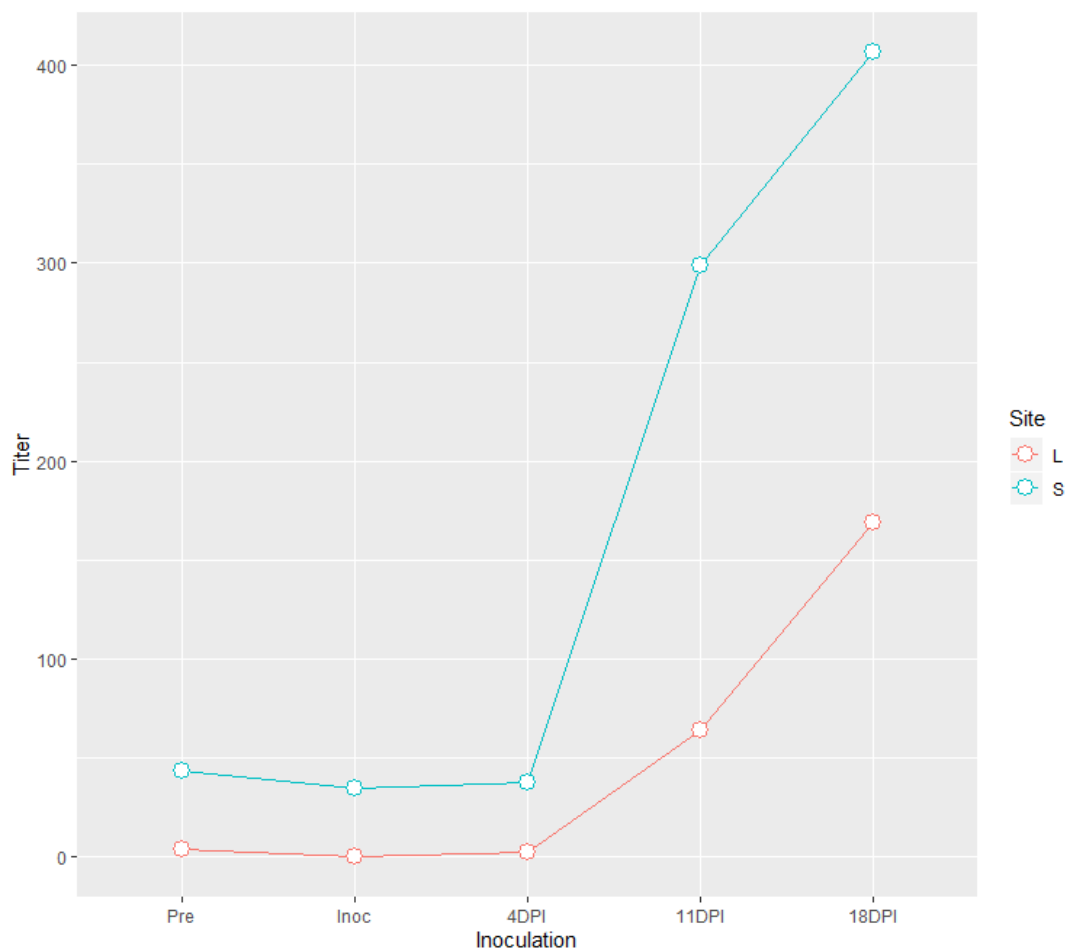


Figure 4.4. Mean change in chicken antibody HI titers against NDV over five sampling periods. Pre and Inoc represent the pre-inoculation period, and the day of inoculation, respectively, while post-inoculation titers are represented by 4 DPI, 11 DPI, and 18 DPI. Site S, the cooped flock, is represented by the blue line, while Site L, the free-range flock, is represented by the red line.

Table 4.1. Mean species NDV HI titers are represented according to species, then further compared by site, whereas Table 4.8 is categorized by site. Surv/Pre mean titers represent the titers performed on serum prior to inoculation. Post mean titers are represented by serum collected from wild birds during the post-inoculation period. Surv/Pre and Post ≥ 16 represent the number of individuals of that species that tested above the threshold of 16. Surv/Pre (n) and post (n) represent the total number of individuals tested during those periods. The superscripts R, B, and W represent resident, breeding, and wintering species, respectively. The *superscript represents a species with a titer that changed or developed from the Surv/Pre period in contrast to the Post period.

Species	Site	Surv/Pre Mean Titers	Post Mean Titers	Surv/Pre ≥ 16	Post ≥ 16	Surv/Pre (n)	Post (n)
American Robin ^{R*}	S	N/A	0.0005	N/A	20%	0	5
Black-and-White Warbler ^B	C	0.0000004	N/A	0	N/A	3	0
Carolina Wren ^{R*}	C	0.0000004	N/A	0	N/A	3	0
	S	0.002	1.927	25%	67%	4	6
	L*	8	40	0	100%	1	1

Chipping Sparrow ^{R*}	S	N/A	5.111	N/A	71%	0	7
Downy	S	N/A	0.0000004	N/A	0%	0	1
Woodpecker ^{R*}	L	57.982	51.720	100%	100%	5	2
Eastern Bluebird ^{R*}	L	N/A	1.372	N/A	50%	0	6
Eastern	S	0.007	64	50%	100%	2	1
Phoebe ^{R*}	L	N/A	32.56	N/A	100%	0	2
Eastern Towhee ^R	C	0.0000004	N/A	0%	N/A	1	0
Eastern-Wood Pewee ^B	S	112	N/A	100%	N/A	1	0
Gray-cheeked Thrush ^M	S	0.0000004	N/A	0%	N/A	1	0
Gray Catbird	C	0.0000004	N/A	0%	N/A	1	0
Hermit Thrush ^{W*}	L	N/A	0.0057	N/A	50%	0	2
House Finch ^R	C	0.0000004	N/A	0%	N/A	3	N/A
Hooded	C	0.0000004	N/A	0%	N/A	2	N/A

Warbler ^B							
Mourning	S	N/A	0.0000004	N/A	0%	0	2
Dove ^R							
Myrtle	L	N/A	0.0002	N/A	33%	0	3
Warbler*							
Northern	C	0.0000004	N/A	0%	N/A	9	N/A
Cardinal ^{R*}	S	0.01	8.15	50%	86%	2	7
	L	0.002	N/A	0%	N/A	2	0
Northern	S	N/A	32	N/A	100%	0	1
Mockingbird ^{R*}							
Slate-colored	S	N/A	80	100%	N/A	0	1
Junco ^{W*}							
Summer	S	0.0000004	N/A	0%	N/A	1	0
Tanager ^B							
Swainson's	L	640	N/A	100%	N/A	1	0
Thrush ^M							
Tufted	C	0.0000004	N/A	0%	N/A	4	N/A
Titmouse ^{R*}	S	0.000186	10.293	33%	66%	3	9
	L	10.293	92.26	66%	100%	3	2

White-eyed	C	0.0000004	N/A	0%	N/A	1	N/A
Vireo ^B							
White-throated	S	N/A	40	N/A	100%	0	1
Sparrow ^{W*}							
Wood Thrush ^B	S	56.57	N/A	100%	N/A	2	0
Yellow-bellied	S	80	N/A	50%	N/A	2	0
Sapsucker ^W							

Table 4.2. Type III analysis of variance table with Satterthwaite's method, using the lme4 and lmerTest packages in R. Site, and Site covaried by NDV sampling date, did not demonstrate a statistically significant difference from the pre-inoculation HI titers in our linear mixed effects model. However, the repeated measures of subsequent NDV HI titers, represented by the variable NDV in the table, did demonstrate a statistically significant difference.

Variable	Sum of Squares	Mean Squares	Numerical DF	Density DF	F-value	p-value
Site	554994	554994	1	87.993	3.5135	0.06419
NDV	3969567	1323189	3	87.999	8.3767	<0.0001*
Site by NDV	1178451	392817	3	87.999	2.4868	0.06576

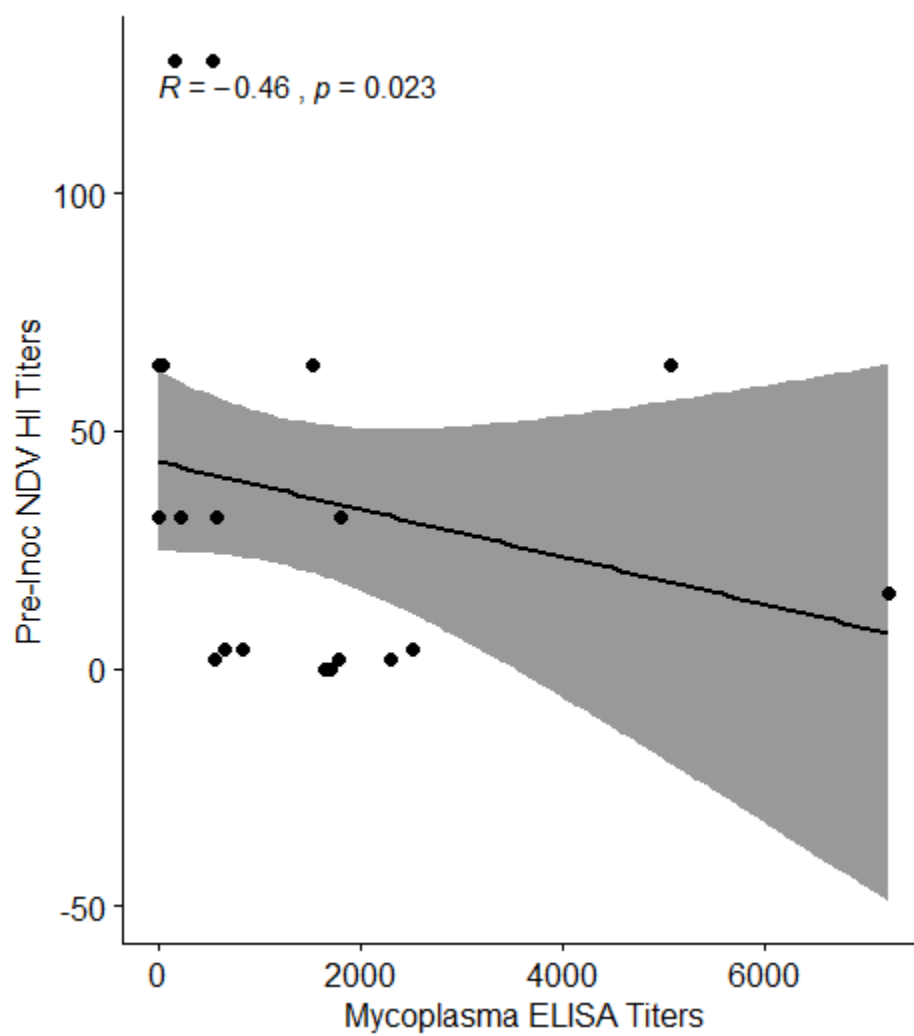


Figure 4.5. Relationship between the Mycoplasma ELISA titers and the NDV HI titers measured at the pre-inoculation period ($r_s(3363.2) = -0.46, p = 0.023$).

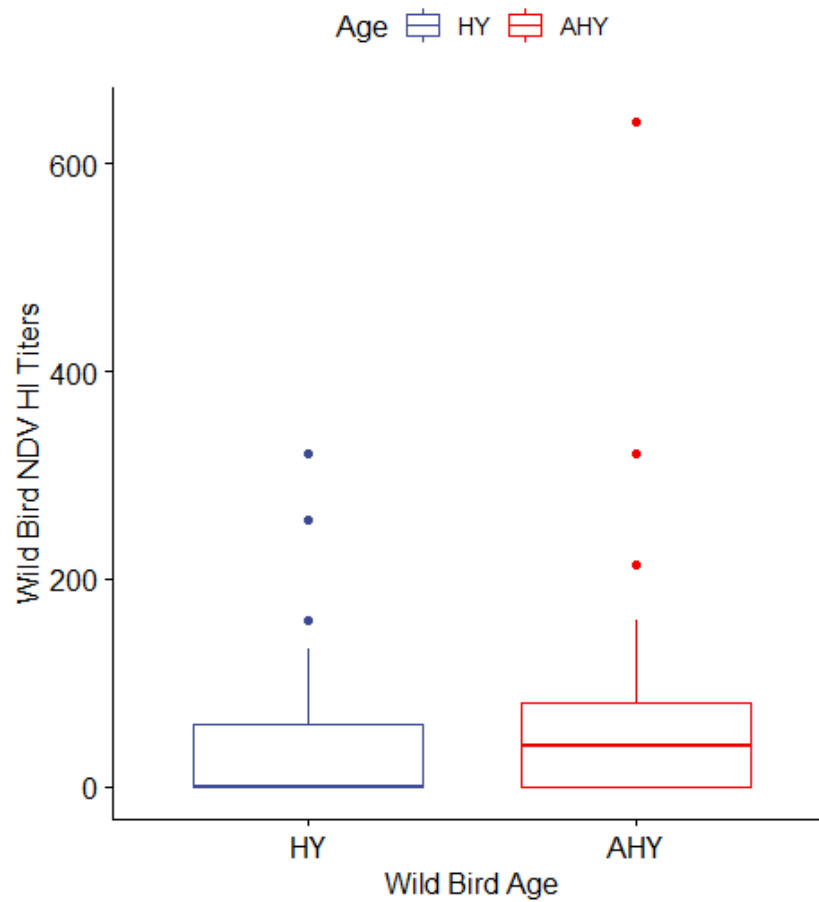


Figure 4.6. Boxplot of mean wild bird HI titers stratified by wild bird age. Across Sites S and L, we found that after-hatch year (AHY) birds had statistically significantly higher NDV HI titers than hatch-year (HY) birds ($\chi^2(1) = 6.5381, p = 0.011$).

Table 4.3. Summary of individual Spearman's Rank Correlation tests that examined the monotonic relationship between each species weight to wing chord ratio and NDV HI titer. Due to insufficient sample size, Hermit Thrushes, Slate-colored Juncos, Northern Mockingbirds, and White-throated Sparrows are not represented here. Eastern Phoebe was the only species that demonstrated a strongly negative, statistically significant relationship between the weight to wing chord ratio and NDV HI titer

Species	Spearman's Rank Correlation (S)	<i>Rho</i> (r_s)	<i>p</i>-value
American Robin	12.108	-0.2108185	0.7892
Carolina Wren	298.69	0.1794337	0.5575
Chipping Sparrow	47.928	0.14415	0.7578
Downy Woodpecker	136.31	-0.6227657	0.0991
Eastern Bluebird	36.015	-0.02898855	0.9565
Eastern Phoebe	40	-1	0.01667
Myrtle Warbler	4	0	1
Northern Cardinal	831.35	0.2707448	0.2622
Tufted Titmouse	1208.4	0.2153338	0.3485

Table 4.4. Mean body condition scores based on subcutaneous fat and pectoral musculature, compared across the flocks at Site S and Site L, for each date of examination. The significance of this data is that it was used as a secondary metric of post-vaccinal reactions.

Flock	Sampling Date	Mean Fat Score	Mean Pectoral m Score
S (n=9)	Pre-inoculation	3.17 ± 1	3.58 ± 0.48
L (n = 15)	Pre-inoculation	2 ± 1.1	2.67 ± 0.65
S (n=8) ^{fat} , (n=9) ^{keel}	0 DPI	1.43 ± 0.79	2.97 ± 0.08
L (n = 14)	0 DPI	1.28 ± 0.83	2.67 ± 0.43
S (n=9)	4 DPI	1.44 ± 0.88	3.19 ± 0.46
L (n=14)	4 DPI	1.92 ± 1.1	2.73 ± 0.58
S (n=9)	11 DPI	$1.89 \pm$	3.25 ± 0.47
L (n=14)	11 DPI	1.29 ± 0.61	2.71 ± 0.32
S (n=9)	18 DPI	1.78 ± 1.09	3.22 ± 0.44
L (n = 15)	18 DPI	1.6 ± 0.63	3.00 ± 0.00

*For each chicken, we scored subcutaneous furculum fat on an eight-point scale from zero to seven, where zero indicated no stored fat in the furculum and seven indicated an excessive fat pocket. We also measured the keel, scoring the pronouncement of the keel on a 6

point, zero to five scale, with a zero indicating no detectable musculature overlaying the sternum, while a five indicated highly robust flight muscles with no discernable keel bone protrusion.

Table 4.5. Summary of wild bird species captured at backyard chicken farms (Sites C, S, and L) between 2016 and 2017, in Athens, Georgia.

Species Captured	Scientific Name	Number per Site	Capture Period	Total Captured	Frequency of Capture (total n = 351)
American Goldfinch	<i>Spinus tristis</i>	C = 1 S = 0 L = 1	Surv, Post	2	0.57%
American Redstart	<i>Setophaga ruticilla</i>	C = 1 S = 0 L = 0	Surv	1	0.28%
American Robin	<i>Turdus migratorius</i>	C = 0 S = 5 L = 0	Post	5	1.42%
Black-and- White Warbler	<i>Mniotilta varia</i>	C = 12 S = 0 L = 0	Surv	12	3.42%
Blue-gray Gnatcatcher	<i>Polioptila caerulea</i>	C = 2 S = 0	Surv	2	0.57%

		L = 0			
Blue	<i>Passerina</i>	C = 1	Surv	1	0.28%
Grosbeak	<i>caerulea</i>	S = 0			
		L = 0			
Carolina	<i>Poecile</i>	C = 23	Surv, Pre,	36	10.26%
Chickadee	<i>carolinensis</i>	S = 8	Post		
		L = 5			
Carolina	<i>Thryothorus</i>	C = 21	Surv,Pre,	44	12.54%
Wren	<i>ludovicianus</i>	S = 17	Post		
		L = 6			
Chipping	<i>Spizella</i>	C = 0	Post	14	3.99%
Sparrow	<i>passerina</i>	S = 14			
		L = 0			
Downy	<i>Picoides</i>	C = 2	Surv, Pre,	12	3.42%
Woodpecker	<i>pubescens</i>	S = 1	Post		
		L = 9			
Eastern	<i>Sialia sialis</i>	C = 0	Post	8	2.28%
Bluebird		S = 0			
		L = 8			

Eastern	<i>Sayornis phoebe</i>	C = 0	Pre, Post	13	3.42%
Phoebe		S = 5			
		L = 8			
Eastern	<i>Pipilo</i>	C = 3	Surv	3	0.85%
Towhee	<i>erythrophthalmus</i>	S = 0			
		L = 0			
Eastern	<i>Contopus virens</i>	C = 0	Pre	1	0.28%
Wood-Pewee		S = 1			
		L = 0			
Golden-	<i>Regulus satrapa</i>	C = 0	Post	2	0.57%
crowned		S = 1			
Kinglet		L = 1			
Gray-	<i>Catharus</i>	C = 0	Pre	1	0.28%
cheeked	<i>minimus</i>	S = 1			
Thrush		L = 0			
Gray Catbird	<i>Dumetella</i>	C = 2	Surv, Pre	4	1.14%
	<i>carolinensis</i>	S = 2			
		L = 0			
Hermit	<i>Catharus</i>	C = 0	Post	4	1.14%

Thrush	<i>guttatus</i>	S = 0			
		L = 4			
House Finch	<i>Haemorhous</i>	C = 6	Surv	6	1.71%
	<i>mexicanus</i>	S = 0			
		L = 0			
Hooded	<i>Setophaga</i>	C = 7	Surv, Pre	8	2.28%
Warbler	<i>citrina</i>	S = 1			
		L = 0			
House Wren	<i>Troglodytes</i>	C = 0	Pre	1	0.28%
	<i>aedon</i>	S = 1			
		L = 0			
Indigo	<i>Passerina cyanea</i>	C = 1	Surv	1	0.28%
Bunting		S = 0			
		L = 0			
Magnolia	<i>Setophaga</i>	C = 1	Surv, Pre	2	0.57%
Warbler	<i>magnolia</i>	S = 1			
		L = 0			
Mourning	<i>Zenaida</i>	C = 3	Surv, Pre,	5	1.42%

Dove	<i>macroura</i>	S = 2	Post		
		L = 0			
Myrtle	<i>Setophaga</i>	C = 0	Post	9	2.56%
Warbler	<i>coronata</i>	S = 0			
		L = 9			
Northern	<i>Cardinalis</i>	C = 17	Surv, Pre,	33	9.40%
Cardinal	<i>cardinalis</i>	S = 14	Post		
		L = 2			
Northern	<i>Mimus</i>	C = 0	Post	1	0.28%
Mockingbird	<i>polyglottos</i>	S = 1			
		L = 0			
Northern	<i>Setophaga</i>	C = 2	Surv	2	0.57%
Parula	<i>Americana</i>	S = 0			
		L = 0			
Ovenbird	<i>Seiurus</i>	C = 1	Surv, Pre	2	0.57%
	<i>aurocapilla</i>	S = 0			
		L = 1			
Pine Warbler	<i>Setophaga pinus</i>	C = 0	Pre, Post	13	3.70%
		S = 0			

		L = 13			
Ruby-crowned Kinglet	<i>Regulus calendula</i>	C = 0	Pre, Post	22	6.27%
		S = 15			
		L = 7			
Slate-colored Junco	<i>Junco hyemalis</i>	C = 0	Post	2	0.57%
		S = 2			
		L = 0			
Summer Tanager	<i>Piranga rubra</i>	C = 0	Pre	1	0.28%
		S = 1			
		L = 0			
Swainson's Thrush	<i>Catharus ustulatus</i>	C = 0	Pre	1	0.28%
		S = 0			
		L = 1			
Tufted Titmouse	<i>Baeolophus bicolor</i>	C = 21	Surv, Pre, Post	50	14.25%
		S = 19			
		L = 10			
White-breasted Nuthatch	<i>Sitta carolinensis</i>	C = 0	Post	4	1.14%
		S = 1			
		L = 3			

White-eyed	<i>Vireo griseus</i>	C = 1	Surv	1	0.28%
Vireo		S = 0			
		L = 0			
Worm-eating	<i>Helmitheros</i>	C = 3	Surv	3	0.85%
Warbler	<i>vermivorum</i>	S = 0			
		L = 0			
Winter Wren	<i>Troglodytes</i>	C = 0	Post	1	0.28%
	<i>hiemalis</i>	S = 0			
		L = 1			
Wood	<i>Hylocichla</i>	C = 0	Pre	3	0.85%
Thrush	<i>mustelina</i>	S = 3			
		L = 0			
White-	<i>Zonotrichia</i>	C = 0	Post	2	0.57%
throated	<i>albicollis</i>	S = 2			
Sparrow		L = 0			
Yellow-	<i>Empidonax</i>	C = 1	Surv	1	0.28%
bellied	<i>flaviventris</i>	S = 0			
Flycatcher		L = 0			
Yellow-	<i>Sphyrapicus</i>	C = 0	Pre	2	0.57%

bellied	<i>varius</i>	S = 2			
Sapsucker		L = 0			
Yellow-	<i>Colaptes auratus</i>	C = 0	Post	1	0.28%
shafted		S = 1			
Flicker		L = 0			
Blue Jay	<i>Cyanocitta</i>	C = 0	Pre	1	0.28%
	<i>cristata</i>	S = 1			
		L = 0			
Brown	<i>Toxostoma rufum</i>	C = 0	Pre	1	0.28%
Thrasher		S = 1			
		L = 0			

Table 4.6. Represented are the number of wild birds captured during each period of the inoculation cycle at each site.

Site	Period	Number Captured	Number Recaptured	Percent Recapture Rate
C	Surveillance	104	24	23.1
S	Pre-inoculation	40	5	12.5
S	Post-inoculation	58	29	33.3
L	Pre-inoculation	32	3	9.4
L	Post-inoculation	45	11	19.6

Table 4.7. Mean species NDV HI titers are represented according to site. Surv/Pre mean titers represent the titers performed on serum prior to inoculation. Post mean titers are represented by serum collected from wild birds during the post-inoculation period. Surv/Pre and Post ≥ 16 represent the number of individuals of that species that tested above the threshold of 16. Surv/Pre (n) and post (n) represent the total number of individuals tested during those periods.

Site	Species	Surv/Pre Mean Titers	Post Mean Titers	Surv/Pre ≥ 16	Post $\geq$ 16	Surv/Pre (n)	Post (n)
C	Black-and- White Warbler ^B	0.0000004	N/A	0	N/A	3	0
	Carolina Wren	0.0000004	N/A	0	N/A	3	0
	Eastern Towhee	0.0000004	N/A	0	N/A	1	0
	Gray Catbird	0.0000004	N/A	0	N/A	1	0
	House Finch	0.0000004	N/A	0	N/A	3	0

	Hooded	0.0000004	N/A	0	N/A	2	0
	Warbler						
	Northern	0.0000004	N/A	0	N/A	9	0
	Cardinal						
	Tufted	0.0000004	N/A	0	N/A	4	0
	Titmouse						
	White-eyed	0.0000004	N/A	0	N/A	1	0
	Vireo						
S	American	N/A	0.0005	N/A	20%	0	5
	Robin						
	Carolina	0.002	1.927	25%	67%	4	6
	Wren						
	Chipping	N/A	5.111	N/A	71%	0	7
	Sparrow						
	Downy	N/A	0.0000004	N/A	0%	0	1
	Woodpecker						
	Eastern	0.007	64	50%	100%	2	1
	Phoebe						
	Eastern-	112	N/A	100%	N/A	1	0

Wood						
Pewee						
Gray-cheeked	0.0000004	N/A	0%	N/A	1	0
Thrush						
Mourning Dove	N/A	0.0000004	N/A	0%	0	2
Northern Cardinal	0.01	8.15	50%	86%	2	7
Northern Mockingbird	N/A	32	N/A	100%	0	1
Slate-colored Junco	N/A	80	100%	N/A	0	1
Summer Tanager	0.0000004	N/A	0%	N/A	1	0
Tufted Titmouse	0.000186	10.293	33%	66%	3	9
White-	N/A	40	N/A	100%	0	1

	throated						
	Sparrow						
	Wood	56.57	N/A	100%	N/A	2	0
	Thrush						
	Yellow-	80	N/A	50%	N/A	2	0
	bellied						
	Sapsucker						
L	Carolina	8	40	0	100%	1	1
	Wren						
	Downy	57.982	51.720	100%	100%	5	2
	Woodpecker						
	Eastern	N/A	1.372	N/A	50%	0	6
	Bluebird						
	Eastern	N/A	32.56	N/A	100%	0	2
	Phoebe						
	Hermit	N/A	0.0057	N/A	50%	0	2
	Thrush						
	Myrtle	N/A	0.0002	N/A	33%	0	3
	Warbler						

Northern	0.002	N/A	0%	N/A	2	0
Cardinal						
Swainson's	640	N/A	100%	N/A	1	0
Thrush						
Tufted	10.293	92.26	66%	100%	3	2
Titmouse						

CHAPTER 5

RISKY BUSINESS IN GEORGIA'S WILD BIRDS: CONTACT RATES WITH BACKYARD CHICKENS MAY INFLUENCE PATHOGEN TRANSMISSION IN PASSERINES ¹

¹ Ayala, A.J., L.K. Haas, B.M. Williams, S.S. Fink, and S.M. Hernandez. To be submitted to *Epidemiology and Infection*.

Abstract

Backyard chickens are increasingly popular pets, and their husbandry varies widely (e.g. the frequency with which backyard chickens can roam, and their housing), and thus, the accessibility of their food and water sources to wild birds also vary, nor are they regulated. These conditions, e.g. free-roaming chickens, and freely accessible resources, increase contact rates between backyard chickens and wild birds, which have implications for pathogen transmission. For instance, avian-orthohavulavirus-1 (Newcastle disease virus), can be transmitted both to and from backyard chickens, and is transmissible from contaminated water or feed. Due to the likelihood of increased contact rates between backyard chickens and wild birds, we identified which wild bird species were most likely to encounter backyard chickens and their resources. To do so, we performed an experimental supplemental feeding experiment followed by systematic observations at three sites associated with backyard chickens in Athens, Georgia. One site allowed their chickens to free-roam during the day, the second site cooped their chickens all day, and the third site served as the control, with the chickens having been recently removed. At each site, we identified the species of wild birds that: a) shared habitat with the chickens, b) the frequency of detection, and c) encountered the chicken coops. For the species that encountered chicken coops, we also measured the contact rate with which they did so. We identified 14 wild bird species of the 72 species recorded that entered the coops to consume supplemental feed and were considered ‘high’ risk for pathogen transmission. Across all sites combined, these 14 high-

risk species had 589 out of 1,574 total contacts with the backyard chicken coops. Chipping Sparrows (*Spizella passerina*), Northern Cardinals (*Cardinalis cardinalis*), and Tufted Titmice (*Baeolophus bicolor*) had the highest contact rates across all three sites. We found that site species richness was more predictive of wild bird contacts with backyard chickens than whether chickens free-roamed or were cooped. Our results provide evidence that contact between wild birds and backyard chickens is a common occurrence, which has crucial implications for pathogens shared among these species.

Introduction

The ownership of backyard chickens is garnering increased interest in the United States [1-3]. Many flock owners believe that backyard chicken rearing results in a ‘healthier’ and more ecologically responsible food source [4, 5]. However, the sale, trade, and husbandry of backyard chickens are poorly regulated, and as they are less likely to receive standard veterinary care, the data to support this perception are lacking [4-6]. Moreover, because backyard chickens often free-range, their contact with native wildlife is also increasing, with implications for pathogen transmission for both groups. An effective contact is defined as an interaction between two hosts that leads to pathogen transmission from one to another [7]. Thus, increasing contact rates between backyard chickens and native wildlife may lead to an increase in successful pathogen transmission events.

Poultry husbandry practices range along a continuum, from densely populated, bio-contained commercial flocks to free-roaming chickens in urban centers [8-10]. Commercial flocks are managed under stringent protocols that eliminate interactions with wild birds, reducing potential pathogen transfer from wildlife [11]. Backyard poultry are maintained under a variety of husbandry practices that influence their contact rates with wild birds [12], often lacking infrastructure to prevent contact with other animals. As a result, contact rates between backyard chickens and wild birds may increase as a result of shared resources [1, 2, 13]. For example, bird feeders, waterers, treat piles, chicken-feed troughs, and fecal matter both attract and are used by multiple individuals from different species [5, 14, 15]. As backyard chickens increase in density and frequency, each additional effective contact with wildlife increases the probability that a successful pathogen transmission event will occur among infectious and susceptible hosts.

Given the inconsistency of husbandry protocols of this emerging ‘backyard chicken-wildlife interface,’ it is difficult to predict which pathogens present the greatest epidemic risks to both susceptible wild birds and backyard chickens in the United States. Previous studies of the backyard-chicken interface suggest pathogen transmission between wild birds and backyard chickens occurs more frequently for viruses such as highly pathogenic avian influenza (HPAI) and avian-orthonavulavirus-1 [Newcastle disease virus (NDV)] [10, 16-19]. In countries where backyard chickens are the primary source of protein, avian-orthonavulavirus-1, is considered an endemic pathogen and can cause up to 100% mortality in affected poultry flocks [20, 21]. In the

United States, backyard chickens have been cited as the source for two prior, and one ongoing, epizootic of virulent NDV in the southwest [22-24].

The epidemiology of Newcastle disease virus has been reviewed in depth [25-30], yet it remains a transboundary pathogen of emerging significance [31-33]. The field of NDV research has recently garnered a broader, interdisciplinary audience in response to increasing viral spillover and spillback events between chickens and wild birds from NDV of velogenic (virulent) and lentogenic (low-virulent) origin [34-42]. While NDV remains a ‘poultry’ pathogen, recent studies suggest that most, if not all, wild bird species are susceptible to the virus [43-45]. More recently, Rock Pigeons, Double-crested Cormorants, and dabblers of the Anatidae have been proposed as wild bird reservoir hosts of this pathogen [33, 44, 46]. Isolate type and affected host species both influence NDV transmission efficiency; however, direct transmission has many routes, including fecal-oral ingestion and the inhalation of infectious respiratory secretions [47-51].

Virulent NDV (vNDV) is one of only two avian viruses that are listed as notifiable to the World Organization for Animal Health (OIE), the other being highly pathogenic avian influenza (HPAI) [52]. It causes significant, acute disease among affected chickens and many species of wild birds [53, 54]. However, the ecology of this virus has gone largely unexplored, specifically concerning which species of wild birds, due to both their susceptibility and probability of contact with backyard chickens may become naturally infected with NDV [55-57]. Outside of estimates

derived from mathematical models [58, 59], contact rates between backyard chickens and wild birds have not been measured in the field. In a previous study, we addressed the question of susceptibility in four native passerines [45] under experimental conditions in BSL-2 isolators, yet, where natural contact rates between chickens and wild birds could not be accurately measured. To address this knowledge gap about contact rates between backyard chickens and free-living birds, we performed systematic observations at three sites associated with backyard chickens in Athens, Georgia, where we identified and quantified the species of wild birds, both at the farm level, as well as those that entered areas that housed chickens. We also measured the frequency of contact for wild birds that came in contact with chickens. We identified the species of wild birds that a) shared habitat with the chickens, (e.g. species richness) b) the frequency of detection at which individual species were detected, and c) encountered the chicken coops. Lastly, for the species that encountered chicken coops, we also measured the frequency or the ‘contact rate’ with which they did so.

Methods

Study Sites

We conducted observations at three backyard chicken farms in Athens, Georgia (33.9519° N, 83.3576° W), which were located at the juncture of three counties, Athens-Clarke, Jackson, and Oconee counties (Figure 5.1). The three farms were similar habitat (mixed hardwood forest surrounding open grassland) and size, (average 0.004 km²); however, their

backyard chicken husbandry methods varied (Table 5.1). Site L allowed all chickens to free-range at all times of the day (n = 15 chickens), Site S primarily kept the chickens penned 24 hours a day (n = 9 chickens), Site C, which we considered our control, had an open coop, where chickens had been recently removed (395 days before beginning experiment; n = 0 chickens). All coops had wire mesh surrounding the pens, which kept out larger predators but did not exclude most passerines smaller than American Robins (*Turdus migratorius*). To ensure that each site had avian communities that were independent of one another, all sites were located at least 10 km apart. All observations were performed between February and May 2018.

Twice per week, all three sites were provided a commercial chicken feed mixed with supplemental foods considered highly palatable to domestic and wild species, including suet, mealworms, *Nyjer* seeds, cracked corn, sunflower seeds, and fruits and nuts [45, 60]. Based on interviews and a review of the literature, the backyard chicken owners fed a wide variety of non-standard, supplemental *ad-libitum* foods to their chickens, including table scraps, fruit rinds, assorted wild bird seeds and nuts, suet, cracked corn, and vegetables [6, 15, 60-64]. Therefore, during this experiment, we asked the owners to allow us to control and manage all feeding to reduce wild bird preference bias. This was essential to the standardization and quantification of wild bird visits to the coops. Lastly, we also placed the supplemental feed in each coop's pen, as opposed to the interior of the coop, so that during observations, we could identify the species and the frequency with which they were foraging. Feeding troughs were placed no higher than 0.3

meters above the ground, sanitized weekly with a 10% bleach solution to mitigate pathogen transmission, and placed within one meter of each pen opening, allowing wild birds to enter and leave freely (Figure 5.2).

Observations of Wild Birds at Backyard Chicken Sites

To determine the frequency of detection for wild birds and contact rates between wild birds and backyard chickens, a total of 60, one-hour, unlimited-radius observations were conducted at our three sites (Table 5.1). Our observations were modeled upon standard avian point count protocols in order to capture site species richness and their frequency of detection. [65]. However, the point count duration and the observer focus were both slightly modified for logistical purposes. Specifically, we utilized hour-long point counts to ensure that cryptic species would be recorded, which is considerably longer than the standard 3-10 minute observations [65]. We also moved the center of the point count from where the observer was standing, to the center of chicken coop and pen, to ensure that wild birds would not avoid the coop due to observer interference (Figure 5.2). Each study site was only visited once per day, and all three locations were visited twice per week.

Observations were performed from dawn to dusk at opportunistically appointed times to ensure that wild bird foraging behaviors were proportionately represented. Upon arrival to each site, the observer waited for ten minutes prior to beginning the census to ensure all birds returned to their natural routines. Birds were then detected both by song and by sight and identified to

species when the observer was confident of identification; otherwise, unknown species were classified according to family. Observers had been previously trained and assessed to identify common Georgia species [66].

Each bird noted was placed into one of five distance classes based on distance observed from the coop: 1) within the coop (distance class one, D1), 2) from the edge of the coop to < 10 m (distance class two, D2), 3) between 10 and < 25 m from the coop (distance class three, D3), 4) between 25 and 50 m from the coop (distance class four, D4), and 5) > 50 meters from the coop (including flyovers) (distance class five, D5) (Figure 5.1). Although uncommon, when individuals were observed in multiple distance classes during the same census, they were assigned to the distance class in which they spent the greatest amount of time in the one-hour observation period. These observations identified 14 species of interest, or high-risk species, for which we calculated the ratio at which each species was detected relative to other species, otherwise known as the frequency of detection (Table 5.2), and the contact rates in which they encountered the chicken coops (Table 5.3). We defined a high-risk species as any bird detected in D1 during the field season.

Calculating frequency of detection and backyard chicken-wild bird contact rates

All detected wild birds were tallied first according to species, then placed into one of the five distance strata (D1-D5) during each census. Unless otherwise specified, frequency of detection was calculated as the number of observations per site, per species of interest, divided

by the total number of wild bird observations across all sites over the course of the field season ($n=1,574$), multiplied by 100. This was critical to ensure all species were tallied, as some species were not detected at all three sites. Moreover, comparing the calculated frequency of detection from the total number of observations allowed us to compare the frequency of detection across each of the three sites. Although we calculated the frequency of detection on all species that were detected throughout the field season, we calculated contact rates only on species which we observed in D1. Those contact rates were then calculated according to their visits to D1 and D2 throughout the field season.

Transmission rates are a function of a per-capita unit of time, thus we used a modified equation from Courtenay and Quinnell [67] to determine our contact rates relative to the encounters that wild birds had with backyard chicken coops. As we were primarily interested in interspecific interactions that would influence viral transmission, we analyzed contact rates only for species that had encounters with backyard chickens and their coops, specifically for species detected within both D1 and D2.

For each observation, the species contact rate c_i (Equation 1) was calculated as the sum of each species detected in distance class one D_1 plus distance class two D_2 for each observation j , divided by the number of minutes per observation N_j . The mean contact rate $\bar{c}_{if}$ (Equation 2) of an individual species i at each farm f was calculated as the mean of the species contact

visit rate c_i (Equation 1). The $\bar{c}_{if}$ for each high-risk species at each farm are summarized in Table 5.3.

$$c_i = \left(\frac{D_{j1} + D_{j2}}{N_j} \right)$$

(Equation 1)

$$\bar{c}_{if} = \frac{1}{N_{fj}} \sum_{i=1}^N c_i$$

(Equation 2)

Newcastle disease virus has been shown to be transmitted both directly and through environmental sources [68-71]; thus, when defining a ‘contact,’ each distance class was assigned a differential NDV exposure risk. For example, species detected in both D1 and D2 throughout the field season were considered directly ‘exposed’ to the virus. Individuals detected in D3, D4, and D5 (Figure 5.2) were all assigned the same exposure risk of zero and, thus, were not used in calculations in Equation 1.

We calculated the average number of minutes c_{mi} between each contact for each species at each site by dividing one by the value for $\bar{c}_{if}$ (Equation 3). Specifically, in Equation 1, the rate at which wild bird species visited the chicken coops was calculated as the mean of

$\left(\frac{D_{j1} + D_{j2}}{N_j}\right)$ over the course of the field season. By calculating the inverse of the rate $\bar{c}_{if}$, which represented the mean number of contacts that occurred for each species i at each farm f , we obtained the average time in minutes between each species' encounter with the chicken coops [7] (Equation 3)

$$c_{mi} = \left(\frac{1}{\bar{c}_{if}}\right)$$

(Equation 3)

We divided the average activity period for the diurnal passerine species included in this study (720 min) by the mean amount of time between visits for each species i at each farm f , which was represented by the variable c_{mi} from Equation 3. Thus, we obtained the average number of chicken coop visits made by each high-risk species per day, at each site, a value represented by the variable c_{di} in Equation 4.

$$c_{di} = \left(\frac{720}{c_{mi}}\right)$$

(Equation 4)

Results

a) *Frequency of Detection*

Over 60 hours of observation, 1,574 individual wild birds were detected across the three sites in Athens, Georgia, comprising 72 species from 24 families (Table 5.2). The mean detection rate was 26.2 ± 13.5 wild birds per observation period, a mean species richness of 12.7, and a mean of 9.3 families. Across all sites, Tufted Titmice (*Baeolophus bicolor*) (15.6%), Northern Cardinals (*Cardinalis cardinalis*) (13.2%), Carolina Chickadees (*Poecile carolinensis*) (7.0%), and Carolina Wrens (6.0%) (*Thryothorus ludovicianus*) were the four most commonly detected species, and the taxa Paridae, Cardinalidae, and Troglodytidae were the most frequently detected families.

Site S, the cooped flock, had the greatest number of overall detections (44.9%), as well as the greatest diversity of species richness—77.8% of the total observed species detected at Site S. Site L, the free-ranging flock, had the fewest number of overall detections, (22.4%), as well as the lowest rank in terms of species richness, with 63.9% of the total species detected. Detections at Site C, the control site, comprised 32.7% of the 1,574 observations and was the median in terms of species richness, with 70.8% of the total species observed at that site.

Site C had a higher frequency of detection for Carolina Chickadees (Paridae) than at Site S, where Chipping Sparrows (Passerellidae) had the highest frequency of detection. Myrtle Warblers (Parulidae) had the highest frequency of detection at Site L. Several species, detected at Site C and S were not detected at all at Site L, including Eastern Towhees, White-throated Sparrows, and Song Sparrows.

The species with the highest frequencies of detection were also more likely to be detected in D1 and D2 than in D3-D5; however, there were some notable exceptions. Northern Cardinals were frequently detected throughout all distance classes (D1-D5), whereas Myrtle Warblers were most commonly detected in D2 and D3. Most high-risk species were more frequently detected in the spring, except for Chipping Sparrows, Tufted Titmice, Carolina Wrens, and White-breasted Nuthatches.

b) Contact Rates

Of the 72 total species recorded in each backyard chicken habitat, 14 species were detected in D1. Across all sites combined, these 14 high-risk species had 589 total contacts with the backyard chicken coops when accounting for visits to D1 and D2. This comprised a contact-based frequency of detection of 37.4% from our 1,574 detections across 60 observation hours. These species were also the most frequently detected overall, comprising 67.3% of all detections, across all distance classes.

We used contact rates to identify the species with the highest visit rates, and to extrapolate the mean number of minutes between each species' visit. With those values, we calculated a mean daily species visit rate for an average 720-minute period of activity. The values for Equations 2-4 are represented in table 5.3 in a species by site table. Across all sites, Tufted Titmice and Northern Cardinals had the highest backyard chicken contact rates. Species

with the highest backyard chicken contact rates at Site S, C, and L also included Chipping Sparrows, White-throated Sparrows, and Myrtle Warblers, respectively .

c) Mean Daily Visits

Chipping Sparrows had the highest number of visits with 85.3 mean daily visits at Site S. At Site L, Myrtle Warblers performed 15 mean daily visits, while Tufted Titmice performed 41.4 mean daily visits at Site C.

Discussion

Although prior studies have quantified contact rates for wildlife to determine pathogen exposure [67, 72-77], we are unaware of similar studies quantifying contact rates between backyard chickens and passerines. In this experiment, we have demonstrated that a minimum of 14 wild bird species in Athens, Georgia, have frequent encounters with chicken coops that may lead to pathogen exposure between backyard chickens.

Chickens spend significant periods of time in a limited area and, thus, it is likely that if pathogens are shed, they are concentrated into their coops and pens [76, 78]. Our results provide evidence that contact between wild birds and chickens is common and frequent, information that has important implications for pathogens shared among these species, such as NDV, HPAI, *Salmonella* spp., and *Mycoplasma* spp. These pathogens have broad avian host ranges, may induce high mortality in affected hosts, and may be transmitted via the fecal-oral route [33, 79-81]. Our results may be extrapolated to similar species with analogous supplementary feeding

habits to the high-risk families we identified in our study, including species in the Paridae, Passerellidae, Cardinalidae, and Parulidae families. For instance, high contact rates between chickens in multiple backyard flocks and wild bird species with natural histories similar to Tufted Titmice, Chipping Sparrows, and Northern Cardinals may be one explanation for the accelerated spread of virulent NDV throughout backyard chickens across the southwestern United States during each vNDV epidemic [24, 38, 82-85].

In general, our high-risk exposure species were among the most commonly detected species across all observations, and it is likely that the frequency of detection for our high-risk birds were influenced by the consistent availability of supplemental feed. It has been demonstrated that birds who utilize provisional resources may undergo altered population dynamics, such as increased productivity and survival rates, and earlier breeding attempts [86-88]. Thus, it is possible that the distributions and abundances of our high-risk species were inflated, in contrast to sites with similar habitats where supplemental feed was not available. Moreover, it has also been demonstrated that species who utilize supplemental feed may influence avian community composition in a manner that negatively impacts the abundances of non-supplemental feeding species [89, 90].

The importance of these points cannot be understated as our frequencies of detection appeared highly correlated to our contact rates for our high-risk species. On the other hand, some low-risk wild birds were detected at frequencies that were comparable to high-risk species. For

example, Red-bellied Woodpeckers had a 4.8% frequency of detection, and Pine Warblers had a 2.2% frequency of detection over the course of the field season. What remains uncertain is whether the frequencies of detection for low-risk species may have also been influenced by the presence of supplemental feed. Given that we did not have a coop without supplemental feed as an experimental unit, it is difficult to make an inference concerning low risk but frequently detected species.

Our results, in conjunction with the literature, suggest that the risk of spillover to these wild bird species is considerable, especially considering the high visitation rates at D1 and D2 for species such as Chipping Sparrows (Table 5.3). However, the timing of surveillance may play a role when sampling wild birds for exposure to viruses in which chickens serve as a reservoir, such as NDV [91]. Our prior experimental work in four wild passerines suggested that NDV shedding in these species may be sporadic and is consistently detectable only up to thirteen days [45]. Thus, viral shedding in free-living songbirds may be under-detected, given the relatively short duration of viral replication. Surveying passerines using serology also has its limitations, as songbirds have relatively short-life spans and frequently move in and out of their territories.[45, 92-95]. Moreover, the decay patterns of antibodies in passerines are also unknown and should be given a great deal more study, given the number of seroprevalence studies performed on pathogen exposure in wild birds.

Prior surveillance and experimental studies of NDV have sampled the same or related birds as the high-risk species identified in our study. For instance, during the outbreak of vNDV in California in 1971-1974, 0.93% of tested Mourning Doves demonstrated NDV infection by virus isolation and a 0.42% NDV prevalence in the American Crow (*Corvus brachyrhynchos*) [96]. Several other species related to the ones in our study were tested for NDV exposure by hemagglutination inhibition assay, but all were negative.

Lastly, although we sought a clear answer regarding the role of husbandry in chicken-wild bird contact rates, our results suggest that the role of husbandry as a factor in the number of daily bird visits was secondary to the role of site species richness and the availability of supplemental feed. Our results also suggest that how frequently wild bird species were detected at a site was more important than chicken husbandry when determining wild bird contact rates with chicken coops. Given that Site S accounted for almost half of the contact rates from our high-risk species, followed by Site C, it appeared that the presence of chickens in the pen was negligible. However, the flock at Site L was somewhat more aggressive than the flock at Site S, which may account for the variability in contact rates and species visits not attributed to richness and the frequency of detection. On the other hand, given the lack of replicates for each treatment type (e.g. cooped, free-ranging, and control), it is difficult to make broader inferences.

In terms of management, targeted surveillance of the high-risk species we identified in this study during an NDV outbreak may elucidate an additional mode of microbial transmission

between backyard chicken flocks that have no other means of contact. We hope that our work will also provide additional information on how to manage the current, ongoing epidemic of NDV in southern California. Our results demonstrate that the removal of chickens does not deter wild birds from the pens where pathogens may have been present – it is the removal of freely accessible feed that would likely reduce visits, in conjunction with biosecurity aimed at preventing the wild bird visits both inside chicken houses and the perimeter. Thus, mandating the usage of automatic feeders that require chickens to step on platforms to open the lids, i.e. treadle feeders, would minimize waste and reduce the spillage that attracts wild birds and pests. In addition, the use of plastic screening over pens, and educating owners on the dangers of feed and water that are also accessible to wild birds, may help interrupt the chain of transmission between wild birds and backyard flocks.

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Figure 5.1. One-hour long point counts were performed from February through May 2017 at each of the three sites highlighted in the upper right Georgia map. At each site, we identified the species of wild birds that a) shared habitat with the chickens b) the frequency of detection and c) encountered the chicken coops. Farm L allowed their chickens to free-roam during the day, Farm S cooped their chickens all day, and Farm C served as the control, with the chickens

having been recently removed. The inset image (lower left) demonstrates the distribution of each farm across the three counties of North Georgia in which they were located: The teal green lines indicate county lines, and Clarke, Jackson, and Oconee counties are labeled on the map. Farm S was located at the juncture of Jackson and Athens-Clarke counties, Farm L was located at the juncture of Oconee and Athens-Clarke Counties, and Farm C was located in Athens-Clarke County

Table 5.1. Site husbandry, flock size, and a summary of the number of birds observed are below.

Site Name	Flock Size	Husbandry Type	Number of Observations Periods	Total Species Detected	Number of Individual Birds Observed	Total Families Detected	Number of Species Exposed (D1 and D2)	Percent of Birds Exposed (D1 and D2)
Site S	n = 9	Cooped	20	56	706	22	12	21.4%
Site L	n = 15	Free-range	20	46	353	20	8	17.4%
Site C	n = 0	Absent	20	52	535	23	10	19.2%

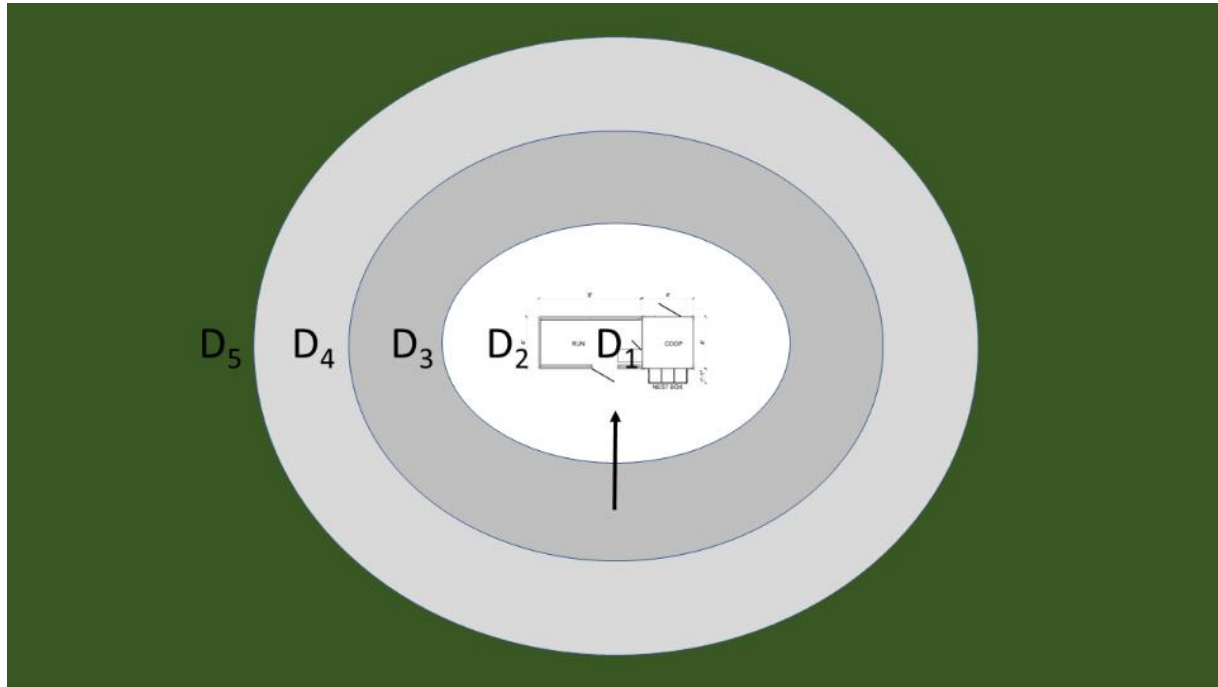


Figure 5.2. Schematic of how observations were performed at each farm. Each D_x corresponds to a distance class. D_1 demarks the inside of the coop and pen, D_2 correlates to the perimeter of the coop, D_3 is 10 to 25 meters from the coop perimeter, D_4 is 25 to 50 meter, and D_5 is > 50 meters from the coop. The black arrow denotes the observer focus.

Table 5.2. Species Richness and detection ratios of all birds detected at Site L, C, and S. The species in this table were observed in the same habitats with backyard chickens across three sites in Athens, Georgia. A total of 72 species are represented in this table. Data in the column titled, “Total Detections Across all Sites’ are represented as pooled observations across all sites in order to provide comparisons for the columns with site specific detections. Rows in which the text are bolded represent our 14 high risk species.

Species	Scientific	Total	Site S	Site L	Site C	Exposed	Detectio
	Name	Bird	Detectio	Detectio	Detectio	Species	n Ratio
		Detected	ns	ns	ns		
		Across					
		all Sites					
Acadian	Empidonax	2	0	2	0	No	0.13%
Flycatcher	x						
	virescens						
American	Corvus	34	16	5	13	No	2.16%
Crow	brachyrhynchos						
American	Spinus	11	2	3	6	No	0.70%
Goldfinch	tristis						

American Robin	Turdus migratorius	6	3	2	1	No	0.38%
Barn Swallow	Hirundo rustica	1	0	0	0	No	0.06%
Black-and-white Warbler	Mniotilta varia	10	4	4	2	No	0.64%
Blue-gray Gnatcatcher	Polioptila caerulea	15	1	1	13	No	0.95%
Brown-headed Cowbird	Molothrus ater	4	2	0	2	No	0.25%
Brown-headed Nuthatch	Sitta pusilla	12	3	7	2	No	0.76%
Blue Jay	Cyanocitta cristata	62	36	20	6	Yes	3.93%

Blackpoll	Setophaga	1	0	1	0	No	0.06%
Warbler	striata						
Black	Coragyps	5	4	0	1	No	0.32%
Vulture	atratus						
Brown	Toxostom	12	12	0	0	No	0.76%
Thrasher	a rufum						
Black-throated	Setophaga	6	3	2	1	No	0.38%
Blue	caerulesce						
Warbler	ns						
Black-throated	Setophaga	1	1	0	0	No	0.06%
Green	virens						
Warbler							
Broad-winged	Buteo	4	0	0	4	No	0.25%
Hawk	platypterus						
	s						
Carolina	Poecile	110	34	19	57	Yes	6.99%
Chickadee	carolinensis						

e	is						
Canada	Branta	1	0	0	1	No	0.06%
Goose	Canadensi						
	s						
Cape May	Setophaga	5	3	1	1	No	0.32%
Warbler	tigrine						
Carolina	Thryotho	94	30	25	39	Yes	5.97%
Wren	rus						
	ludovicia						
	nus						
Cedar	Bombycill	10	0	10	0	No	0.64%
Waxwing	a						
	cedrorum						
Chipping	Spizella	156	152	2	2	Yes	9.91%
Sparrow	passerina						
Chimney	Chaetura	7	4	0	3	No	0.45%
Swift	pelagica						
Common	Quiscalus	7	7	0	0	No	0.45%
Grackle	quiscula						

Common	Geothlypis	2	1	0	1	No	0.13%
Yellowthroat	trichas						
Dark-eyed Junco	Junco hyemalis	3	3	0	0	No	0.19%
Downy Woodpecker	Picoides pubescens	25	5	11	9	No	1.59%
Eastern Bluebird	Sialia sialis	10	6	3	1	No	0.64%
Eastern Phoebe	Sayornis phoebe	43	9	16	18	Yes	2.73%
Eastern Towhee	Pipilo erythrophthalmus	19	1	0	15	Yes	1.21%
Eastern Wood-Pewee	Contopus virens	3	2	0	1	No	0.19%
Fish Crow	Corvus	16	11	0	5	No	1.02%

	ossifragus						
Great-crested Flycatcher	Myiarchus crinitus	10	6	2	2	No	0.64%
Golden-crowned Kinglet	Regulus satrapa	6	2	1	3	No	0.38%
Gray Catbird	Dumetella carolinensis	3	1	0	2	No	0.19%
Hairy Woodpecker	Leuconotus villosus	3	1	2	0	No	0.19%
Hermit Thrush	Catharus guttatus	1	0	0	1	No	0.06%
House Finch	Haemorhous mexicanus	17	13	4	0	Yes	1.08%

	s						
Hooded	Setophaga	7	0	1	6	No	0.45%
Warbler	citrina						
Indigo	Passerina	8	8	0	0	No	0.51%
Bunting	cyanea						
Louisiana	Parkesia	1	0	1	0	No	0.06%
Waterthru	motacilla						
sh							
Mississipp	Ictinia	1	1	0	0	No	0.06%
i Kite	mississipp						
	iensis						
Mourning	Zenaida	17	8	8	1	Yes	1.08%
Dove	macroura						
Myrtle	Setophag	60	17	24	19	Yes	3.81%
Warbler	a						
	coronata						
Northern	Cardinali	207	94	47	66	Yes	13.15%
Cardinal	s						
	cardinalis						

Northern	Mimus	9	6	0	3	No	0.57%
Mockingbird	polyglottos						
Northern	Setophaga	8	1	3	4	No	0.51%
Parula	Americana						
Northern	Stelgidopteryx	2	0	1	1	No	0.13%
Rough-winged Swallow	serripennis						
Ovenbird	Seiurus	4	2	1	1	No	0.25%
	aurocapilla						
Pine Warbler	Setophaga pinus	35	17	12	6	No	2.22%
Pileated Woodpecker	Dryocopus pileatus	1	0	1	0	No	0.06%
Prairie Warbler	Setophaga discolor	2	1	1	0	No	0.13%

Prothonota	Protonotar	2	0	0	2	No	0.13%
ry Warbler	ia citrea						
Red-	Melanerpe	76	26	19	31	No	4.83%
bellied	s carolinus						
Woodpeck							
er							
Ruby-	Regulus	30	14	5	11	No	1.91%
crowned	calendula						
Kinglet							
Red-eyed	Vireo	6	3	1	2	No	0.38%
Vireo	olivaceus						
Red-	Buteo	5	0	1	2	No	0.32%
shouldered	lineatus						
Hawk							
Red-tailed	Buteo	1	1	0	0	No	0.06%
Hawk	jamaicensi						
	s						
Ruby-	Archiloch	6	1	3	2	No	0.38%
throated	us colubris						

Humming							
bird							
Scarlet	Piranga	3	1	2	0	No	0.19%
Tanager	olivacea						
Song	Melospiza	6	4	0	2	Yes	0.38%
Sparrow	melodia						
Summer	Piranga	1	1	0	0	No	0.06%
Tanager	rubra						
Tufted	Baeoloph	237	93	51	93	Yes	15.06%
Titmouse	us bicolor						
Turkey	Cathartes	33	10	5	18	No	2.09%
Vulture	aura						
White-	Sitta	12	8	4	0	Yes	0.76%
breasted	carolinens						
Nuthatch	is						
White-	Vireo	4	0	0	4	No	0.25%
eyed Vireo	griseus						
White-	Zonotrich	19	0	0	19	Yes	1.21%
throated	ia						

Sparrow	albicollis						
Wood	Hylocichla	2	0	1	1	No	0.13%
Thrush	mustelina						
Yellow-	Sphyrapic	8	4	1	3	No	0.51%
bellied	us varius						
Sapsucker							
Yellow-	Colaptes	12	1	10	1	No	0.76%
shafted	auratus						
Flicker							
Yellow-	Vireo	1	0	1	0	No	0.06%
throated	flavifrons						
Vireo							
Yellow-	Setophaga	1	1	0	0	No	0.06%
throated	dominica						
Warbler							

Supplementary Table 2. Listed in this table are the contact rates, mean minutes between each contact, and mean number of contacts per day for the 14 high risk species. The equation results are broken down according to Sites S, L, and C.

Species	Site S	Site L	Site C	Site S	Site L	Site C	Site S	Site L	Site C
	Mean	Mean	Mean	Mean	Mean	Mean	Mean	Mean	Mean
	Exposure	Exposure	Exposure	Minutes	Minutes	Minutes	Number of	Number of	Number of
	Contact	Contact	Contact	Between	Between	Between	Contacts Per	Contacts Per	Contacts Per
	Rate	Rate	Rate	Days	Days	Days	Day	Day	Day
	$\bar{c}_{if}$	$\bar{c}_{if}$	$\bar{c}_{if}$	Contacts	Contacts	Contacts	c_{di}	c_{di}	c_{di}
				c_{mi}	c_{mi}	c_{mi}			
Chipping	0.1184	0.0000	0.0000	8.4	0.0	0.0	85.3	0.0	0.0
g									
Sparrow									
w									
Tufted	0.0575	0.0102	0.0575	17.4	98.2	17.4	41.4	7.3	41.4
Titmouse									

e

Northern	0.0500	0.0142	0.0258	20	70.6	38.7	36	10.2	18.6
----------	---------------	--------	--------	----	------	------	----	------	------

n

Cardinal

l

Myrtle	0.0217	0.0208	0.0148	46.2	48.0	67.5	15.6	15.0	10.7
--------	--------	--------	--------	------	------	------	------	-------------	------

Warbler

Carolina	0.0198	0.0069	0.0438	50	144.0	22.9	14.3	5.0	31.5
----------	--------	--------	--------	----	-------	------	------	-----	------

a

Chickadee

ee

Mourning	0.0194	0.0042	0.0000	51.4	240	0.0	14	3.0	0.0
----------	--------	--------	--------	------	-----	-----	----	-----	-----

ng

Dove

House	0.0150	0.0056	0.0000	66.7	180	0.0	10.8	4.0	0.0
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Finch

Carolina	0.0118	0.0000	0.0074	85	0.0	135	8.5	0.0	5.3
----------	--------	--------	--------	----	-----	-----	-----	-----	-----

a Wren

Blue	0.0108	0.0042	0.0042	92.7	240	240	7.8	3.0	3.0
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Jay

White-	0.0095	0.0042	0.000	105	240	0.0	6.9	3.0	0.0
--------	--------	--------	-------	-----	-----	-----	-----	-----	-----

breasted

Nuthatch

h

Song	0.0083	0.0000	0.0083	120	0.0	120	6	0.0	6.0
------	--------	--------	--------	-----	-----	-----	---	-----	-----

Sparrow

w

Eastern	0.0028	0.0050	0.0071	360	200	140	2	3.6	5.1
---------	--------	--------	--------	-----	-----	-----	---	-----	-----

Phoebe

White-	0.0000	0.0000	0.0500	0.0	0.0	20	0.0	0.0	36
--------	--------	--------	--------	-----	-----	----	-----	-----	----

throated

Sparrow

w

Eastern	0.0000	0.0000	0.0050	0.0	0.0	200	0.0	0.0	3.6
---------	--------	--------	--------	-----	-----	-----	-----	-----	-----

Towhee

C

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

This work was inspired by my Master's research, in which we found evidence that Rock Pigeons in Atlanta, Georgia were infected with the Hitchner B1 vaccine in 2012. Those findings fueled this dissertation, upon the realization that I could safely utilize a low-virulent NDV vaccine virus to disentangle how lentogenic viruses are transmitted from competent hosts (backyard chickens) to spillover hosts (wild birds). In the United States, there have been three major and one minor outbreak of virulent Newcastle disease virus. Millions of birds were culled, and billions of dollars were spent in the stamping out and containment of the virus. Although passerines seemed to be a mechanism of spread, evidence to support this hypothesis has been scant.

In this dissertation, I present one synthesis review, and three important findings that may be added to the increasing body of literature concerning the ecology of Newcastle disease virus in wild birds. Specifically, in Chapter 2, we thoroughly reviewed the literature on spillover events between backyard chickens and wild birds. We also expanded our review to include 'poultry' pathogens in addition to Newcastle disease virus, notably infectious bronchitis virus, Marek's disease virus, and *Mycoplasma synoviae*. Within that review, we discussed the importance of transboundary emerging and re-emerging infectious as part of the One Health paradigm. We also surveyed what is currently known concerning pathogen emergence at the global livestock-wildlife interface. Within chapter one, we also discussed unique facets of

poultry agriculture, specifically how it pertains to wild birds, and provided a critical analysis of pathogen transmission events between wild and domestic birds. We also provided a risk assessment of backyard chickens and provided recommendations on how to mitigate spillover events into wild birds.

Chapter 3 presented three experiments performed at the U.S. National Poultry Research Center that were designed to examine how four passerine species respond to different routes of exposure of our lentogenic Newcastle disease virus strain, the LaSota vaccine virus. Within experiment I, the direct inoculation study, we directly inoculated Northern Cardinals, Brown-headed Cowbirds, House Finches, and American Goldfinches with the LaSota vaccine virus. In experiment II, the transmission study from one inoculated chicken to wild birds, we co-housed one group of House Finches and one group of Brown-headed Cowbirds for fourteen days with a single SPF chicken. In experiment III, the transmission study from two inoculated chicken to wild birds, we co-housed two groups of Northern Cardinals, and House Finches, and one group each of American Goldfinches and Brown-headed Cowbirds with two inoculated SPF chickens.

In experiment I, we found that Northern Cardinals, followed by Brown-headed Cowbirds demonstrated the greatest susceptibility to direct inoculation by seroconverting with higher antibody titers and shedding larger amounts of virus as determined by virus isolation. In experiment II, we found that House Finches and Brown-headed Cowbirds were not susceptible to transmission of the vaccine from a single infectious chicken. In experiment III however, one

House Finch and one American Goldfinch demonstrated evidence of transmission as a result of exposure to two infectious chickens for fourteen days. Overall, we provided evidence that at least four passerine species are susceptible to direct inoculation with the LaSota vaccine virus. Future work should further examine whether the transmission of lentogenic Newcastle disease virus to wild birds is dependent upon the number of infectious chickens, given our comparative results between experiment II and experiment III. Future work should also examine the susceptibility and transmission potential of the Fringillidae, the New World finches, to Newcastle disease virus. Our results suggest that passerine family may be susceptible to infection with the LaSota vaccine virus as a result of exposure to shedding chickens. On the other hand, Northern Cardinals (Cardinalidae) and Brown-headed Cowbirds (Icteridae) did not demonstrate the same susceptibility when exposed to chickens that were shedding the LaSota vaccine virus.

In Chapter 4, we examined the spillover dynamics of Newcastle disease virus using a field inoculation experiment, by vaccinating two backyard chicken flocks with the LaSota vaccine virus and surveying a third site as a control. In this work, we identified 13 wild bird species that demonstrated evidence of seroconversion in response to the inoculation of backyard chickens with the LaSota vaccine virus. We defined seroconversion as a response that occurred in response to field challenge with the LaSota vaccine virus as one that occurred under two distinct conditions. The first condition was fulfilled by species that demonstrated a measurable change in antibody titers from the pre-inoculation to the post inoculation period. The second was

defined by a species that demonstrated evidence of antibody titers during the post-inoculation period, but who were not captured during the pre-inoculation period as a result of migration. Across our two experimental sites and single control site, these species had 197 out of 351 captures (56%) in areas associated with backyard chickens and their coops. Given their capture rate and seroconversion status, we indicated that these species should be designated as ‘high’ risk for NDV transmission from backyard chickens. Interestingly, Chipping Sparrows, Northern Cardinals, and Tufted Titmice demonstrated the highest antibody titers as measured by hemagglutination inhibition assays – with our field results on Northern Cardinals complementing our results from our laboratory analyses.

Overall, the results of this experiment demonstrate that the transmission of lentogenic NDV between passerines and backyard chickens likely occurs more frequently than previously believed. It also demonstrates that inhibiting spillover events may also be crucial to the management of NDV outbreaks in backyard chickens, specifically those that share habitat with wild birds. Future work should replicate our field experiment in areas that have undergone consistent exposure to virulent NDV outbreaks, such as the southwestern United States, in order to determine which species may travel between backyard chicken flocks, and thus serves as bridge hosts of the virus.

Given the results of chapters 3 and 4, we performed an experimental supplemental feeding experiment to determine why wild birds were attracted to chicken coops, where they can

become infected with the virus. Specifically, we examined whether wild birds were attracted to chickens or chicken feed within the backyard chicken coops. To answer that question, we performed observations at three sites associated with backyard chickens in Athens, Georgia. We chose our backyard chicken sites based on how the chickens were managed, e.g. Site L allowed their chickens to free-roam during the day, Site S site maintained their chickens in the coops during the day, and Site C was the control, with the chickens having been recently removed. During the four-month field season, we identified the species of wild birds shared the same habitat as the chickens or chicken coops and their frequency of detection at each site. Most importantly, we quantified the species and individuals that encountered the chicken coops. For the species that encountered chicken coops, we also measured their contact rate with the coops. Of the 72 species recorded sharing the same habitat with backyard chickens, we identified 14 species of wild birds that entered the coops to consume supplemental feed. We designated these 14 species as ‘high’ risk for pathogen transmission, specifically from lentogenic Newcastle disease virus.

Interestingly, the same species that demonstrated increased antibody titers in Chapter 4, Chipping Sparrows, Northern Cardinals, and Tufted Titmice, were also the same species that had the highest contact rates across all three sites in Chapter 5. Future work should examine the role that Chipping Sparrows, Northern Cardinals, and Tufted Titmice play in the transmission of lentogenic Newcastle disease virus.